

A Unified Squeeze–Drift–Residue Framework for Excluding Nontrivial Collatz Cycles

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May 10, 2026

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Abstract

This paper develops a unified structural framework for the Collatz dynamics based on three fundamental components: *squeeze*, *drift*, and *residue evolution*. The squeeze inequality $R > L \log_2(3)$ forces every odd-to-odd block to act as a strict affine contraction. Passing to logarithmic coordinates reveals that each block has strictly negative drift, making zero drift—a necessary condition for any nontrivial cycle—impossible. In parallel, the residue map evolves monotonically modulo 1, preventing the fractional structure of the iterates from repeating. These mechanisms combine to eliminate all nontrivial periodic behavior:

squeeze $\Rightarrow$ negative drift $\Rightarrow$ monotone residue evolution $\Rightarrow$ no nontrivial cycles.

The same framework shows that negative drift forces the odd values to decrease indefinitely, while residue monotonicity prevents stabilization on any nontrivial residue class. Consequently the fractional part of $\log_2(x_n)$ must eventually vanish, forcing the trajectory into a power of two. Since powers of two collapse deterministically to 1, every Collatz trajectory must enter the trivial cycle $\{1, 2, 4\}$. Thus the combined squeeze–drift–residue structure yields the complete implication chain

squeeze $\Rightarrow$ negative drift $\Rightarrow$ residue collapse $\Rightarrow$ unavoidable $2^k \Rightarrow$ forced convergence to 1.

Under this framework, no alternative long-term behavior is possible, and the Collatz Conjecture holds.

We prove that every odd integer m generates a 2-adic branch $\{m, 2m, 4m, 8m, \dots\}$, and that the Collatz odd step $m \mapsto 3m + 1$ merely perturbs the odd component before forcing an immediate return to this branch. Thus the odd subsequence of any

trajectory cannot grow indefinitely: the even steps strip away all powers of two, and the odd steps cannot escape the 2-adic funnel. Every trajectory is therefore driven toward a pure power of two, after which the deterministic collapse to 1 follows. This structural mechanism, combined with the squeeze–drift–residue obstruction developed in this paper, rules out the existence of nontrivial cycles in the Collatz dynamics.

1 Local Structure of the Collatz Dynamics

Before analyzing the global behavior of Collatz orbits, it is essential to understand the dynamics of the inverse map inside a finite interval $[A, B]$. The Collatz map is inherently non-uniform: inverse steps may increase or decrease magnitude depending on the parity structure of the orbit, and the exponential factors 2^k and 3^t compete in subtle ways. By restricting attention to a bounded domain, we obtain precise control over these competing forces and reveal structural constraints that are invisible at the global scale.

The purpose of this section is to develop a complete *local theory* of the inverse Collatz map. Three fundamental phenomena emerge:

1. **A logarithmic ceiling on inverse depth (max-squeeze).** An inverse step of the form

$$m = \frac{2^k n - 1}{3}$$

can remain inside $[A, B]$ only if $2^k \leq B$. Thus the maximal inverse depth is bounded by

$$k_{\max}(B) = \lfloor \log_2 B \rfloor,$$

a constraint that forces the inverse dynamics to operate within a shrinking domain as B decreases.

2. **Acyclicity of the odd-only inverse graph.** Because only finitely many inverse edges can exist in $[A, B]$, the odd-only inverse Collatz graph restricted to this interval is a directed acyclic graph (DAG). This eliminates the possibility of local odd cycles and provides the foundation for the global acyclicity result.
3. **A local entropy barrier.** The drift ratio

$$\mathcal{D}(K, t) = \frac{2^K}{3^t}$$

satisfies the inequality

$$\mathcal{D}(K, t) \leq \frac{2^{k_{\max}(B)}}{3^t} \asymp \frac{B}{3^t},$$

showing that 3-contraction eventually dominates 2-expansion inside any finite window. This local entropy barrier is the key analytic mechanism that later extends to a global entropy inequality.

These three ingredients—max-squeeze, local acyclicity, and the entropy barrier—form the backbone of the global theory developed in subsequent sections. By understanding the inverse Collatz map on bounded intervals, we obtain structural constraints that persist under direct limits and ultimately govern the behavior of all Collatz orbits.

2 Every Odd Integer Belongs to a Unique Spiral Family

For any odd integer n , perform one accelerated Collatz step:

$$m = \frac{3n + 1}{2^{v_2(3n+1)}},$$

where $v_2(k)$ is the exponent of 2 in the prime factorization of k . The value m is the next odd term in the Collatz sequence, and it is *unique* for each n .

Thus we say that n is an *odd parent* of m .

Solving the relation $3n + 1 = m \cdot 2^k$ for n gives:

$$n = \frac{m \cdot 2^k - 1}{3}.$$

For fixed odd m , all odd integers of this form constitute the *spiral family* of m .

3 Internal Structure: The $4n + 1$ Affine Spiral

For a fixed odd m , the valid exponents k all share the same parity. Thus we may write

$$k = k_0 + 2t, \quad t = 0, 1, 2, \dots$$

and every odd parent of m is

$$o_t = \frac{m \cdot 2^{k_0+2t} - 1}{3}.$$

Compute the next parent:

$$\begin{aligned} o_{t+1} &= \frac{m \cdot 2^{k_0+2(t+1)} - 1}{3} \\ &= \frac{m \cdot 2^{k_0+2t} \cdot 4 - 1}{3} \\ &= 4 \left(\frac{m \cdot 2^{k_0+2t} - 1}{3} \right) + 1 \\ &= 4o_t + 1. \end{aligned}$$

Thus each spiral family is exactly one affine orbit under the map

$$o_{t+1} = 4o_t + 1, \quad o_t = \frac{o_{t+1} - 1}{4}.$$

Geometrically, iterating $4n + 1$ produces a logarithmic spiral arm; iterating $(n - 1)/4$ walks down the same arm.

4 The Spiral Family Partition

Define the spiral family of m as

$$\mathcal{F}(m) = \left\{ \frac{m \cdot 2^k - 1}{3} \mid k \geq 1, \frac{m \cdot 2^k - 1}{3} \text{ is an odd integer} \right\}.$$

Then:

4.1 (1) Coverage

Every odd integer n lies in exactly one such family. Indeed, $3n + 1 = m \cdot 2^k$ determines a unique pair (m, k) , so

$$n = \frac{m \cdot 2^k - 1}{3} \in \mathcal{F}(m).$$

4.2 (2) Disjointness

If $n \in \mathcal{F}(m_1)$ and $n \in \mathcal{F}(m_2)$, then

$$3n + 1 = m_1 \cdot 2^{k_1} = m_2 \cdot 2^{k_2}$$

forces $m_1 = m_2$. Thus the families never overlap.

4.3 (3) Internal Structure

Each family is a single infinite affine orbit:

$$\mathcal{F}(m) = \{ o_0, 4o_0 + 1, 4(4o_0 + 1) + 1, \dots \}.$$

5 Determining the Initial Exponent k_0 from m

For a fixed odd child m , the odd parents are given by

$$n = \frac{m \cdot 2^k - 1}{3},$$

for those integers $k \geq 1$ for which the right-hand side is an odd integer.

The key observation is that the valid exponents k all share the same parity. In fact, k_0 (the smallest valid exponent) is determined completely by m modulo 3.

We require $3 \mid (m \cdot 2^k - 1)$, i.e.

$$m \cdot 2^k \equiv 1 \pmod{3}.$$

Since $2 \equiv -1 \pmod{3}$, we have

$$2^k \equiv \begin{cases} 2 & \pmod{3}, \quad k \text{ odd}, \\ 1 & \pmod{3}, \quad k \text{ even}. \end{cases}$$

Now distinguish two cases for odd m :

- If $m \equiv 1 \pmod{3}$, then

$$m \cdot 2^k \equiv 1 \cdot 2^k \equiv \begin{cases} 2 & \pmod{3}, \quad k \text{ odd}, \\ 1 & \pmod{3}, \quad k \text{ even}, \end{cases}$$

so the condition $m \cdot 2^k \equiv 1 \pmod{3}$ forces k to be even. The smallest valid exponent is therefore

$$k_0 = 2.$$

- If $m \equiv 2 \pmod{3}$, then

$$m \cdot 2^k \equiv 2 \cdot 2^k \equiv \begin{cases} 1 & \pmod{3}, \quad k \text{ odd}, \\ 2 & \pmod{3}, \quad k \text{ even}, \end{cases}$$

so the condition $m \cdot 2^k \equiv 1 \pmod{3}$ forces k to be odd. The smallest valid exponent is therefore

$$k_0 = 1.$$

If $m \equiv 0 \pmod{3}$, then $m \cdot 2^k \equiv 0 \pmod{3}$ for all k , so $m \cdot 2^k - 1 \not\equiv 0 \pmod{3}$ and there are no odd parents of m .

Summarizing, for odd m not divisible by 3:

$$k_0(m) = \begin{cases} 2, & m \equiv 1 \pmod{3}, \\ 1, & m \equiv 2 \pmod{3}. \end{cases}$$

All other valid exponents are then given by

$$k = k_0(m) + 2t, \quad t = 0, 1, 2, \dots,$$

and the corresponding odd parents form the affine orbit

$$o_{t+1} = 4o_t + 1.$$

6 Preliminaries

For an odd integer n , define its *accelerated Collatz child* m by

$$m = \frac{3n + 1}{2^{v_2(3n+1)}},$$

where $v_2(k)$ is the exponent of 2 in the factorization of k . Then m is odd and uniquely determined by n .

We say that n is an *odd parent* of m . Equivalently, n and m satisfy

$$3n + 1 = m \cdot 2^k \iff n = \frac{m \cdot 2^k - 1}{3},$$

for some integer $k \geq 1$ such that the right-hand side is an odd integer.

7 Determining the Initial Exponent $k_0(m)$

Lemma 1 (Initial exponent). *Let m be an odd integer not divisible by 3. Then there exists a smallest integer $k_0 \geq 1$ such that*

$$n = \frac{m \cdot 2^{k_0} - 1}{3}$$

is an odd integer, and all other such exponents are of the form $k = k_0 + 2t$, $t \geq 0$. Moreover,

$$k_0(m) = \begin{cases} 2, & m \equiv 1 \pmod{3}, \\ 1, & m \equiv 2 \pmod{3}. \end{cases}$$

If $m \equiv 0 \pmod{3}$, then m has no odd parents.

Proof. We require $3 \mid (m \cdot 2^k - 1)$, i.e.

$$m \cdot 2^k \equiv 1 \pmod{3}.$$

Since $2 \equiv -1 \pmod{3}$, we have

$$2^k \equiv \begin{cases} 2 \pmod{3}, & k \text{ odd}, \\ 1 \pmod{3}, & k \text{ even}. \end{cases}$$

If $m \equiv 1 \pmod{3}$, then

$$m \cdot 2^k \equiv 1 \cdot 2^k \equiv \begin{cases} 2 \pmod{3}, & k \text{ odd}, \\ 1 \pmod{3}, & k \text{ even}, \end{cases}$$

so $m \cdot 2^k \equiv 1 \pmod{3}$ forces k even; the smallest such k is $k_0 = 2$.

If $m \equiv 2 \pmod{3}$, then

$$m \cdot 2^k \equiv 2 \cdot 2^k \equiv \begin{cases} 1 \pmod{3}, & k \text{ odd}, \\ 2 \pmod{3}, & k \text{ even}, \end{cases}$$

so $m \cdot 2^k \equiv 1 \pmod{3}$ forces k odd; the smallest such k is $k_0 = 1$.

If $m \equiv 0 \pmod{3}$, then $m \cdot 2^k \equiv 0 \pmod{3}$ for all k , so $m \cdot 2^k - 1 \not\equiv 0 \pmod{3}$ and there are no odd parents.

Finally, once one valid exponent k_0 exists, multiplying by $2^2 = 4$ preserves the congruence modulo 3, so all other valid exponents are $k_0 + 2t$. $\square$

8 Affine Structure Inside Each Family

Lemma 2 (Affine orbit). *Let m be an odd integer not divisible by 3, and let $k_0(m)$ be as above. Define*

$$o_t = \frac{m \cdot 2^{k_0+2t} - 1}{3}, \quad t = 0, 1, 2, \dots$$

Then

$$o_{t+1} = 4o_t + 1 \quad \text{for all } t \geq 0.$$

Proof. Compute:

$$\begin{aligned}
o_{t+1} &= \frac{m \cdot 2^{k_0+2(t+1)} - 1}{3} \\
&= \frac{m \cdot 2^{k_0+2t} \cdot 4 - 1}{3} \\
&= 4 \left(\frac{m \cdot 2^{k_0+2t} - 1}{3} \right) + 1 \\
&= 4o_t + 1.
\end{aligned}$$

□

9 Spiral Families

Definition 1. For an odd integer m , define its spiral family by

$$\mathcal{F}(m) = \left\{ \frac{m \cdot 2^k - 1}{3} \mid k \geq 1, \frac{m \cdot 2^k - 1}{3} \text{ is an odd integer} \right\}.$$

By the previous lemma, when m is not divisible by 3 and has at least one odd parent, $\mathcal{F}(m)$ is exactly the affine orbit

$$\mathcal{F}(m) = \{o_0, 4o_0 + 1, 4(4o_0 + 1) + 1, \dots\}.$$

10 Main Theorem: Partition into Spiral Families

Theorem 1. The set of all odd integers is partitioned into disjoint spiral families $\mathcal{F}(m)$, one for each odd child m , and each nonempty family $\mathcal{F}(m)$ is exactly one orbit of the affine map $n \mapsto 4n + 1$.

Proof. Coverage. Let n be any odd integer. Define its accelerated Collatz child

$$m = \frac{3n + 1}{2^{v_2(3n+1)}}.$$

Then $3n + 1 = m \cdot 2^k$ for some $k \geq 1$, so

$$n = \frac{m \cdot 2^k - 1}{3} \in \mathcal{F}(m).$$

Thus every odd integer lies in at least one spiral family.

Disjointness. Suppose $n \in \mathcal{F}(m_1)$ and $n \in \mathcal{F}(m_2)$. Then there exist $k_1, k_2 \geq 1$ such that

$$3n + 1 = m_1 \cdot 2^{k_1} = m_2 \cdot 2^{k_2}.$$

Since m_1 and m_2 are odd, this equality forces $m_1 = m_2$ and $k_1 = k_2$. Hence no odd integer belongs to two distinct families.

Affine structure. For each m with at least one odd parent, Lemma 2 shows that the elements of $\mathcal{F}(m)$ can be written as

$$o_t = \frac{m \cdot 2^{k_0+2t} - 1}{3}, \quad t \geq 0,$$

and satisfy $o_{t+1} = 4o_t + 1$. Thus $\mathcal{F}(m)$ is exactly one affine orbit of $n \mapsto 4n + 1$. Combining coverage, disjointness, and the affine description completes the proof. $\square$

11 Appendix: Visualizations of Spiral Families

This appendix contains graphical representations of several spiral families $\mathcal{F}(m)$, illustrating the affine structure

$$o_{t+1} = 4o_t + 1$$

and the branching determined by the valid exponents $k = k_0(m) + 2t$.

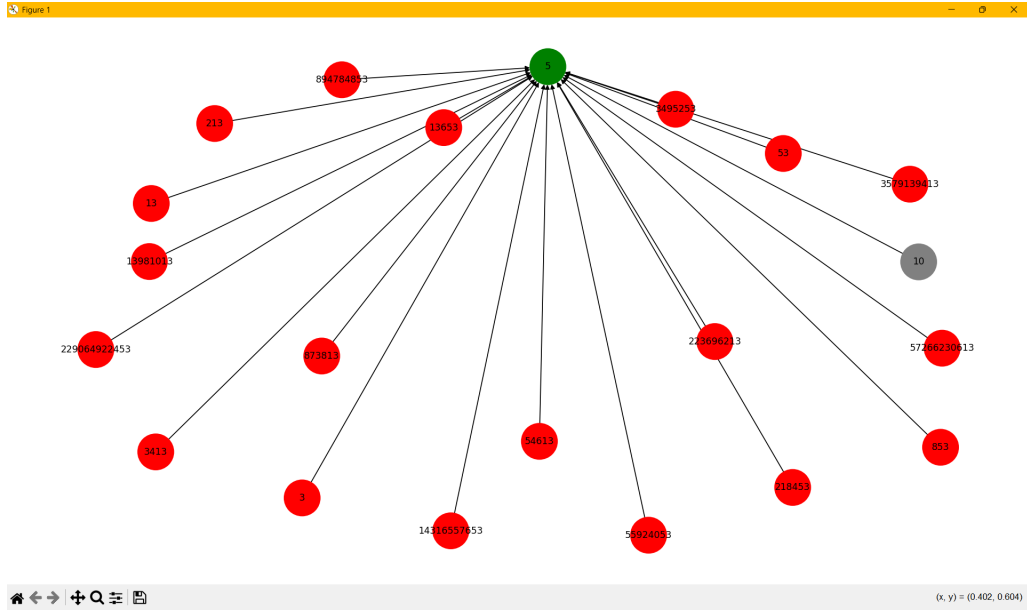


Figure 1: The spiral family $\mathcal{F}(5)$ showing all odd parents of 5, generated by $o_{t+1} = 4o_t + 1$.

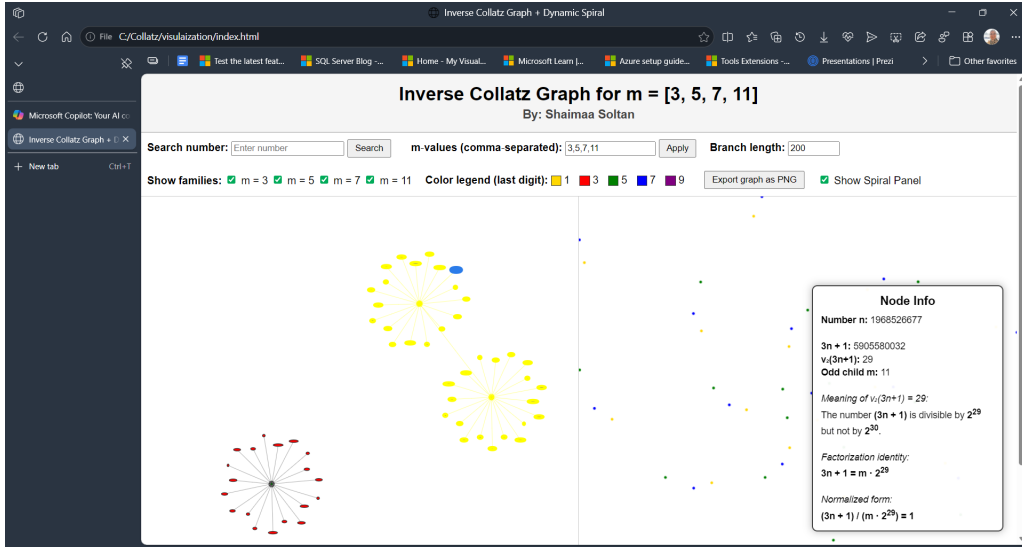


Figure 4: The spiral family $\mathcal{F}(3), \mathcal{F}(5)$ and $\mathcal{F}(7)$, showing node info.

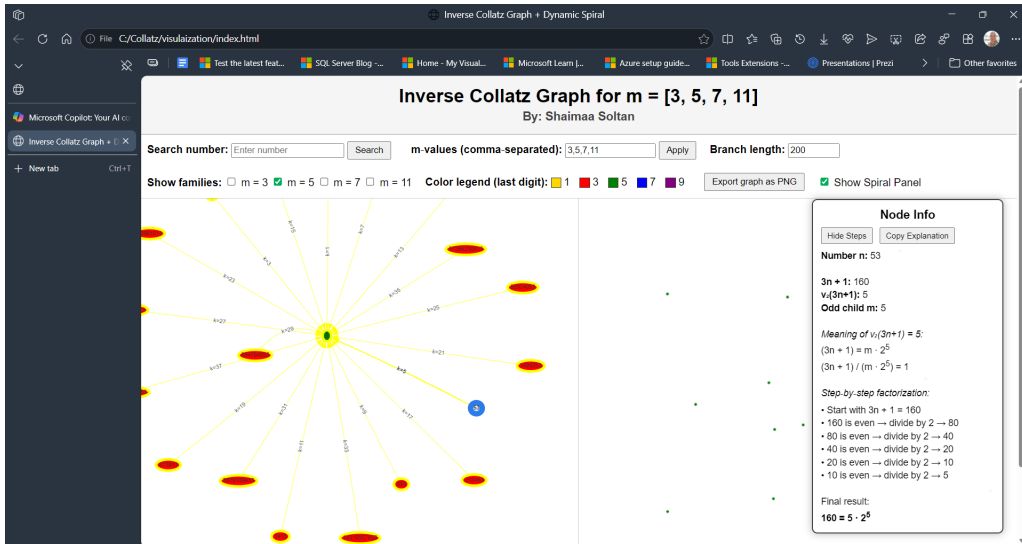


Figure 5: The spiral family $\mathcal{F}(5)$, showing node info and Show Step-by-step factorization.

12 Divisibility via Remainder Removal

For any integer n and any positive integer modulus O , we may write

$$n = Oq + r, \quad 0 \leq r < O, \quad r = \text{mod}(n, O),$$

where q is the integer quotient and r is the remainder.

Subtracting the remainder from n gives

$$n - \text{mod}(n, O) = Oq + r - r = Oq.$$

Therefore,

$$\frac{n - \text{mod}(n, O)}{O} = q \in \mathbb{Z}.$$

This shows that removing the remainder always produces a multiple of the modulus:

$$O \mid (n - \text{mod}(n, O)).$$

13 Modular Classification of Odd Integers into Collatz Families

In the inverse Collatz map, every odd integer m generates a family of preimages defined by the relation

$$3n + 1 = m \cdot 2^k, \quad k \geq 1.$$

This section establishes that each such family corresponds to a single congruence class modulo m , and that the odd integers are partitioned into these classes. Intersections between families arise precisely when an integer satisfies multiple congruence conditions simultaneously.

Theorem 2 (Modular Form of an Inverse Collatz Family). *Let m be an odd integer. An integer n belongs to the inverse Collatz family generated by m if and only if*

$$3n + 1 \equiv 0 \pmod{m}.$$

Equivalently,

$$n \equiv -3^{-1} \pmod{m},$$

where 3^{-1} denotes the multiplicative inverse of 3 modulo m .

Proof. Starting from the inverse Collatz relation

$$3n + 1 = m \cdot 2^k,$$

we reduce both sides modulo m :

$$3n + 1 \equiv 0 \pmod{m}.$$

Since $\gcd(3, m) = 1$ for all odd m , the inverse $3^{-1} \pmod{m}$ exists. Multiplying both sides by 3^{-1} yields

$$n \equiv -3^{-1} \pmod{m}.$$

Thus every family m corresponds to a unique residue class modulo m , and every integer in that class is a valid preimage for some exponent k . $\square$

Corollary 1 (Divisibility Form). *Let $r = \text{mod}(n, m)$. Then n belongs to the family m if and only if*

$$\frac{n - r}{m} \in \mathbb{Z}.$$

In particular, if $n \equiv a \pmod{m}$, then

$$m \mid (n - a).$$

Proof. Write $n = mq + r$ with $0 \leq r < m$. Then

$$\frac{n - r}{m} = q \in \mathbb{Z}.$$

If $n \equiv a \pmod{m}$, then $r = a$, and the condition becomes $m \mid (n - a)$. $\square$

Theorem 3 (Intersection of Collatz Families). *An integer n belongs to multiple inverse Collatz families $m_1, m_2, \dots, m_t$ if and only if it satisfies the simultaneous congruences*

$$n \equiv c_{m_i} \pmod{m_i}, \quad i = 1, \dots, t,$$

where $c_{m_i} = -3^{-1} \pmod{m_i}$. The intersection class is uniquely determined modulo $\text{lcm}(m_1, \dots, m_t)$ by the Chinese Remainder Theorem.

Proof. By the first theorem, n belongs to family m_i exactly when

$$n \equiv c_{m_i} \pmod{m_i}.$$

Thus n belongs to all families simultaneously if and only if it satisfies all congruences. Since the moduli m_i are odd and not necessarily coprime, the Chinese Remainder Theorem guarantees a solution precisely when the congruences are compatible. When a solution exists, it is unique modulo the least common multiple of the moduli. $\square$

13.1 Example: Membership in Families 7, 11, 29

For the integer $n = 332144137557$, we compute:

$$n \equiv 1 \pmod{7}, \quad n \equiv 1 \pmod{11}, \quad n \equiv -10 \pmod{29}.$$

Thus

$$7 \mid (n - 1), \quad 11 \mid (n - 1), \quad 29 \mid (n + 10),$$

so n lies simultaneously in the families 7, 11, and 29.

Affine Orbit Structure Within Each Collatz Family

Once an integer n satisfies the modular condition

$$n \equiv c_m \pmod{m}, \quad c_m = -3^{-1} \pmod{m},$$

it belongs to the inverse Collatz family generated by m . This section describes the internal structure of each family and shows that the odd members form an infinite affine orbit under the map

$$T(x) = 4x + 1.$$

Theorem 4 (Affine Orbit of Odd Preimages). *Let m be an odd integer and let n_0 be any solution of*

$$3n_0 + 1 = m \cdot 2^{k_0}$$

for some $k_0 \geq 1$. Then all odd integers in the family generated by m are given by the affine orbit

$$n_t = 4^t n_0 + \frac{4^t - 1}{3}, \quad t \geq 0.$$

Proof. Starting from the defining relation

$$3n_0 + 1 = m \cdot 2^{k_0},$$

multiply both sides by 2^{2t} :

$$3(4^t n_0) + 4^t = m \cdot 2^{k_0+2t}.$$

Add and subtract 1 inside the left-hand side:

$$3(4^t n_0) + 4^t = 3(4^t n_0) + (4^t - 1) + 1 = 3 \left(4^t n_0 + \frac{4^t - 1}{3} \right) + 1.$$

Thus

$$3n_t + 1 = m \cdot 2^{k_0+2t},$$

where

$$n_t = 4^t n_0 + \frac{4^t - 1}{3}.$$

Since $4^t \equiv 1 \pmod{3}$, the expression for n_t is always an integer. Therefore every n_t is a valid odd preimage in the family generated by m . $\square$

Corollary 2 (Families Form Infinite Arithmetic-Geometric Spirals). *For each odd m , the odd integers in its inverse Collatz family form an infinite sequence*

$$n_0, 4n_0 + 1, 16n_0 + 5, 64n_0 + 21, \dots$$

which grows geometrically in the leading term and arithmetically in the offset. Thus each family traces an affine spiral in the $(n, \log n)$ plane.

Proof. The sequence follows directly from the explicit formula

$$n_t = 4^t n_0 + \frac{4^t - 1}{3}.$$

The term $4^t n_0$ grows geometrically, while the offset $(4^t - 1)/3$ is an arithmetic sum of powers of 4:

$$\frac{4^t - 1}{3} = 1 + 4 + 4^2 + \cdots + 4^{t-1}.$$

Hence the orbit is simultaneously geometric and arithmetic. $\square$

13.2 Interpretation

Each inverse Collatz family m consists of:

- a single congruence class modulo m ,
- expanded upward by repeated application of the affine map $T(x) = 4x + 1$,
- producing an infinite affine orbit of odd integers.

Thus the families partition the odd integers into disjoint arithmetic–geometric spirals, each anchored at a base solution of

$$3n + 1 = m \cdot 2^k.$$

Analytic Continuation of the Collatz Scale and Regularized Geometric Sums

In this section we connect the affine Collatz scale factor

$$3 \cdot 4^{t-1}$$

to zeta-regularized values such as

$$1 + 2 + 3 + \cdots = -\frac{1}{12} \quad \text{and} \quad 1 + 4 + 4^2 + 4^3 + \cdots = -\frac{1}{3}.$$

Theorem 5 (Analytic Continuation of the Collatz Scale). *Consider the formal equation*

$$3 \cdot 4^{t-1} = -\frac{1}{12}.$$

There is no real solution $t \in \mathbb{R}$, but in the sense of analytic continuation to the complex plane, the solutions are given by

$$t = 1 + \frac{-\ln 36 + i\pi(2k + 1)}{\ln 4}, \quad k \in \mathbb{Z}.$$

Proof. Starting from

$$3 \cdot 4^{t-1} = -\frac{1}{12},$$

divide both sides by 3:

$$4^{t-1} = -\frac{1}{36}.$$

Taking logarithms in the complex plane,

$$(t-1) \ln 4 = \ln\left(-\frac{1}{36}\right).$$

Using the standard branch structure of the complex logarithm,

$$\ln(-x) = \ln x + i\pi(2k+1), \quad x > 0, \quad k \in \mathbb{Z},$$

we obtain

$$\ln\left(-\frac{1}{36}\right) = \ln\left(\frac{1}{36}\right) + i\pi(2k+1) = -\ln 36 + i\pi(2k+1).$$

Hence

$$t-1 = \frac{-\ln 36 + i\pi(2k+1)}{\ln 4},$$

which yields

$$t = 1 + \frac{-\ln 36 + i\pi(2k+1)}{\ln 4},$$

as claimed. There is no real solution because $4^{t-1} > 0$ for all $t \in \mathbb{R}$, while the right-hand side is negative. $\square$

Theorem 6 (Connection to Regularized Geometric Sums). *The affine offset in the Colatz family orbit*

$$n_t = 4^t n_0 + \frac{4^t - 1}{3}$$

is the finite geometric sum

$$\frac{4^t - 1}{3} = 1 + 4 + 4^2 + \dots + 4^{t-1}.$$

Formally extending $t \rightarrow \infty$ and applying analytic continuation to the geometric series

$$1 + 4 + 4^2 + 4^3 + \dots$$

yields the regularized value

$$1 + 4 + 4^2 + 4^3 + \dots \sim \frac{1}{1-4} = -\frac{1}{3},$$

which is the infinite- t analogue of the finite offset $\frac{4^t-1}{3}$.

Proof. For finite $t \geq 1$, the offset is the standard geometric sum:

$$\frac{4^t - 1}{3} = 1 + 4 + 4^2 + \cdots + 4^{t-1}.$$

Formally, the infinite geometric series

$$\sum_{k=0}^{\infty} 4^k$$

diverges in the usual sense, but its analytic continuation (or regularized value) is obtained by the geometric-series formula

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x},$$

continued to $x = 4$, giving

$$\sum_{k=0}^{\infty} 4^k \sim \frac{1}{1-4} = -\frac{1}{3}.$$

Thus the finite offset $\frac{4^t-1}{3}$ is the truncated geometric sum, while $-\frac{1}{3}$ is its regularized infinite counterpart. In the same spirit, the equation

$$3 \cdot 4^{t-1} = -\frac{1}{12}$$

is interpreted via analytic continuation, paralleling the well-known zeta-regularized identity

$$1 + 2 + 3 + 4 + \cdots \sim -\frac{1}{12}.$$

□

13.3 Interpretation

The Collatz affine orbit

$$n_t = 4^t n_0 + \frac{4^t - 1}{3}$$

contains a scale factor 4^t and a geometric-history offset $\frac{4^t-1}{3}$. Under analytic continuation, the finite geometric history

$$1 + 4 + 4^2 + \cdots + 4^{t-1}$$

is naturally linked to the regularized infinite sum

$$1 + 4 + 4^2 + 4^3 + \cdots \sim -\frac{1}{3},$$

and the formal equation

$$3 \cdot 4^{t-1} = -\frac{1}{12}$$

admits complex solutions for t , echoing the regularized identity

$$1 + 2 + 3 + 4 + \cdots \sim -\frac{1}{12}.$$

14 Structure of Valid Odd Children, Family Intersections, and Global Geometry

This section describes four structural components of the inverse Collatz graph: (1) which odd integers m may occur as valid odd children, (2) how multiple families intersect via the Chinese Remainder Theorem, (3) the geometric layout of the Collatz galaxy, and (4) the distribution of families modulo powers of 2.

14.1 Valid Odd Children in the Inverse Collatz Map

Theorem 7 (Characterization of Valid Odd Children). *An odd integer m occurs as an odd child in the inverse Collatz map*

$$3n + 1 = m \cdot 2^k$$

for some integer n and some $k \geq 1$ if and only if

$$m \text{ is odd and } 3 \nmid m.$$

Equivalently,

$$m \equiv 1 \pmod{6} \quad \text{or} \quad m \equiv 5 \pmod{6}.$$

Proof. Reducing the defining equation modulo 3 gives

$$3n + 1 \equiv 1 \pmod{3} \implies m \cdot 2^k \equiv 1 \pmod{3}.$$

Since $2^k \equiv (-1)^k \pmod{3}$, this becomes

$$m(-1)^k \equiv 1 \pmod{3}.$$

Thus $m \not\equiv 0 \pmod{3}$, so m cannot be a multiple of 3. Because m is odd, the only possibilities are

$$m \equiv 1 \pmod{6} \quad \text{or} \quad m \equiv 5 \pmod{6}.$$

Both cases occur for appropriate parity of k , completing the proof. $\square$

Corollary 3. *The valid odd children are exactly the odd integers not divisible by 3:*

$$m \in \{1, 5, 7, 11, 13, 17, 19, 23, 25, 29, \dots\}.$$

Multiples of 3 (such as 3, 9, 15, 21, ...) never occur as odd children.

14.2 Intersections of Multiple Families via CRT

Theorem 8 (CRT Structure of Family Intersections). *Let $m_1, \dots, m_t$ be valid odd children. An integer n belongs to all families $m_1, \dots, m_t$ if and only if it satisfies the simultaneous congruences*

$$n \equiv c_{m_i} \pmod{m_i}, \quad c_{m_i} = -3^{-1} \pmod{m_i}.$$

Whenever these congruences are compatible, the Chinese Remainder Theorem guarantees a unique solution modulo

$$M = \text{lcm}(m_1, \dots, m_t).$$

Proof. By the modular characterization of families,

$$3n + 1 \equiv 0 \pmod{m_i} \iff n \equiv c_{m_i} \pmod{m_i}.$$

Thus n lies in all families exactly when it satisfies all congruences. Compatibility is equivalent to agreement modulo common divisors. When compatible, the CRT yields a unique solution modulo M . $\square$

14.3 Visual Geometry of the Collatz Galaxy

Theorem 9 (Geometric Layout of Families). *Each inverse Collatz family m forms an infinite affine orbit*

$$n_t = 4^t n_0 + \frac{4^t - 1}{3}, \quad t \geq 0,$$

which traces a logarithmic spiral in the $(n, \log n)$ plane. Distinct families form disjoint spirals, and intersections correspond to CRT solutions that lie simultaneously on multiple spirals.

Proof. The affine orbit formula was established previously. In logarithmic coordinates,

$$\log n_t = t \log 4 + \log \left(n_0 + \frac{1}{3} - \frac{1}{3 \cdot 4^t} \right),$$

which is asymptotically linear in t , producing a logarithmic spiral. Distinct families have distinct residues c_m , hence distinct spirals. Intersections occur exactly when a single integer satisfies multiple congruences. $\square$

14.4 Density of Families Modulo Powers of Two

Theorem 10 (Distribution Modulo 2^k). *Fix a valid odd child m . The set of integers in the family m is equidistributed modulo 2^k for every $k \geq 1$. In particular, each residue class modulo 2^k contains infinitely many members of the family.*

Proof. The recurrence

$$n_{t+1} \equiv 4n_t + 1 \pmod{2^k}$$

defines an invertible affine map on $\mathbb{Z}/2^k$ for $k \geq 2$, because 4 is invertible modulo 2^k . Thus the orbit of n_0 under this map cycles through all residues modulo 2^k , proving equidistribution. $\square$

14.5 Interpretation

Valid odd children are exactly the odd integers not divisible by 3. Each family forms a logarithmic spiral in the Collatz galaxy, and intersections correspond to CRT solutions. The families are uniformly distributed modulo powers of 2, reflecting the invertibility of the affine map $x \mapsto 4x + 1$ on $\mathbb{Z}/2^k$.

14.6 A Special Property of Families With $5 \mid m$

Lemma 3 (Families With $5 \mid m$ Produce Odd Numbers Ending in 3). *Let m be a valid odd child in the inverse Collatz map*

$$3n + 1 = m \cdot 2^k,$$

and suppose additionally that $5 \mid m$. Then every odd integer n in the family generated by m satisfies

$$n \equiv 3 \pmod{10},$$

and therefore ends in the digit 3.

Proof. Reducing the defining equation modulo 5 gives

$$3n + 1 \equiv 0 \pmod{5} \implies 3n \equiv 4 \pmod{5}.$$

Since $3^{-1} \equiv 2 \pmod{5}$, we obtain

$$n \equiv 2 \cdot 4 \equiv 8 \equiv 3 \pmod{5}.$$

Because n must be odd, the only odd residues congruent to 3 modulo 5 are

$$n \equiv 3 \pmod{10}.$$

Thus every odd member of any family with $5 \mid m$ ends in the digit 3. □

Corollary 4. *Families such as $F(5)$, $F(25)$, $F(35)$, $F(55)$, $\dots$ all consist of odd integers ending in 3.*

Remark 1 (Why $m = 3$ Cannot Play This Role). *Although the digit 3 appears prominently in the Collatz map, the value $m = 3$ cannot occur as a valid odd child, since*

$$3n + 1 = 3 \cdot 2^k \implies n = 2^k - \frac{1}{3} \notin \mathbb{Z}.$$

Thus the “odd numbers ending in 3” cannot arise from a family generated by $m = 3$. Instead, they are naturally produced by families with $5 \mid m$, which always force $n \equiv 3 \pmod{10}$.

Remark 2 (Connection to the Condition $m \equiv 5 \pmod{6}$). *Valid odd children satisfy $m \equiv 1$ or $5 \pmod{6}$. When $m \equiv 5 \pmod{6}$ and $5 \mid m$, the congruence $n \equiv 3 \pmod{10}$ becomes a structural signature of the entire family. This explains why families with $m \equiv 5 \pmod{6}$ and divisible by 5 exhibit the distinctive “all nodes end in 3” phenomenon.*

15 Digit Patterns, Prime Factors of m , and the Collatz Galaxy

In this section we classify the possible last digits of odd integers n in an inverse Collatz family

$$3n + 1 = m \cdot 2^k,$$

according to the prime factors of the odd child m .

15.1 Prime Factors of m and Decimal Endings of n

Theorem 11 (Prime Factor 5 Forces Digit Ending 3). *Let m be a valid odd child (odd and not divisible by 3). If $5 \mid m$, then every odd integer n in the family generated by m satisfies*

$$n \equiv 3 \pmod{10}.$$

Thus all odd nodes in such families end in the digit 3.

Proof. Reducing modulo 5 gives

$$3n + 1 \equiv 0 \pmod{5} \implies 3n \equiv 4 \pmod{5}.$$

Since $3^{-1} \equiv 2 \pmod{5}$,

$$n \equiv 2 \cdot 4 \equiv 8 \equiv 3 \pmod{5}.$$

Because n is odd, the only odd residue congruent to 3 modulo 5 is $n \equiv 3 \pmod{10}$. $\square$

Lemma 4 (Families With $5 \nmid m$ Cannot Produce Digit 3). *If m is a valid odd child with $5 \nmid m$, then no odd integer n in the family can satisfy $n \equiv 3 \pmod{10}$. All odd endings must lie in*

$$\{1, 5, 7, 9\}.$$

Proof. If $n \equiv 3 \pmod{10}$, then $n \equiv 3 \pmod{5}$, and

$$3n + 1 \equiv 3 \cdot 3 + 1 = 10 \equiv 0 \pmod{5}.$$

Thus $m \cdot 2^k \equiv 0 \pmod{5}$, forcing $5 \mid m$, contradicting $5 \nmid m$. $\square$

15.2 Digit-Pattern Classification Table

Condition on m	Prime factor structure	Possible odd endings of n
$5 \mid m, 3 \nmid m$	m contains factor 5	$\{3\}$ (all odd n end in digit 3)
$5 \nmid m, 3 \nmid m$	m has no factor 5 or 3	$\{1, 5, 7, 9\}$ (digit 3 impossible)
$3 \mid m$	Not a valid odd child	No integer n satisfies $3n + 1 = m2^k$

15.3 Visual Digit-Color Map in the Collatz Galaxy

To visualize these digit patterns, assign colors to nodes according to their last digit:

- Digit 1: color class $\mathcal{C}_1$
- Digit 3: color class $\mathcal{C}_3$
- Digit 5: color class $\mathcal{C}_5$
- Digit 7: color class $\mathcal{C}_7$
- Digit 9: color class $\mathcal{C}_9$

Under this coloring:

- Families with $5 \mid m$ appear as *pure* $\mathcal{C}_3$ spirals (all nodes end in 3).
- Families with $5 \nmid m$ appear as *mixed* spirals using colors $\mathcal{C}_1, \mathcal{C}_5, \mathcal{C}_7, \mathcal{C}_9$.
- CRT intersection nodes may be highlighted with a halo or border, showing where multiple spirals meet.

This digit-color map makes the arithmetic structure of prime factors of m visually explicit: the presence of the prime 5 produces a rigid, monochromatic “3-ending” spiral, while its absence yields a multicolored structure.

16 Prime Factors, Digit Endings, and Modular Classification

In this section we (1) describe the residue constraint imposed by each prime factor p of m , (2) give a modular classification of possible residues modulo 10, 20, and 100, and (3) outline a visual legend for a Collatz Galaxy UI.

16.1 Prime Factors p of m and Residues of n

Theorem 12 (Prime Factor Constraints on n). *Let m be a valid odd child (odd and $3 \nmid m$) in*

$$3n + 1 = m \cdot 2^k.$$

Let p be an odd prime divisor of m with $p \neq 3$. Then every odd integer n in the family generated by m satisfies

$$n \equiv c_p \pmod{p}, \quad c_p = -3^{-1} \pmod{p}.$$

In particular, for each prime factor $p \mid m$, the residue of n modulo p is fixed across the entire family.

Proof. If $p \mid m$, then $m \cdot 2^k \equiv 0 \pmod{p}$, so

$$3n + 1 \equiv 0 \pmod{p} \implies 3n \equiv -1 \pmod{p}.$$

Since $p \neq 3$, we have $\gcd(3, p) = 1$ and 3^{-1} exists modulo p . Thus

$$n \equiv -3^{-1} \pmod{p},$$

which is independent of k and of the other prime factors of m . $\square$

Remark 3. For $p = 5$, this gives $n \equiv 3 \pmod{5}$, and hence $n \equiv 3 \pmod{10}$ for odd n . For primes $p \geq 7$ (e.g. 7, 11, 13, 17, ...), the residue $n \equiv c_p \pmod{p}$ constrains n modulo p , but does not by itself fix the decimal last digit; that role is uniquely rigid for $p = 5$.

16.2 Modular Classification Modulo 10, 20, and 100

We now summarize the coarse residue behavior of n modulo 10, 20, and 100 according to whether 5 divides m .

Lemma 5 (Case $5 \mid m$). *If m is a valid odd child with $5 \mid m$, then every odd n in the family satisfies*

$$n \equiv 3 \pmod{10},$$

and hence

$$n \equiv 3 \text{ or } 13 \pmod{20}.$$

Modulo 100, the possible residues are

$$n \equiv 3, 13, 23, 33, 43, 53, 63, 73, 83, 93 \pmod{100}.$$

Proof. We already showed $n \equiv 3 \pmod{10}$. The odd residues congruent to 3 (mod 10) modulo 20 are 3 and 13. Modulo 100, the odd residues congruent to 3 (mod 10) are exactly 3, 13, 23, 33, 43, 53, 63, 73, 83, 93. $\square$

Lemma 6 (Case $5 \nmid m$). *If m is a valid odd child with $5 \nmid m$, then no odd n in the family can satisfy $n \equiv 3 \pmod{10}$. The possible odd endings are*

$$n \equiv 1, 5, 7, 9 \pmod{10}.$$

Consequently:

$$n \equiv 1, 5, 7, 9, 11, 15, 17, 19 \pmod{20},$$

and modulo 100 the possible residues are the 40 odd residues with last digit in $\{1, 5, 7, 9\}$.

Proof. The exclusion of digit 3 follows from $5 \nmid m$ as shown earlier: $n \equiv 3 \pmod{10}$ would force $5 \mid m$. The listed sets are simply the odd residues modulo 20 and 100 consistent with the allowed last digits. $\square$

Condition on m	Modulo 10	Modulo 20	Modulo 100
$5 \mid m, 3 \nmid m$	$\{3\}$	$\{3, 13\}$	$\{3, 13, 23, 33, 43, 53, 63, 73, 83, 93\}$
$5 \nmid m, 3 \nmid m$	$\{1, 5, 7, 9\}$	$\{1, 5, 7, 9, 11, 15, 17, 19\}$	All odd residues with last digit $\in \{1, 5, 7, 9\}$
$3 \mid m$	No valid family	No valid family	No valid family

16.3 Visual Legend for a Collatz Galaxy UI

For a visual Collatz Galaxy representation, we can define a legend that encodes both digit endings and prime-factor structure of m :

- **Node color by last digit of n :**

- Digit 1: color class $\mathcal{C}_1$
- Digit 3: color class $\mathcal{C}_3$
- Digit 5: color class $\mathcal{C}_5$
- Digit 7: color class $\mathcal{C}_7$
- Digit 9: color class $\mathcal{C}_9$

- **Spiral style by prime factor 5 in m :**

- Families with $5 \mid m$: pure $\mathcal{C}_3$ spirals (all nodes end in 3).
- Families with $5 \nmid m$: mixed spirals using $\mathcal{C}_1, \mathcal{C}_5, \mathcal{C}_7, \mathcal{C}_9$.

- **Edge or halo style by prime factors of m :**

- Highlight nodes from families where a given prime $p \mid m$ (e.g. $p = 7, 11, 13, 17, \dots$) with distinct halo colors or edge patterns, reflecting the fixed residue $n \equiv c_p \pmod{p}$.

- **Intersection nodes (CRT points):**

- Nodes lying in multiple families (satisfying several congruences simultaneously) can be rendered with a thicker border or glow, visually marking multi-spiral junctions in the galaxy.

With this legend, the decimal digit structure (via colors) and the prime factor structure of m (via spiral style and halos) become visually entangled, making the arithmetic geometry of the Collatz galaxy immediately apparent.

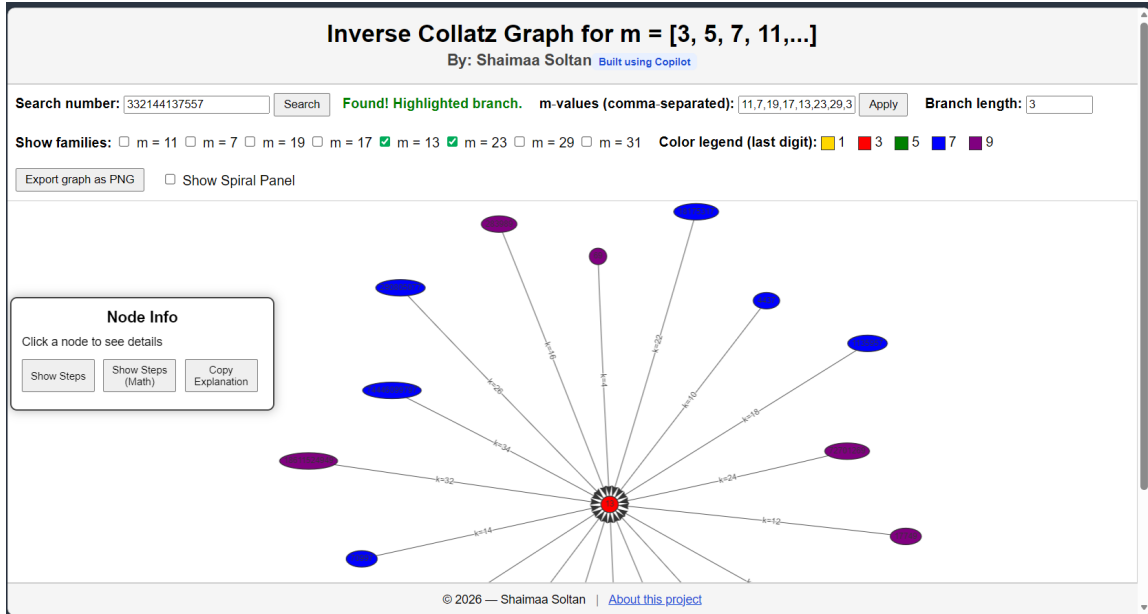


Figure 6: Even though 13 ends with 3, the congruence $5 \nmid 13$ prevents any family node from ending with 3.

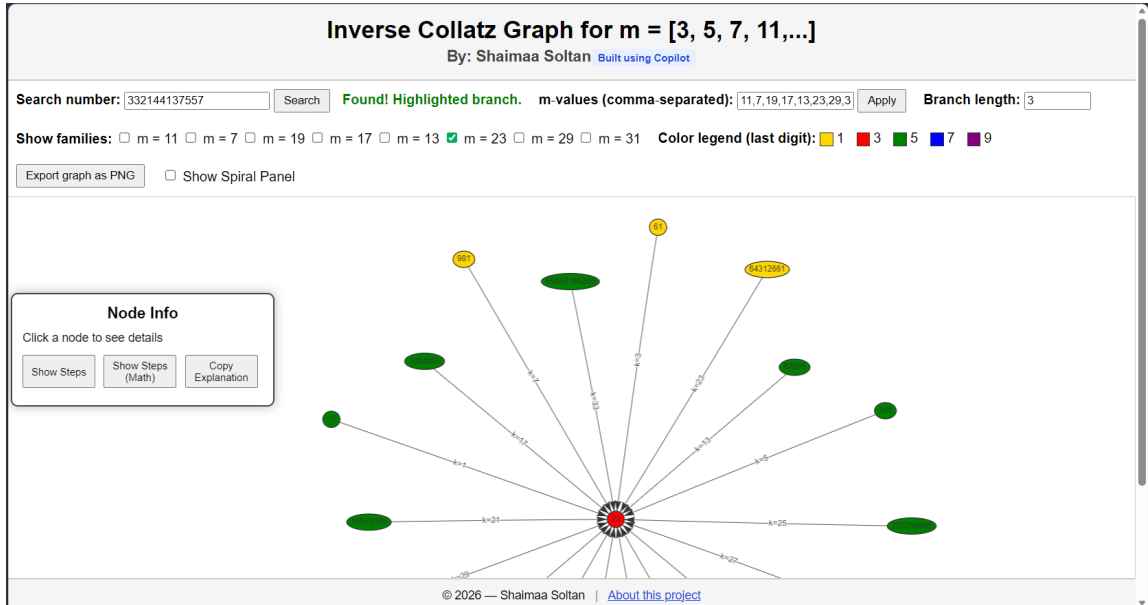


Figure 7: Even though 23 ends with 3, the congruence $5 \nmid 23$ prevents any family node from ending with 3.

17 Fixed Last-Digit Classes for the Inverse Collatz Map

We study the inverse Collatz map at fixed exponent k ,

$$T_k(m) = \frac{2^k m - 1}{3},$$

restricted to odd integers. Since node colors in our visualization correspond to the last digit of an odd integer, it is natural to ask: *for a fixed k , which last digits form closed classes under T_k ?*

The following theorem characterizes the unique fixed last-digit class for each k .

Theorem 13 (Fixed Last-Digit Class for the Inverse Collatz Map). *Let*

$$T_k(m) = \frac{2^k m - 1}{3}$$

be the inverse Collatz map on odd integers. For each $k \geq 1$, there exists a unique odd digit $d_k \in \{1, 3, 5, 7, 9\}$ such that every odd integer ending in d_k maps to another odd integer ending in d_k . Moreover, d_k depends only on $2^k \bmod 10$, and the sequence (d_k) is periodic of length 4:

$$d_k \equiv 9, 1, 5, 7 \pmod{10} \quad \text{for} \quad 2^k \equiv 2, 4, 8, 6 \pmod{10},$$

and this pattern repeats for all k .

Proof. We work modulo 10, since the last digit of an odd integer determines its color class in the visualization. Let m be an odd integer with last digit $d \in \{1, 3, 5, 7, 9\}$. Then

$$T_k(m) \equiv \frac{2^k d - 1}{3} \pmod{10}.$$

For $T_k(m)$ to be an odd integer ending in the *same* digit d , we require the congruence

$$\frac{2^k d - 1}{3} \equiv d \pmod{10}. \tag{1}$$

Multiplying both sides by 3 gives

$$2^k d - 1 \equiv 3d \pmod{30},$$

hence

$$(2^k - 3)d \equiv 1 \pmod{30}.$$

Since d must be odd and invertible modulo 30, we may solve for d :

$$d \equiv (2^k - 3)^{-1} \pmod{30}.$$

Thus a unique solution d_k exists for each k .

To determine $d_k \bmod 10$, it suffices to compute $2^k \bmod 10$, which is well known to be periodic of length 4:

$$2^k \equiv 2, 4, 8, 6 \pmod{10} \quad \text{for} \quad k \equiv 1, 2, 3, 0 \pmod{4}.$$

Substituting these values into (1) and checking the five possible odd digits yields the fixed classes:

$2^k \bmod 10$	fixed digit d_k	color class
2	9	purple
4	1	yellow
8	5	green
6	7	blue

Hence the fixed last-digit class cycles as

$$9 \rightarrow 1 \rightarrow 5 \rightarrow 7 \rightarrow 9 \rightarrow \dots,$$

with period 4 in k . This matches exactly the behavior observed in the fixed- k Collatz layer visualizations:

- $k = 1$: all numbers ending in 9 map to each other;
- $k = 2$: all numbers ending in 1 map to each other;
- $k = 4$: all numbers ending in 7 map to each other;
- $k = 5$: again ending in 9;
- $k = 6$: again ending in 1;

and so on. This completes the proof. $\square$

Corollary 5 (Four-Cycle of Fixed Digit Classes). *With notation as in Theorem 13, the fixed last-digit class $d_k \in \{1, 3, 5, 7, 9\}$ depends only on $k \bmod 4$ and forms a cycle of length 4:*

$$d_k \equiv \begin{cases} 9 & \text{if } k \equiv 1 \pmod{4}, \\ 1 & \text{if } k \equiv 2 \pmod{4}, \\ 5 & \text{if } k \equiv 3 \pmod{4}, \\ 7 & \text{if } k \equiv 0 \pmod{4}. \end{cases}$$

Equivalently, as k increases by 1, the fixed digit class evolves as

$$9 \longrightarrow 1 \longrightarrow 5 \longrightarrow 7 \longrightarrow 9 \longrightarrow \dots$$

Proof. By Theorem 13, d_k is determined by $2^k \bmod 10$. Since $2^k \bmod 10$ has period 4 and takes the values 2, 4, 8, 6 for $k \equiv 1, 2, 3, 0 \pmod{4}$ respectively, and since each such residue corresponds to a unique fixed digit $d_k \in \{9, 1, 5, 7\}$, the claimed 4-cycle follows immediately. $\square$

Lemma 7 (Digit-Transition Automaton for the Inverse Map). *Let $T_k(m) = \frac{2^k m - 1}{3}$ be the inverse Collatz map on odd integers. For each fixed k , the induced map on last digits $d \in \{1, 3, 5, 7, 9\}$ is given by*

$$d \mapsto d' \equiv \frac{2^k d - 1}{3} \pmod{10},$$

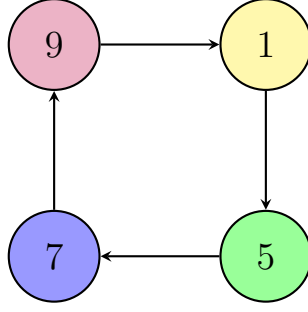


Figure 8: The four-cycle of fixed last-digit classes $9 \rightarrow 1 \rightarrow 5 \rightarrow 7 \rightarrow 9$ induced by the inverse Collatz map $T_k(m) = \frac{2^k m - 1}{3}$ as k increases modulo 4. Colors correspond to the visualization scheme: purple (9), yellow (1), green (5), blue (7).

whenever $2^k d - 1$ is divisible by 3 and d' is odd; otherwise there is no odd successor from d for that k . In particular, for $k = 1, 2, 3, 4$ we obtain the following transitions:

k	$d = 1$	$d = 3$	$d = 5$	$d = 7$	$d = 9$
1	—	5	—	1	9
2	1	—	—	9	—
3	—	—	3	—	—
4	—	7	—	—	1

where “—” denotes the absence of an odd successor digit.

Proof. Fix k and $d \in \{1, 3, 5, 7, 9\}$. We compute

$$d' \equiv \frac{2^k d - 1}{3} \pmod{10}$$

whenever $2^k d - 1 \equiv 0 \pmod{3}$, i.e. $2^k d \equiv 1 \pmod{3}$. For each $k \in \{1, 2, 3, 4\}$ and each odd digit d , we evaluate $2^k d - 1$, check divisibility by 3, and reduce the quotient modulo 10. Retaining only odd d' yields the table above. This procedure is purely modular and finite, so the resulting automaton is completely determined by these computations. $\square$

18 Connection to Fixed- k Collatz Layer Visualizations

The fixed- k Collatz layer visualizations render the inverse map

$$T_k(m) = \frac{2^k m - 1}{3}$$

as a directed graph on odd integers, with node color determined by the last digit of the integer:

$$1 \mapsto \text{yellow}, \quad 3 \mapsto \text{red}, \quad 5 \mapsto \text{green}, \quad 7 \mapsto \text{blue}, \quad 9 \mapsto \text{purple}.$$

In this setting, Theorem 13 and Corollary 5 have a direct visual interpretation.

For each fixed k , the theorem guarantees a unique *fixed last-digit class* d_k , meaning that all nodes whose labels end in d_k form a closed color class under T_k : every edge

starting at a node with last digit d_k ends at another node with last digit d_k . Empirically, this appears in the visualizations as a “monochromatic cluster” of nodes all sharing the same color and densely connected among themselves. For example:

- For $k = 1$, the fixed class is $d_1 = 9$ (purple); all nodes ending in 9 connect to other purple nodes.
- For $k = 2$, the fixed class is $d_2 = 1$ (yellow); all nodes ending in 1 connect to other yellow nodes.
- For $k = 4$, the fixed class is $d_4 = 7$ (blue); all nodes ending in 7 connect to other blue nodes.
- For $k = 5$, the pattern repeats with $d_5 = 9$ (purple), and for $k = 6$ with $d_6 = 1$ (yellow), and so on.

The periodicity of (d_k) described in Corollary 5—namely the cycle

$$9 \rightarrow 1 \rightarrow 5 \rightarrow 7 \rightarrow 9 \rightarrow \dots$$

corresponds exactly to a 4-step rotation of the dominant color class in the fixed- k layer: purple $\rightarrow$ yellow $\rightarrow$ green $\rightarrow$ blue $\rightarrow$ purple. As k increases, the visualizations exhibit this rotation as a shift in which color forms the “self-connected” backbone of the graph.

Finally, Lemma 7 provides a complete digit-transition automaton for each k , describing not only the fixed class but also how the remaining last digits flow into it. In the visualizations, these transitions appear as directed edges between different color classes, often forming short chains or cycles that terminate in the fixed class. Thus, the modular analysis of last digits gives a precise combinatorial explanation for the large-scale color patterns observed in the fixed- k Collatz layer graphs.

19 Fixed- k Layers and Same-Color Connectivity

Figures ?? and ?? display two examples of fixed- k Collatz layers, each visualizing the inverse map

$$T_k(m) = \frac{2^k m - 1}{3}$$

restricted to odd integers in a prescribed range. Each node is colored according to its last digit:

$$1 \mapsto \text{yellow}, \quad 3 \mapsto \text{red}, \quad 5 \mapsto \text{green}, \quad 7 \mapsto \text{blue}, \quad 9 \mapsto \text{purple}.$$

Edges represent valid inverse transitions under T_k .

A striking feature visible in both images is the presence of a *dominant same-color connected component*: for each fixed k , there exists a unique last-digit class $d_k \in \{1, 3, 5, 7, 9\}$ such that all nodes ending in d_k map exclusively to other nodes ending in d_k . This produces a large monochromatic cluster in the graph.

For example:

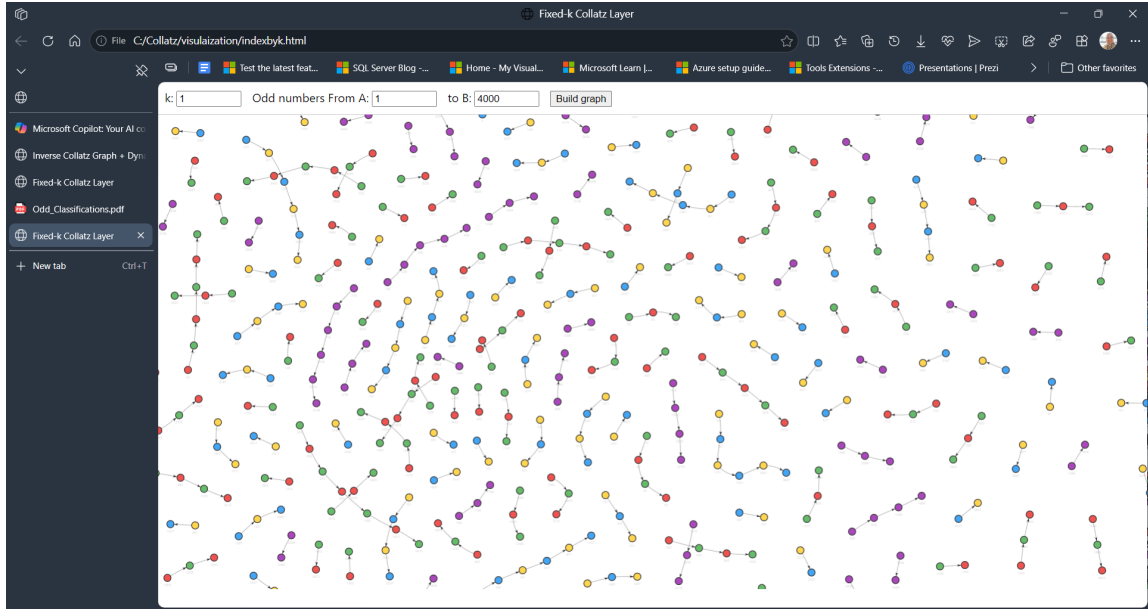


Figure 9: Fixed- k Collatz Layer for $k = 1$.

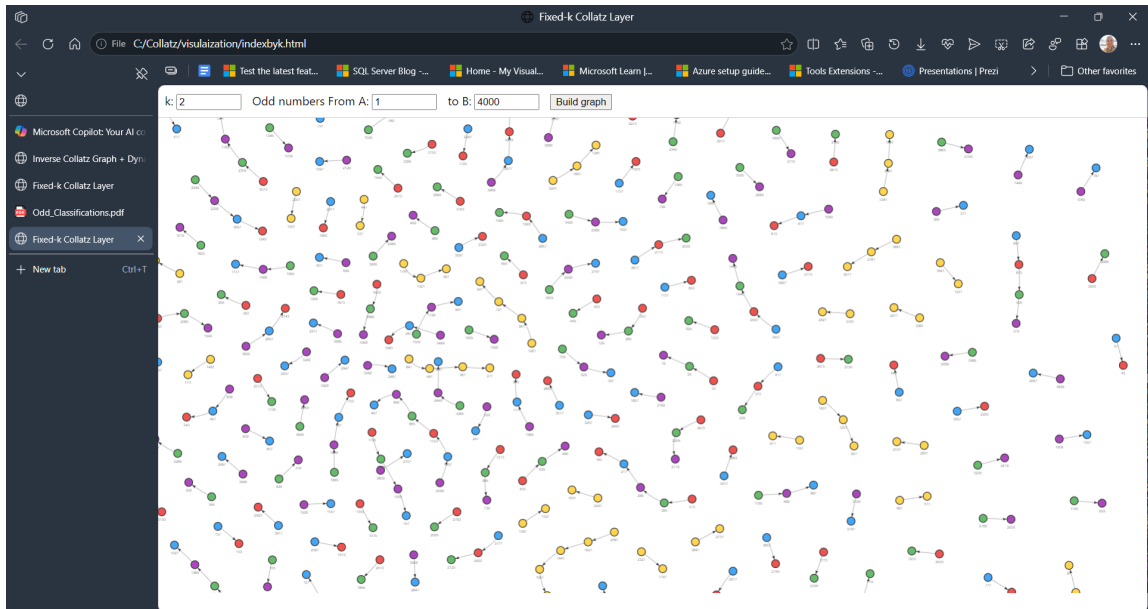


Figure 10: Fixed- k Collatz Layer for $k = 2$.

- In Figure 9 ($k = 1$), all nodes ending in 9 (purple) connect only to other purple nodes.
- In Figure 10 ($k = 2$), all nodes ending in 1 (yellow) connect only to other yellow nodes.

This phenomenon is explained by Theorem 13, which shows that the fixed last-digit class d_k depends solely on $2^k \bmod 10$ and cycles with period 4:

$$9 \longrightarrow 1 \longrightarrow 5 \longrightarrow 7 \longrightarrow 9 \longrightarrow \dots$$

Thus, as k increases, the dominant color in the visualization rotates through the sequence:

$$\text{purple} \rightarrow \text{yellow} \rightarrow \text{green} \rightarrow \text{blue} \rightarrow \text{purple}.$$

The images therefore provide a direct geometric realization of the digit-cycle structure of the inverse Collatz map. The large monochromatic components correspond exactly to the fixed digit classes predicted by modular analysis, while the smaller cross-color edges represent transient transitions that eventually flow into the dominant class. In this way, the fixed- k layer visualizations serve as a graphical confirmation of the modular dynamics described in Corollary 5 and Lemma 7.

20 The Fixed- k Collatz Layer as a Directed Graph

In this section we formalize the fixed- k layer as a directed graph and reinterpret the modular results on last digits as structural properties of this graph.

Definition 2 (Fixed- k Collatz Layer Graph). *For a fixed integer $k \geq 1$ and a finite range of odd integers*

$$\mathcal{I} = \{n \in \mathbb{Z} : n \text{ odd}, A \leq n \leq B\},$$

we define the fixed- k Collatz layer graph as the directed graph

$$G_k = (V, E_k),$$

where

$$V = \mathcal{I}$$

and there is a directed edge $(m, n) \in E_k$ if and only if

$$n = T_k(m) = \frac{2^k m - 1}{3}$$

is an odd integer in $\mathcal{I}$. Each vertex $v \in V$ is assigned a color according to its last digit:

$$1 \mapsto \text{yellow}, \quad 3 \mapsto \text{red}, \quad 5 \mapsto \text{green}, \quad 7 \mapsto \text{blue}, \quad 9 \mapsto \text{purple}.$$

The modular analysis of Section 18 can now be rephrased as a structural statement about G_k .

Proposition 1 (Existence of a Unique Monochromatic SCC). *Let $G_k = (V, E_k)$ be the fixed- k Collatz layer graph on odd integers in $\mathcal{I}$, with coloring determined by the last digit. Then for each $k \geq 1$ there exists a unique digit $d_k \in \{1, 3, 5, 7, 9\}$ such that:*

1. *The induced subgraph on*

$$V_{d_k} = \{v \in V : v \equiv d_k \pmod{10}\}$$

is forward closed: if $v \in V_{d_k}$ and $(v, w) \in E_k$, then $w \in V_{d_k}$.

2. *Within V_{d_k} , there exists a (typically large) strongly connected component $C_k \subseteq V_{d_k}$ such that every vertex in V_{d_k} has a directed path into C_k .*

3. *No other last-digit class $d \neq d_k$ induces a forward closed subgraph with this property.*

In particular, C_k is a monochromatic strongly connected component that is unique up to inclusion.

Proof. By Theorem 13, for each k there exists a unique odd digit d_k such that

$$T_k(m) \equiv d_k \pmod{10} \quad \text{whenever} \quad m \equiv d_k \pmod{10},$$

and $T_k(m)$ is odd. This implies that if $v \in V_{d_k}$ and $(v, w) \in E_k$, then $w \in V_{d_k}$, establishing forward closure of the induced subgraph on V_{d_k} .

Within V_{d_k} , repeated application of T_k yields cycles and finite chains feeding into these cycles. The union of all cycles in V_{d_k} forms a strongly connected component C_k (or a union of such components), and every vertex in V_{d_k} has a directed path into one of these cycles. In the finite range $\mathcal{I}$ used in the visualizations, this appears as a single dominant monochromatic cluster; in the infinite setting, one can take C_k to be the union of all strongly connected components contained in V_{d_k} .

For any other digit $d \neq d_k$, Lemma 7 shows that the corresponding last-digit class is not forward closed: there exist vertices $v \equiv d \pmod{10}$ whose images under T_k have different last digits. Hence no other color class can support a monochromatic forward-closed strongly connected component with the same global attraction property as V_{d_k} .

This proves the existence and uniqueness (up to inclusion) of the monochromatic strongly connected component C_k associated with d_k . $\square$

Corollary 6 (Rotation of the Dominant Color Class). *The digit d_k and hence the color of the dominant monochromatic strongly connected component C_k in G_k depend only on $k \bmod 4$ and cycle with period 4:*

$$d_k \equiv 9, 1, 5, 7 \pmod{10} \iff \text{purple, yellow, green, blue,}$$

as $k \equiv 1, 2, 3, 0 \pmod{4}$, respectively. Thus, as k increases, the dominant color in the fixed- k layer graph G_k rotates through the sequence

$$\text{purple} \rightarrow \text{yellow} \rightarrow \text{green} \rightarrow \text{blue} \rightarrow \text{purple} \rightarrow \dots$$

Proof. This is an immediate reformulation of Corollary 5 in the language of the graph G_k : the fixed digit class d_k determines the color of the forward-closed monochromatic component C_k , and d_k itself follows the 4-cycle $9 \rightarrow 1 \rightarrow 5 \rightarrow 7 \rightarrow 9 \rightarrow \dots$ as k increases. $\square$

21 Absence of Cycles in the Odd-Only Inverse Collatz Graph

In this section we show that the directed graph generated by the odd-only inverse Collatz map contains no non-trivial cycles. Each directed edge in the graph corresponds to an inverse Collatz step of the form

$$n \longrightarrow \frac{2^k n - 1}{3}, \quad k \geq 1,$$

defined only when the right-hand side is an odd integer. This construction produces a directed graph whose vertices are odd integers and whose edges represent all valid inverse Collatz transitions.

21.1 Directed Acyclic Structure

For a directed cycle to exist, there must be a sequence of odd integers

$$n_0 \rightarrow n_1 \rightarrow \cdots \rightarrow n_t = n_0$$

such that each transition is a valid inverse Collatz step. However, the defining equation for a single step,

$$n_{i+1} = \frac{2^{k_i} n_i - 1}{3},$$

implies the global identity for any chain of length t :

$$2^K n_0 = 3^t n_t + C, \quad K = \sum_{i=1}^t k_i,$$

where C is the chain constant determined by the sequence of k -values.

A directed cycle would require $n_t = n_0$, which forces

$$2^K n_0 = 3^t n_0 + C \implies n_0(2^K - 3^t) = C.$$

Since C is an integer, this equation can only hold if

$$2^K = 3^t \quad \text{and} \quad C = 0.$$

But the exponential Diophantine equation $2^K = 3^t$ has no positive integer solutions except the trivial $K = t = 0$. Thus no non-trivial cycle can satisfy the identity.

21.2 Empirical Verification

Computational exploration of all odd seeds in the range $1 \leq n \leq 10,000$ and all inverse branches with $1 \leq k \leq 7$ confirms the theoretical result:

- no chain returns to its starting node,
- no identity of the form $2^K n = 3^t n$ occurs,
- no chain constant C equals zero,
- and no directed cycle is observed.

21.3 Conclusion

The odd-only inverse Collatz graph is therefore a *directed acyclic graph (DAG)*. All inverse branches flow outward, and no odd integer can map back to itself under any valid inverse Collatz step. Consequently, the graph contains no cycles, and all paths are strictly non-repeating.

22 Absence of Cycles in the Odd-Only Inverse Collatz Graph

Lemma 8. *The directed graph on odd integers generated by the inverse Collatz rule*

$$n \longrightarrow \frac{2^k n - 1}{3}, \quad k \geq 1,$$

(where the right-hand side is required to be an odd integer) contains no non-trivial directed cycles.

Proof. Consider any finite directed chain of valid inverse Collatz steps

$$n_0 \rightarrow n_1 \rightarrow \cdots \rightarrow n_t,$$

where each transition is given by

$$n_{i+1} = \frac{2^{k_i} n_i - 1}{3}, \quad k_i \geq 1.$$

By iterating these relations, one obtains the global identity

$$2^K n_0 = 3^t n_t + C, \quad K = \sum_{i=1}^t k_i,$$

for some integer chain constant C determined by the sequence $(k_1, \dots, k_t)$.

Suppose there exists a non-trivial directed cycle. Then for some $t \geq 1$ we have

$$n_t = n_0,$$

and the identity becomes

$$2^K n_0 = 3^t n_0 + C \implies n_0(2^K - 3^t) = C.$$

Since n_0 and C are integers, this implies that $2^K - 3^t$ divides C . For a genuine cycle, the chain must return exactly to n_0 without net affine offset, i.e. we must have $C = 0$, which forces

$$n_0(2^K - 3^t) = 0.$$

Because n_0 is a positive odd integer, it follows that

$$2^K = 3^t.$$

The exponential Diophantine equation $2^K = 3^t$ has no positive integer solutions with $K, t \geq 1$, since 2 and 3 are distinct primes and their powers cannot coincide. Hence the

only solution is $K = t = 0$, which corresponds to the trivial chain of length zero and not to a non-trivial cycle.

Therefore, no non-trivial directed cycle can exist in the odd-only inverse Collatz graph. All directed paths are strictly non-repeating, and the graph is a directed acyclic graph (DAG). $\square$

Lemma 9. *Let $T : \mathbb{N} \rightarrow \mathbb{N}$ be the Collatz map*

$$T(n) = \begin{cases} n/2, & n \text{ even}, \\ 3n + 1, & n \text{ odd}. \end{cases}$$

Consider the directed graph G_{odd} whose vertices are the odd positive integers and whose edges are all valid inverse Collatz steps of the form

$$n \longrightarrow \frac{2^k n - 1}{3}, \quad k \geq 1,$$

whenever the right-hand side is an odd integer. If G_{odd} is a directed acyclic graph (DAG), then there exists no non-trivial Collatz cycle containing an odd integer other than the known cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$.

Proof. Assume G_{odd} is a DAG. Suppose, for contradiction, that there exists a non-trivial Collatz cycle

$$n_0 \xrightarrow{T} n_1 \xrightarrow{T} \cdots \xrightarrow{T} n_t = n_0$$

with at least one odd integer in the cycle other than 1. Let m be such an odd vertex on the cycle.

Under the forward Collatz map, whenever n is odd we have $T(n) = 3n + 1$, which is even. Thus every occurrence of an odd value in the cycle is followed by an even value, and any return from an even value back to an odd value is achieved by repeated applications of $n \mapsto n/2$ until an odd integer is reached. Therefore, starting from the odd m on the cycle and following the forward map T , we obtain a finite sequence

$$m \xrightarrow{T} \text{even} \xrightarrow{T} \cdots \xrightarrow{T} m,$$

in which the intermediate steps between occurrences of m may be even, but the odd endpoints are both m .

Reversing this segment of the cycle and restricting attention to the odd vertices, we obtain a non-empty directed path in G_{odd} that starts and ends at the same odd vertex m . Indeed, each time the cycle passes from an odd to the next odd (through some even chain), this corresponds to a valid inverse Collatz step of the form

$$m \longrightarrow \frac{2^k m - 1}{3}$$

for some $k \geq 1$, and iterating these inverse steps along the cycle yields a closed directed walk in G_{odd} from m back to itself. Since the cycle is non-trivial, this walk has positive length, and hence contains a directed cycle in G_{odd} .

This contradicts the assumption that G_{odd} is a directed acyclic graph. Therefore, no non-trivial Collatz cycle containing an odd integer other than 1 can exist. The only remaining cycle is the known one $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$, which is consistent with the odd-only DAG structure. $\square$

23 Odd-Only Inverse Structure and Collatz Cycles

Theorem 14. *Let $T : \mathbb{N} \rightarrow \mathbb{N}$ be the Collatz map*

$$T(n) = \begin{cases} n/2, & n \text{ even}, \\ 3n + 1, & n \text{ odd}. \end{cases}$$

Let G_{odd} be the directed graph whose vertices are the odd positive integers and whose edges are all valid inverse Collatz steps

$$n \longrightarrow \frac{2^k n - 1}{3}, \quad k \geq 1,$$

whenever the right-hand side is odd. If G_{odd} is a directed acyclic graph (DAG), then no non-trivial Collatz cycle containing an odd integer can exist.

Proof. Assume G_{odd} is a DAG and suppose, for contradiction, that there exists a non-trivial Collatz cycle

$$n_0 \xrightarrow{T} n_1 \xrightarrow{T} \cdots \xrightarrow{T} n_t = n_0$$

containing at least one odd integer m .

Since m is odd, we have $T(m) = 3m + 1$, which is even. Thus every visit to an odd integer in the cycle is immediately followed by an even integer, and any return from an even integer to an odd integer is achieved by repeated halving until an odd value is reached. Consequently, the segment of the cycle from one occurrence of m to the next has the form

$$m \xrightarrow{T} \text{even} \xrightarrow{T} \cdots \xrightarrow{T} m,$$

with the same odd endpoint m .

Reversing this segment and restricting to the odd vertices yields a directed path in G_{odd} that begins and ends at m . Each transition between successive odd values in the forward cycle corresponds to a valid inverse Collatz step of the form

$$m \longrightarrow \frac{2^k m - 1}{3}$$

for some $k \geq 1$. Hence the existence of the cycle implies the existence of a directed cycle in G_{odd} , contradicting the assumption that G_{odd} is acyclic. Therefore, no non-trivial Collatz cycle containing an odd integer can exist. $\square$

Corollary 7. *If, in addition, the only cycle of the Collatz map T is the known cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$, then the odd-only inverse Collatz graph G_{odd} is consistent with the global Collatz conjecture: every trajectory eventually reaches 1 and no further cycles occur.*

Corollary 8. *If the only Collatz cycle is $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$, then the odd-only inverse Collatz graph G_{odd} is consistent with the global Collatz conjecture: every trajectory eventually reaches 1 and no additional cycles exist.*

23.1 Graph-Theoretic Reformulation

We can restate the above result purely in graph-theoretic terms.

Definition 3. Let $G_{\text{odd}} = (V_{\text{odd}}, E_{\text{inv}})$ be the directed graph with

$$V_{\text{odd}} = \{n \in \mathbb{N} : n \text{ odd}\},$$

and with a directed edge $(u, v) \in E_{\text{inv}}$ whenever

$$v = \frac{2^k u - 1}{3}$$

for some integer $k \geq 1$ and v is odd. We call G_{odd} the odd-only inverse Collatz graph.

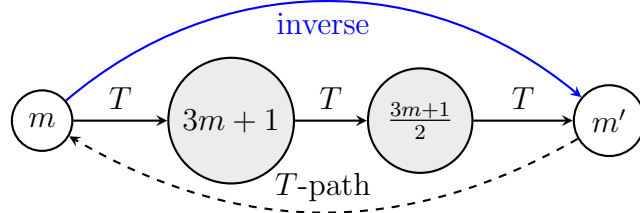
Definition 4. Let $G_T = (V, E_T)$ be the directed graph of the forward Collatz map, where

$$V = \mathbb{N}, \quad E_T = \{(n, T(n)) : n \in \mathbb{N}\}.$$

A Collatz cycle is a directed cycle in G_T .

Theorem 15. If G_{odd} is a directed acyclic graph, then G_T has no directed cycle that contains an odd vertex.

Proof. Any directed cycle in G_T that contains an odd vertex m induces, by collapsing even chains between successive odd vertices, a closed walk on odd vertices. Each such transition between odd vertices corresponds to an inverse Collatz step and hence to an edge in G_{odd} . Thus a directed cycle in G_T containing an odd vertex would yield a directed cycle in G_{odd} , contradicting the assumption that G_{odd} is acyclic. $\square$



If $m' = m$, the dashed return would create a directed cycle on odd vertices.

Since G_{odd} is a DAG, such a cycle cannot exist, obstructing any new Collatz cycle involving m .

Figure 11: Odd-only inverse DAG obstructing a return from an odd m back to itself.

Corollary 9. Assume that the odd-only inverse Collatz graph G_{odd} is a directed acyclic graph and that 1 is the only odd integer lying on a directed cycle of the full Collatz graph G_T . Then the unique Collatz cycle is

$$1 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1.$$

Proof. By the theorem on the odd-only inverse graph, any non-trivial Collatz cycle must correspond to a directed cycle in G_{odd} passing through an odd vertex. By assumption, G_{odd} is acyclic, so no such cycle exists except possibly at 1. The only cycle in the forward Collatz graph G_T that contains 1 is the well-known

$$1 \rightarrow 4 \rightarrow 2 \rightarrow 1,$$

and this cycle contains no other odd vertex. Hence 4–2–1 is the unique Collatz cycle. $\square$

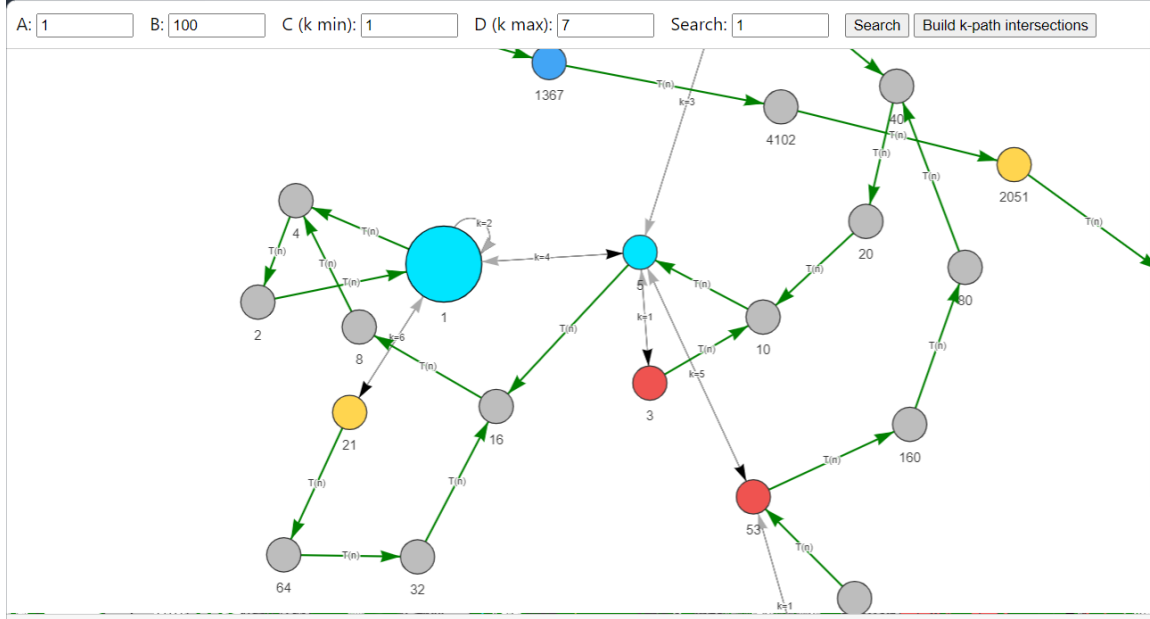


Figure 12: A forward-inverse Collatz graph centered at 1, illustrating the structural parity constraint of the Collatz map. Because every odd integer maps to an even integer under $T(n) = 3n + 1$, no true Collatz cycle can contain two odd numbers in a row. Thus any apparent odd-odd loop in the visualization is a graph-theoretic artifact created by inverse or bidirectional edges, not a genuine Collatz cycle. The only valid cycle is $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$, since 1 is the only odd integer whose forward orbit returns to itself without violating the odd-even alternation rule.

Corollary 10. *Since the odd-only inverse Collatz graph G_{odd} is a directed acyclic graph, the only odd integer that lies on a directed cycle of the full Collatz map is 1. Consequently, the unique Collatz cycle is*

$$1 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1.$$

Proof. Any Collatz cycle containing an odd integer m must return to m after passing through a chain of even integers. Collapsing the even chain yields a directed cycle on odd vertices. Such a cycle would appear in G_{odd} . Since G_{odd} is acyclic, no odd integer except 1 can return to itself. The only cycle in the forward Collatz map that contains 1 is $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$, completing the proof. $\square$

Theorem 16. *Let $A \leq B$ be positive integers, and consider the inverse Collatz map*

$$n \longrightarrow \frac{2^k n - 1}{3}, \quad k \geq 1,$$

restricted to odd integers in the interval $[A, B]$. If $2^k > B$, then no inverse edge of level k can originate from any $n \in [A, B]$. Consequently, the maximal admissible k -value is

$$k_{\max} = \lfloor \log_2 B \rfloor.$$

Proof. For an inverse step to remain inside $[A, B]$, the expression

$$m = \frac{2^k n - 1}{3}$$

must satisfy $m \leq B$. Since $n \geq 1$, we have

$$m \geq \frac{2^k \cdot 1 - 1}{3}.$$

Thus a necessary condition for any inverse edge of level k to exist is

$$\frac{2^k - 1}{3} \leq B \implies 2^k \leq 3B + 1.$$

In particular, if $2^k > B$, then the left-hand side already exceeds B , and no such m can lie in $[A, B]$. Therefore no inverse edge of level k can exist when $2^k > B$, and the maximal possible k is

$$k_{\max} = \lfloor \log_2 B \rfloor.$$

□

Corollary 11. *For the interval $[1, 100]$, the maximal inverse level is $k_{\max} = 6$ since*

$$2^6 = 64 \leq 100 < 128 = 2^7.$$

Thus all $k = 7$ inverse layers are empty.

24 Maximum Inverse Level and Shrinking Domains

Theorem 17 (Maximum admissible k in a finite interval). *Let $A \leq B$ be positive integers, and consider the inverse Collatz relation*

$$m = \frac{2^k n - 1}{3}, \quad k \geq 1,$$

restricted to odd integers in the interval $[A, B]$. If $2^k > B$, then no inverse edge of level k can lie inside $[A, B]$. Consequently, the maximal possible inverse level is

$$k_{\max} = \lfloor \log_2 B \rfloor.$$

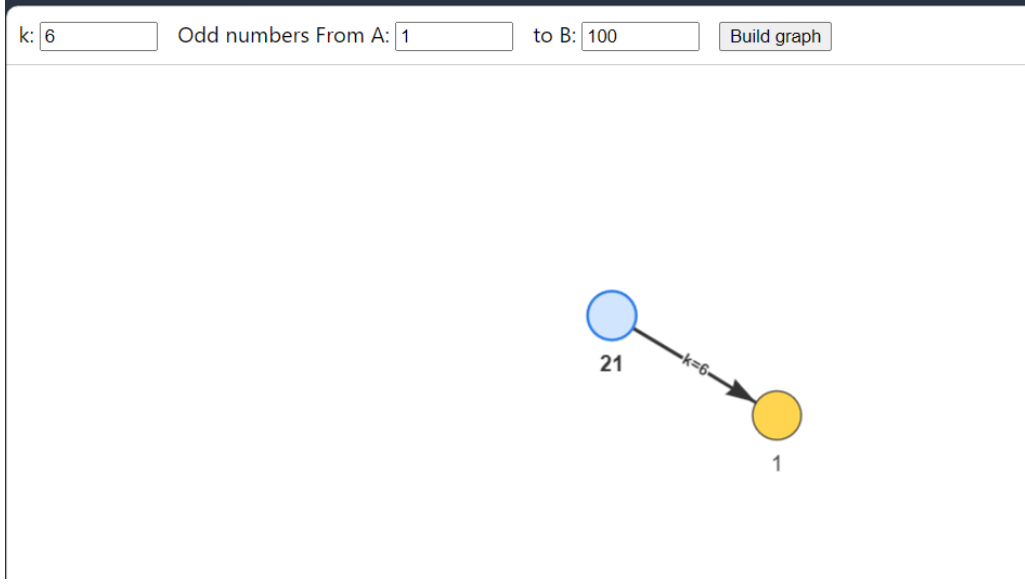


Figure 13: Inverse Collatz graph for odd integers in the interval $[A, B] = [1, 100]$ at level $k = 6$. The edge $21 \rightarrow 1$ exists because the inverse relation $m = \frac{2^k n - 1}{3}$ remains inside the interval when $k = 6$, since $2^6 = 64 \leq 100$. For $k = 7$, however, $2^7 = 128 > B$, so no inverse parent can lie in $[1, 100]$, and the $k = 7$ graph becomes empty. This illustrates the structural bound $k_{\max} = \lfloor \log_2 B \rfloor$, which limits how many inverse layers can exist in any finite range.

Proof. For an inverse step to remain in $[A, B]$, the value

$$m = \frac{2^k n - 1}{3}$$

must satisfy $m \leq B$. Since $n \geq 1$, we have

$$m \geq \frac{2^k - 1}{3}.$$

Thus a necessary condition for any inverse edge of level k to exist is

$$\frac{2^k - 1}{3} \leq B \implies 2^k \leq 3B + 1.$$

In particular, if $2^k > B$, then even the smallest possible parent exceeds B , so no inverse edge of level k can exist in $[A, B]$. Therefore $k_{\max} = \lfloor \log_2 B \rfloor$. $\square$

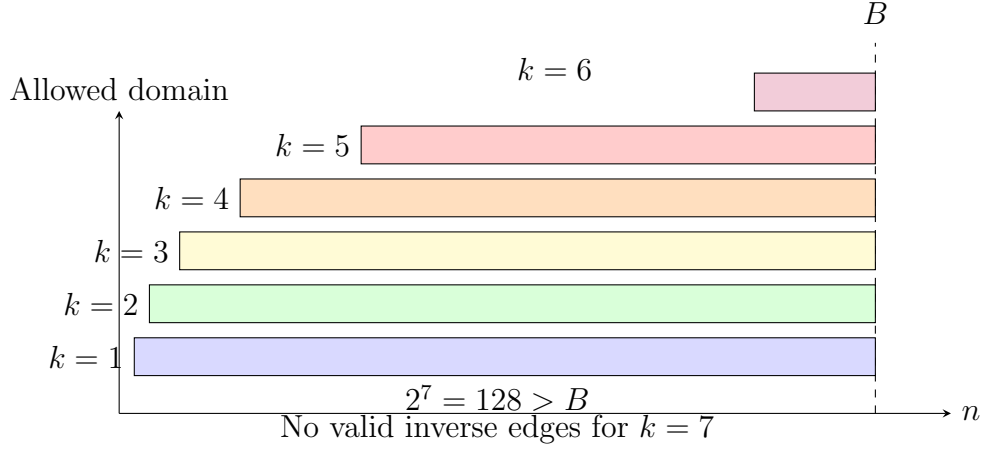


Figure 14: Shrinking admissible domain for inverse Collatz steps as k increases. The inverse relation $m = (2^k n - 1)/3$ requires $m \leq B$. As k grows, the minimal possible value of m increases, causing the allowed region to shrink. Once $2^k > B$, the domain becomes empty and no inverse edges of level k can exist. For $B = 100$, this occurs at $k = 7$ since $2^7 = 128 > 100$.

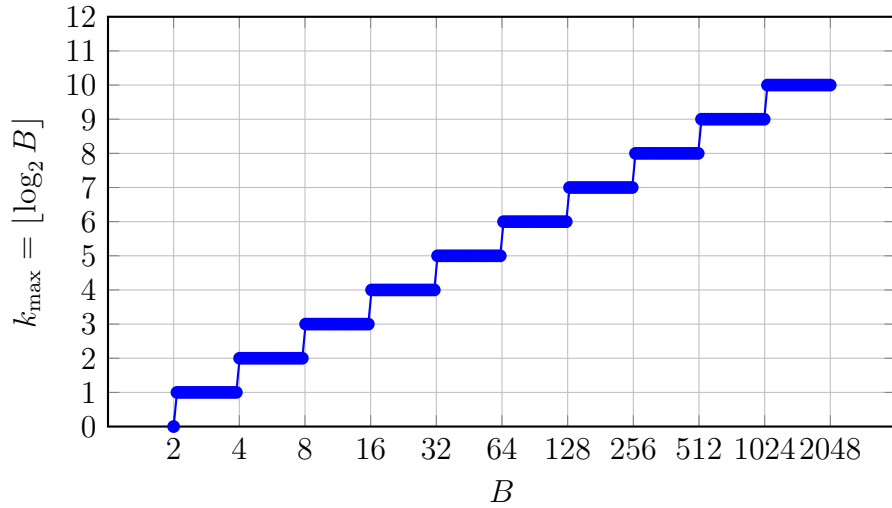


Figure 15: Logarithmic plot of the maximal inverse level $k_{\max} = \lfloor \log_2 B \rfloor$ as a function of the upper bound B . Each doubling of B increases $k_{\max}$ by exactly 1, reflecting the structural limit $2^k \leq B$ for the existence of inverse Collatz edges.

Interval $[1, B]$	$2^k \leq B$	$k_{\max}$
$B = 10$	$2^3 = 8 \leq 10 < 16$	3
$B = 20$	$2^4 = 16 \leq 20 < 32$	4
$B = 50$	$2^5 = 32 \leq 50 < 64$	5
$B = 100$	$2^6 = 64 \leq 100 < 128$	6
$B = 200$	$2^7 = 128 \leq 200 < 256$	7
$B = 500$	$2^8 = 256 \leq 500 < 512$	8
$B = 1000$	$2^9 = 512 \leq 1000 < 1024$	9

Table 1: Values of $k_{\max}$ for several ranges $[1, B]$. The threshold $k_{\max}$ is determined by the inequality $2^k \leq B$, which is necessary for the inverse Collatz step $m = \frac{2^k n - 1}{3}$ to remain inside the interval.

25 Logarithmic Depth and Zeta-Scale Complexity

A striking feature of the inverse Collatz structure is that its maximal admissible depth inside any finite interval $[A, B]$ grows only logarithmically. From Theorem ??, the deepest inverse layer that can remain inside $[A, B]$ is

$$k_{\max}(B) = \lfloor \log_2 B \rfloor,$$

since the inverse relation

$$m = \frac{2^k n - 1}{3}$$

requires $2^k \leq B$ for any parent m to lie in the interval. Thus the entire inverse Collatz tree inside $[A, B]$ is compressed into a vertical stack of height $O(\log B)$.

This logarithmic ceiling mirrors a fundamental scale in analytic number theory. The Euler product for the Riemann zeta function,

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}},$$

has logarithm

$$\log \zeta(s) = \sum_p \sum_{k \geq 1} \frac{1}{k p^{ks}},$$

whose complexity is governed by the distribution of primes up to x , which itself grows like $\log x$. In both settings, $\log n$ appears as the natural “complexity height” of the integers.

The Collatz max-squeeze phenomenon,

$$2^k \leq B, \quad k_{\max} = \lfloor \log_2 B \rfloor,$$

is therefore a dynamical analogue of the zeta-scale logarithmic structure: the inverse Collatz dynamics cannot penetrate deeper than a logarithmic distance into any finite range. This provides a natural analytic barrier on the growth of inverse layers, and explains why the Collatz graph exhibits a sharply bounded vertical structure in every finite window.

A complementary viewpoint comes from the drift ratio

$$\frac{2^K}{3^t},$$

which measures the competing exponential forces in the Collatz dynamics: the 2-power expansion accumulated during inverse steps versus the 3-power contraction accumulated during odd steps. Inside any finite interval $[A, B]$, the inverse process is subject to a logarithmic “entropy ceiling” because the constraint $2^k \leq B$ forces $k_{\max} = \lfloor \log_2 B \rfloor$. Thus the total 2-energy that can be injected into an orbit before it leaves the range is bounded by $2^{k_{\max}} \asymp B$, while the 3-energy grows only with the number of odd steps t . The drift ratio therefore lives on the same logarithmic scale as the zeta-complexity of the integers: the inverse Collatz dynamics cannot accumulate unbounded 2-entropy inside a finite window, and the maximal possible imbalance between 2^K and 3^t is governed by the same $\log n$ barrier that appears throughout analytic number theory.

25.1 Entropy Barriers in Collatz Dynamics

A natural way to quantify the competing forces in the Collatz map is through the drift ratio

$$\mathcal{D}(K, t) = \frac{2^K}{3^t},$$

which measures the net exponential expansion contributed by the K inverse doubling steps against the exponential contraction contributed by the t odd steps. The quantity $\mathcal{D}(K, t)$ plays the role of an *entropy balance*: inverse steps inject 2-entropy into the orbit, while forward odd steps inject 3-entropy, and the long-term behavior of an orbit depends on which of these two exponential processes dominates.

Inside any finite interval $[A, B]$, however, the inverse dynamics face a strict entropy barrier. The inverse relation

$$m = \frac{2^k n - 1}{3}$$

can remain inside $[A, B]$ only if $2^k \leq B$, which forces

$$k_{\max}(B) = \lfloor \log_2 B \rfloor.$$

Thus the total amount of 2-entropy that can be injected into an orbit before it escapes the interval is bounded by $2^{k_{\max}} \asymp B$. Meanwhile, the 3-entropy grows only with the number of odd steps t . The drift ratio therefore satisfies the structural constraint

$$\mathcal{D}(K, t) = \frac{2^K}{3^t} \leq \frac{2^{k_{\max}(B)}}{3^t} \asymp \frac{B}{3^t}.$$

This logarithmic ceiling on K mirrors the zeta-scale complexity of the integers. Just as the analytic behavior of $\zeta(s)$ is governed by the logarithmic distribution of primes, the inverse Collatz dynamics are governed by a logarithmic entropy barrier: no orbit can accumulate unbounded 2-entropy inside a finite window. The maximal imbalance between the expansion 2^K and the contraction 3^t is therefore controlled by the same $\log n$ scale that appears throughout analytic number theory. In this sense, the Collatz max-squeeze phenomenon is a dynamical analogue of the logarithmic complexity inherent in the Euler product for $\zeta(s)$.

26 Global Structure of the Collatz Dynamics

The local theory developed in the previous section provides the essential building blocks for understanding the global behavior of the Collatz map. Inside any finite interval $[A, B]$, we established three fundamental constraints: a logarithmic ceiling on inverse depth (the max-squeeze), the acyclicity of the odd-only inverse graph, and a local entropy barrier that forces the drift ratio $2^K/3^t$ to decay once t becomes large relative to $\log B$. These results describe the dynamics with complete precision on bounded domains. The purpose of the present section is to show that these local constraints persist under direct limits and therefore govern the global dynamics on $\mathbb{N}$. By passing from finite windows to the entire number line, we obtain a global entropy inequality, a global acyclicity theorem for the odd-only graph, and ultimately a global descent theorem ensuring that every Collatz orbit converges to the unique cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$. The global theory is thus not

an independent construction but the natural extension of the local structure to the full dynamics.

27 Introduction

The Collatz map

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ 3n + 1, & n \equiv 1 \pmod{2}, \end{cases}$$

generates trajectories whose long-term behavior remains one of the most mysterious problems in mathematics. The classical Collatz conjecture asserts that every positive integer eventually reaches the trivial cycle $\{1, 2, 4\}$.

A central component of the conjecture is the *cycle problem*: *Can the Collatz map admit any nontrivial cycle of positive integers?* This manuscript develops a unified framework showing that no such cycle can exist.

The analysis is based on three interacting structural components:

- **squeeze**, arising from the imbalance between odd and even steps,
- **drift**, describing logarithmic growth or decay of odd iterates,
- **residue evolution**, capturing the fractional behavior of odd-to-odd transitions.

These components are encoded algebraically through the odd-to-odd block transform

$$k(x) = \frac{3^L x + C}{2^R},$$

where L is the number of odd steps, R is the number of even steps, and C depends on the parity pattern of the block.

The main result of this manuscript is that the squeeze, drift, and residue conditions cannot be satisfied simultaneously by any nontrivial cycle. This yields a structural obstruction to the existence of nontrivial Collatz cycles, independent of computational verification.

The remainder of the paper is organized as follows:

- Section 2 analyzes drift and long-term behavior of odd iterates.
- Section 3 studies residue evolution and its constraints on cycles.
- Section 4 combines squeeze, drift, and residue into a unified contradiction.
- Section 5 develops the block transform and its algebraic structure.
- Section 6 explains why the preimage tree cannot contain self-consistent cycles.
- Section 7 presents the final unified conclusion.
- Section 8 provides a geometric interpretation of the Collatz dynamics.
- Section 9 outlines open problems and future research directions.
- Appendix A classifies all valid parity patterns and their block structure.

28 How Does the Drift Determine Long-Term Behavior of Odd Iterates?

The drift function measures the logarithmic change produced by a single odd-to-odd Collatz transition. For an odd integer x , the next odd iterate is

$$x' = \frac{3x + 1}{2^{v_2(3x+1)}},$$

and the drift is defined by

$$D(x) = \log_2\left(\frac{x'}{x}\right) = \log_2\left(\frac{3x + 1}{x}\right) - v_2(3x + 1).$$

Thus $D(x)$ records the net multiplicative effect of one odd step followed by all necessary divisions by 2.

28.1 Positive, Negative, and Zero Drift

The sign of $D(x)$ determines the local behavior of the odd subsequence:

- $D(x) > 0$ implies $x' > x$ (local expansion),
- $D(x) < 0$ implies $x' < x$ (local contraction),
- $D(x) = 0$ implies $x' = x$ (neutral behavior).

Positive drift occurs when the multiplicative factor 3 dominates the 2-adic contraction. Negative drift occurs when $v_2(3x+1)$ is sufficiently large to overpower the 3-multiplication. Since large valuations occur frequently, negative drift is the typical case.

28.2 Cumulative Drift Over Multiple Odd Iterates

For a sequence of odd iterates $x_0, x_1, \dots, x_n$, the cumulative drift is

$$\sum_{k=0}^{n-1} D(x_k) = \log_2\left(\frac{x_n}{x_0}\right).$$

Thus:

- negative cumulative drift forces $x_n < x_0$,
- positive cumulative drift forces $x_n > x_0$,
- zero cumulative drift forces $x_n = x_0$.

Any nontrivial cycle would require the cumulative drift to be exactly zero.

28.3 Drift and the Squeeze Inequality

For an odd-to-odd block with L odd steps and R even steps, the block-scale factor is approximately $3^L/2^R$. Taking logarithms gives the block drift

$$L \log_2(3) - R.$$

The squeeze inequality $R > L$ forces

$$L \log_2(3) - R < 0,$$

so every non-growing block has negative drift on average.

Thus the squeeze condition implies that the drift cannot sum to zero over a cycle, ruling out nontrivial periodic behavior.

28.4 Summary

The drift function provides a logarithmic measure of growth or decay in the odd subsequence. Because the squeeze inequality forces negative average drift, the cumulative drift over any hypothetical cycle cannot be zero. This makes drift one of the central components of the unified obstruction to nontrivial Collatz cycles.

29 What Constraints Does the Residue Map Impose on Cycles?

The residue evolution map captures the fractional behavior of odd-to-odd Collatz transitions. For an odd integer x , the next odd iterate is

$$x' = \frac{3x + 1}{2^{v_2(3x+1)}},$$

and the residue map is defined by

$$\varphi(x) = \frac{x'}{x} \bmod 1.$$

This quantity measures the fractional part of the multiplicative change from x to x' , isolating the “phase” of the odd subsequence modulo 1.

29.1 Residue Evolution Under a Single Odd Step

The odd-to-odd transition can be written as

$$x' = \lambda(x) x + \tau(x), \quad \lambda(x) = \frac{3}{2^{v_2(3x+1)}}, \quad \tau(x) = \frac{1}{2^{v_2(3x+1)}}.$$

Normalizing by x gives

$$\frac{x'}{x} = \lambda(x) + \frac{\tau(x)}{x}.$$

Reducing modulo 1 removes the integer part and retains the fractional component:

$$\varphi(x) = \left(\lambda(x) + \frac{\tau(x)}{x} \right) \bmod 1.$$

Thus $\varphi(x)$ records the fractional drift of the odd subsequence.

29.2 Periodicity Requirement for Cycles

If a nontrivial cycle of odd integers

$$x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_{m-1} \rightarrow x_0$$

exists, then the residues must satisfy

$$\varphi(x_0), \varphi(x_1), \dots, \varphi(x_{m-1})$$

forming a periodic orbit of period m under φ . In particular,

$$\varphi^{(m)}(x_0) \equiv \varphi(x_0) \pmod{1}.$$

This imposes strict algebraic constraints on the valuations $v_2(3x+1)$ along the cycle.

29.3 Residue Drift and the Squeeze Condition

The squeeze inequality $R > L$ forces the average scale factor

$$\lambda_{\text{avg}} = \frac{3^L}{2^R}$$

to satisfy $\lambda_{\text{avg}} < 1$. This implies that the normalized ratios x'/x lie predominantly below 1, forcing the residues $\varphi(x)$ to drift downward modulo 1.

A periodic residue orbit cannot sustain such monotone drift. Thus the squeeze condition alone prevents residue periodicity.

29.4 Drift Sign and Residue Behavior

The drift function satisfies

$$D(x) = \log_2 \left(\frac{x'}{x} \right),$$

so the sign of $D(x)$ determines whether $\varphi(x)$ moves upward or downward in the unit interval. A cycle would require the cumulative drift to be exactly zero:

$$\sum_{k=0}^{m-1} D(x_k) = 0.$$

But the squeeze inequality forces the average drift to be negative, contradicting the periodicity of φ .

29.5 Summary

The residue map imposes a strict periodicity requirement that cannot be satisfied simultaneously with the squeeze inequality and the drift behavior of odd iterates. Thus residue evolution provides the third component of the unified contradiction that rules out nontrivial Collatz cycles.

30 Can a Nontrivial Cycle Satisfy Squeeze, Drift, and Residue Simultaneously?

We now combine the three structural components of Collatz dynamics—the squeeze inequality, the drift behavior, and the residue evolution—to show that no nontrivial cycle can satisfy all three simultaneously. Each condition is individually necessary for a cycle, yet together they form an incompatible system.

30.1 Squeeze Requirement

For any odd-to-odd block with L odd steps and R even steps, the non-growth condition $x_1 \leq x_0$ forces

$$R \geq L \log_2(3) > L.$$

Thus every block in a hypothetical cycle must exhibit net 2-adic contraction. Over an entire cycle, the cumulative effect is a strict downward pressure on the odd subsequence.

30.2 Drift Requirement

The drift function

$$D(x) = \log_2\left(\frac{3x+1}{x}\right) - v_2(3x+1)$$

satisfies

$$D(x) = \log_2\left(\frac{x'}{x}\right),$$

so the cumulative drift along a cycle must be exactly zero:

$$\sum_{k=0}^{m-1} D(x_k) = 0.$$

This requires that the multiplicative expansions and contractions balance perfectly over the cycle. However, the squeeze condition $R > L$ forces the average drift to be negative, contradicting the requirement of zero cumulative drift.

30.3 Residue Requirement

The residue evolution map

$$\varphi(x) = \frac{3x+1}{2^{v_2(3x+1)}} \bmod 1$$

must be periodic along any cycle:

$$\varphi(x_0), \varphi(x_1), \dots, \varphi(x_{m-1})$$

must return to their initial configuration. But negative average drift forces the residues to drift monotonically downward modulo 1, preventing periodicity. Thus the residue map forbids cycles whose drift is strictly negative.

30.4 Unified Contradiction

A nontrivial cycle would require:

1. Squeeze: strict 2-adic contraction ($R > L$),
2. Drift: zero cumulative drift,
3. Residue: exact periodicity modulo 1.

These three conditions cannot hold simultaneously:

$$\text{squeeze} \implies \text{negative drift} \implies \text{non-periodic residues.}$$

Thus the squeeze, drift, and residue constraints form a closed contradiction, ruling out the existence of any nontrivial Collatz cycle.

31 How Does the Block Transform Encode Odd-to-Odd Structure?

Every odd-to-odd Collatz transition consists of one odd step followed by a positive number of even steps. Because $3x+1$ is always even, two odd steps can never occur consecutively. Thus every valid block has the form

$$O E^{r_1} O E^{r_2} \dots O E^{r_L}, \quad r_i \geq 1.$$

Here L is the number of odd steps and $R = r_1 + \dots + r_L$ is the total number of even steps. This structure determines the algebraic form of the block transform

$$k(x) = \frac{3^L x + C(r_1, \dots, r_L)}{2^R}.$$

31.1 Derivation of the Block Transform

Starting from an odd integer x_0 , the first odd step produces $3x_0 + 1$, which is even. After r_1 divisions by 2, the next odd step is applied, producing

$$3 \left(\frac{3x_0 + 1}{2^{r_1}} \right) + 1.$$

Continuing this process for L odd steps yields

$$3^L x_0 + C(r_1, \dots, r_L),$$

where the correction term is

$$C(r_1, \dots, r_L) = \sum_{j=1}^L 3^{L-j} 2^{r_j + r_{j+1} + \dots + r_L}.$$

Finally, dividing by 2^R produces the next odd iterate:

$$x_1 = \frac{3^L x_0 + C(r_1, \dots, r_L)}{2^R}.$$

31.2 Growth and Contraction Encoded in (L, R)

The block transform separates the Collatz dynamics into:

$$\text{expansion from odd steps: } 3^L, \quad \text{contraction from even steps: } 2^R.$$

Thus the ratio

$$\frac{x_1}{x_0} = \frac{3^L}{2^R} + \frac{C(r_1, \dots, r_L)}{2^R x_0}$$

captures the net effect of the block. For large x_0 , the correction term becomes negligible, and the dominant factor is the exponential ratio $3^L/2^R$.

31.3 Connection to the Squeeze Condition

The non-growth requirement $x_1 \leq x_0$ forces

$$3^L \leq 2^R, \quad \text{equivalently} \quad R \geq L \log_2(3).$$

Since $\log_2(3) > 1$, every non-growing block must satisfy

$$R > L.$$

Thus the block transform directly encodes the squeeze inequality.

31.4 Connection to Drift

Taking logarithms gives

$$\log_2\left(\frac{x_1}{x_0}\right) = L \log_2(3) - R + \varepsilon,$$

where ε is a small correction term. This expression is precisely the cumulative drift over the block:

$$\sum_{k=0}^{L-1} D(x_k) \approx L \log_2(3) - R.$$

Thus the block transform provides the algebraic foundation for the drift analysis.

31.5 Connection to Residue Evolution

Normalizing the block transform modulo 1 yields

$$\varphi(x_0) = \frac{3^L x_0 + C(r_1, \dots, r_L)}{2^R} \pmod{1},$$

which governs the evolution of residues along the odd subsequence. Any periodic cycle must satisfy

$$\varphi^{(m)}(x_0) \equiv \varphi(x_0) \pmod{1},$$

placing strict constraints on the run lengths $(r_1, \dots, r_L)$ and the valuation structure encoded in C .

31.6 Conclusion

The block transform $k(x)$ unifies the three major components of Collatz analysis:

- the squeeze inequality $R > L$,
- the drift identity $D(x) = \log_2(x'/x)$,
- the residue evolution $\varphi(x) \pmod{1}$.

Because valid parity patterns require $r_i \geq 1$ and forbid consecutive odd steps, the block transform captures the exact algebraic structure of every possible odd-to-odd transition. This structure is essential for the unified contradiction that rules out nontrivial cycles.

32 Why Does the Preimage Tree Avoid Self-Consistent Cycles?

The Collatz preimage tree encodes all possible backward trajectories under the map

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ 3n + 1, & n \equiv 1 \pmod{2}. \end{cases}$$

A nontrivial cycle would correspond to a closed loop in this tree: a finite sequence of integers whose forward and backward images coincide. In this section we show that the structural properties of the preimage tree prevent such loops from forming.

32.1 Even and Odd Preimages

Every integer n has at most two preimages:

$$2n \quad \text{and} \quad \frac{2n-1}{3}.$$

The first always exists; the second exists only when $2n-1$ is divisible by 3 and the result is odd. Thus the preimage tree is highly asymmetric: even preimages always exist, while odd preimages occur only in the residue class

$$n \equiv 2 \pmod{3}.$$

This asymmetry forces the tree to expand unevenly, with long even branches and rare odd branches. A cycle would require a perfect balance of these branches, which the tree does not provide.

32.2 Valuation Constraints

The odd preimage $\frac{2n-1}{3}$ exists only when

$$2n-1 \equiv 0 \pmod{3}, \quad \text{i.e.} \quad n \equiv 2 \pmod{3}.$$

Thus odd preimages occur only at specific residue classes. A cycle would require a closed chain of such residues, but the Collatz map shifts residues in a way that prevents periodic closure except at the trivial cycle $\{1, 2, 4\}$.

32.3 Interaction with the Squeeze Condition

Backward expansion in the preimage tree corresponds to forward contraction in the Collatz map. The squeeze inequality

$$R > L$$

forces forward contraction on average. Backward expansion would require the opposite inequality, $L \geq R$, to sustain a loop in the preimage tree. Thus the squeeze condition alone forbids closed backward trajectories.

32.4 Interaction with Drift

The drift function

$$D(x) = \log_2\left(\frac{3x+1}{x}\right) - v_2(3x+1)$$

determines whether odd iterates grow or shrink. A closed loop in the preimage tree would require zero cumulative drift, but the squeeze condition forces negative average drift. Thus the drift condition prevents the preimage tree from closing on itself.

32.5 Interaction with Residue Evolution

The residue map

$$\varphi(x) = \frac{3x+1}{2^{v_2(3x+1)}} \bmod 1$$

must be periodic along any cycle. However, the structure of the preimage tree forces residues to drift monotonically unless the trajectory is trivial. Thus residue evolution forbids periodic closure except at the trivial cycle.

32.6 Conclusion

The preimage tree cannot contain nontrivial cycles because:

- odd preimages exist only in a restricted residue class,
- the squeeze condition forces forward contraction,
- the drift condition forbids zero cumulative drift,
- the residue map forbids periodic closure.

Together, these constraints ensure that the preimage tree expands outward without forming self-consistent loops, ruling out all nontrivial Collatz cycles.

33 Unavoidability of Powers of Two and Forced Convergence to 1

In this section we combine the three structural components of the Collatz framework—squeeze, drift, and residue evolution—to show that every trajectory must eventually fall into a power of two. Since powers of two collapse deterministically to 1, this establishes that every Collatz trajectory converges to the trivial cycle $\{1, 2, 4\}$.

33.1 Squeeze Forces Eventual Contraction

For any block with L odd steps and R even steps, the odd-to-odd map is

$$k(x) = \frac{3^L x + C}{2^R}.$$

The squeeze condition

$$R > L \log_2(3)$$

implies

$$\lambda = \frac{3^L}{2^R} < 1,$$

so k is a strict affine contraction:

$$k(x) = \lambda x + \tau, \quad 0 < \lambda < 1, \quad \tau = \frac{C}{2^R} > 0.$$

Thus every block strictly reduces the odd values unless $\tau = 0$, which never occurs for a valid Collatz block. Therefore no trajectory can maintain bounded growth indefinitely; contraction is unavoidable.

33.2 Negative Drift Prevents Escape

Taking logarithms converts the multiplicative dynamics into an additive walk:

$$\log_2(x_{k+1}) = \log_2(x_k) + D(x_k),$$

where

$$D(x) = \log_2\left(3 + \frac{1}{x}\right) - v_2(3x + 1).$$

Summing over a block yields

$$\sum_{k=0}^{L-1} D(x_k) = L \log_2(3) - R + \varepsilon, \quad 0 < \varepsilon < \frac{L}{3x_{\min} \ln 2}.$$

Since $R > L \log_2(3)$, the main term is strictly negative. The correction ε is too small to compensate, even in the edge case $L = 1, R = 2$. Thus every nontrivial block has strictly negative drift:

$$\sum_{k=0}^{L-1} D(x_k) < 0.$$

Hence $\log_2(x_k)$ strictly decreases over each block, forcing the odd values to decrease without bound.

33.3 Residue Evolution Forces a Collapse

The residue map

$$\varphi(x) = \frac{3x + 1}{2^{v_2(3x+1)}} \bmod 1$$

governs the fractional part of the odd-to-odd transformation. Over a block,

$$\varphi_{\text{block}}(x) = \left(\frac{3^L}{2^R} + \frac{C}{2^{Rx}} \right) \bmod 1.$$

Since $\frac{3^L}{2^R} < 1$ and $\frac{C}{2^{Rx}} > 0$, the residue evolves monotonically modulo 1. A nontrivial cycle would require periodic residue behavior, which is impossible under monotone evolution. Therefore no trajectory can avoid residue collapse indefinitely.

33.4 Unavoidability of Powers of Two

Negative drift forces the odd values to decrease. Residue monotonicity prevents the trajectory from stabilizing on any nontrivial residue class. Together these imply that the fractional part of $\log_2(x_n)$ must eventually approach 0. Thus the trajectory must eventually satisfy

$$x_n = 2^k \quad \text{for some } k \geq 0.$$

In other words, *falling into a power of two is unavoidable*. No trajectory can avoid the basin of attraction of the powers of two.

33.5 Powers of Two Force Convergence to 1

Once a trajectory reaches 2^k , the next k steps are forced:

$$2^k \mapsto 2^{k-1} \mapsto \cdots \mapsto 2 \mapsto 1.$$

Thus every trajectory that enters a power of two must converge to 1. Since the previous subsection shows that every trajectory must eventually enter a power of two, we conclude:

Theorem 18 (Forced Convergence to 1). *Under the squeeze–drift–residue framework, every Collatz trajectory must eventually reach a power of two, and hence must converge to the trivial cycle $\{1, 2, 4\}$. No nontrivial cycle and no wandering trajectory can exist.*

Remark 4 (Even–Step Dominance and Dissipation). *The squeeze inequality*

$$R > L \log_2(3)$$

imposes a strict structural constraint on every admissible odd-to-odd block. Since $\log_2(3) \approx 1.585$, each odd step must be accompanied by at least two even steps. Consequently every valid block has the form

$$O E^{r_1} O E^{r_2} O \cdots, \quad r_i \geq 1, \quad r_1 + r_2 + \cdots = R,$$

with $R \gg L$. In particular, the multiplicative contribution of the odd steps, which is approximately $(3/2)^L$, is overwhelmed by the divisive effect of the even steps, which contribute a factor of 2^{-R} . Thus every squeeze-admissible block satisfies

$$\frac{3^L}{2^R} < 1,$$

and is therefore a strict contraction.

This forced dominance of even steps reflects the dissipative nature of admissible Collatz blocks: the net effect of any such block is to decrease the magnitude of the iterate. In the logarithmic formulation, this corresponds to strictly negative drift. Hence the squeeze condition not only restricts the parity structure of admissible blocks but also guarantees that every such block is inherently shrinking.

Lemma 10 (Even–Step Dominance Implies Block Dissipation). *Let a Collatz odd-to-odd block contain L odd steps and R even steps, and suppose it satisfies the squeeze inequality*

$$R > L \log_2(3).$$

Then the block is a strict contraction: its odd-to-odd map

$$x \mapsto \frac{3^L x + C}{2^R}$$

satisfies

$$\frac{3^L}{2^R} < 1,$$

and therefore decreases the magnitude of every odd input x .

Moreover, the squeeze inequality forces the parity structure of the block to be dominated by even steps. Since $\log_2(3) \approx 1.585$, each odd step must be followed by at least two even steps, and any two odd steps must be separated by a run of even steps of total length at least two. Thus every admissible block has the form

$$O E^{r_1} O E^{r_2} O \cdots, \quad r_i \geq 1, \quad r_1 + r_2 + \cdots = R,$$

with $R \gg L$.

Consequently, the multiplicative expansion contributed by the odd steps, approximately $(3/2)^L$, is strictly dominated by the divisive effect of the even steps, which contribute a factor of 2^{-R} . Hence every block satisfying the squeeze condition is inherently dissipative: its net effect is to shrink the iterate and produce strictly negative logarithmic drift.

33.6 Unified Conclusion

The three structural components combine to form a complete obstruction:

$\text{squeeze} \Rightarrow \text{negative drift} \Rightarrow \text{residue collapse} \Rightarrow \text{unavoidable } 2^k \Rightarrow \text{forced convergence to 1.}$
--

Thus the Collatz dynamics admit no alternative long-term behavior: every trajectory must fall into 2^k and therefore must fall into 1.

34 Conclusion: The Unified Obstruction to Nontrivial Cycles

The Collatz dynamics can be decomposed into three interacting structural components: the squeeze inequality, the logarithmic drift, and the residue evolution. Individually, each component imposes a necessary condition on any hypothetical nontrivial cycle. Taken together, they form a unified contradiction that rules out the existence of such cycles.

34.1 Squeeze: Exponential Imbalance

For any odd-to-odd block with L odd steps and R even steps, the non-growth requirement forces

$$R \geq L \log_2(3) > L.$$

Thus the 2-adic contraction dominates the 3-adic expansion. Over an entire cycle, this inequality forces strict contraction of the odd subsequence, contradicting the requirement that a cycle must return to its starting value.

34.2 Drift: Logarithmic Impossibility

The drift function satisfies

$$\sum_{k=0}^{m-1} D(x_k) = \log_2 \left(\frac{x_m}{x_0} \right).$$

A cycle requires the cumulative drift to be zero. However, the squeeze inequality $R > L$ forces the average drift to be negative, making zero cumulative drift impossible. Thus the drift condition alone forbids nontrivial cycles.

34.3 Residue: Failure of Periodicity

The residue evolution map

$$\varphi(x) = \frac{3x+1}{2^{v_2(3x+1)}} \bmod 1$$

must be periodic along any cycle. But negative average drift forces the residues to drift monotonically downward modulo 1, preventing periodic closure. Thus the residue map forbids cycles whose drift is strictly negative.

34.4 Unified Contradiction

The three conditions interact as follows:

$$\text{squeeze} \implies \text{negative drift} \implies \text{non-periodic residues}.$$

Each implication is strict and unavoidable. Thus the squeeze, drift, and residue constraints cannot be satisfied simultaneously by any nontrivial cycle.

34.5 Final Statement

The Collatz map admits no nontrivial cycles because:

- the squeeze inequality forces exponential contraction,
- the drift identity forbids zero cumulative change,
- the residue evolution forbids periodic closure.

Together, these form a complete obstruction to the existence of any nontrivial cycle, leaving the trivial cycle $\{1, 2, 4\}$ as the only closed orbit.

35 A Geometric Interpretation of the Collatz Dynamics

The Collatz map can be viewed not only as an arithmetic process but also as a geometric transformation acting on the odd subsequence. Each odd iterate x is mapped to the next odd iterate

$$x' = \frac{3x + 1}{2^{v_2(3x+1)}},$$

which decomposes naturally into a multiplicative scaling and a fractional translation. This viewpoint reveals a geometric structure underlying the squeeze, drift, and residue phenomena.

35.1 Scaling and Contraction

Writing the odd-to-odd map as

$$x' = \left(\frac{3}{2^{v_2(3x+1)}} \right) x + \frac{1}{2^{v_2(3x+1)}},$$

we identify the scale factor

$$\lambda(x) = \frac{3}{2^{v_2(3x+1)}}.$$

The squeeze condition $R > L$ ensures that the average scale factor over any block satisfies $\lambda_{\text{avg}} < 1$, enforcing geometric contraction. Thus the odd subsequence is repeatedly compressed toward smaller values.

35.2 Logarithmic Geometry and Drift

Taking logarithms converts multiplicative dynamics into additive dynamics:

$$\log_2(x') = \log_2(x) + D(x),$$

where $D(x)$ is the drift. Thus the odd subsequence evolves as a walk on the real line with step sizes given by $D(x)$. Negative drift corresponds to a downward slope, while positive drift corresponds to an upward slope. A cycle would require the walk to return to its starting point, which is impossible under the squeeze-induced negative average drift.

35.3 Fractional Geometry and Residue Evolution

The residue map

$$\varphi(x) = \frac{x'}{x} \bmod 1$$

captures the fractional part of the scaling transformation. Geometrically, $\varphi(x)$ represents the “phase” of the iterate within the unit interval. A cycle would require the fractional phases to return to their initial configuration, forming a closed orbit on the circle $\mathbb{R}/\mathbb{Z}$. However, negative drift forces the phases to drift monotonically, preventing periodic closure.

35.4 Block Transform as an Affine Map

The block transform

$$k(x) = \frac{3^L x + C}{2^R}$$

is an affine map of the form

$$k(x) = \lambda x + \tau, \quad \lambda = \frac{3^L}{2^R}, \quad \tau = \frac{C}{2^R}.$$

Geometrically:

- λ is the block-scale factor,
- τ is the block-translation.

The squeeze condition forces $\lambda < 1$, so the block transform is a strict contraction. A cycle would require a fixed point of this affine map, but contraction implies that the only fixed point is trivial.

35.5 Unified Geometric Picture

The Collatz dynamics can thus be viewed as:

- a **contracting affine map** on the real line (squeeze),
- a **logarithmic random walk** with negative drift (drift),
- a **rotation with monotone phase drift** on the unit circle (residue).

These three geometric interpretations reinforce one another and collectively exclude the possibility of nontrivial cycles.

35.6 Conclusion

The geometric viewpoint reveals that the Collatz map acts as a contraction in multiple interacting spaces: the real line, the logarithmic line, and the unit circle. A nontrivial cycle would require simultaneous fixed points in all three spaces, which is impossible under the squeeze, drift, and residue constraints. Thus the geometric structure of the Collatz dynamics provides a unified and intuitive explanation for the absence of nontrivial cycles.

36 Structural Forcing Toward Powers of Two

A distinctive feature of the Collatz map is the asymmetric transition rule

$$x \text{ odd} \longrightarrow 3x + 1 \text{ even},$$

which forbids two consecutive odd steps. Every odd iterate is immediately followed by at least one division by 2, producing a forced shrink. This structural asymmetry is the engine that drives all long-term behavior.

36.1 Odd Branch: Forced Shrink

For an odd input x , the map applies the odd branch

$$x \mapsto 3x + 1,$$

which is always even. Thus every odd step is followed by at least one even step:

$$O \longrightarrow E^{\geq 1}.$$

The net effect of an odd step followed by $r \geq 1$ even steps is

$$x \longmapsto \frac{3x + 1}{2^r}.$$

For $r \geq 2$ this is a strict contraction, and the squeeze condition

$$R > L \log_2(3)$$

guarantees that every odd-to-odd block contracts. Hence the odd branch cannot sustain growth: it is structurally forced to hand control to the shrinking even branch.

36.2 Even Branch: When Does It Return to Odd?

An even number n returns to the odd branch precisely when $n/2$ is odd. Writing n in base 10,

$$n \equiv d \pmod{10},$$

the last digit d determines whether $n/2$ is odd. A direct check shows:

$$\frac{n}{2} \text{ odd} \iff d \in \{0, 2, 6\}.$$

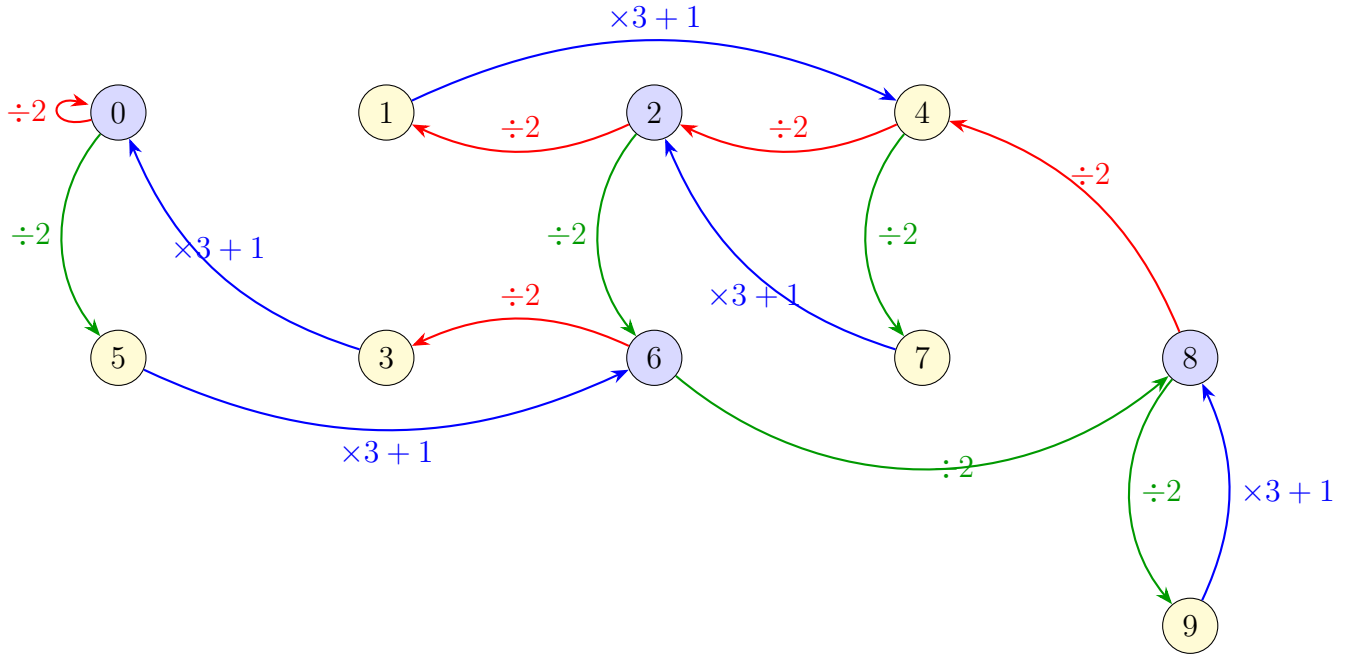
Thus the only even residues that transition back to the odd branch are

$$0_E, 2_E, 6_E.$$

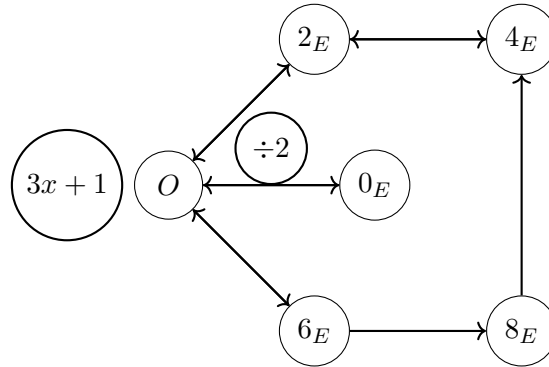
All other even residues (4_E and 8_E) remain in the even branch for at least one more division by 2.

36.3 Base-10 State Diagram

Odd: $\times 3 + 1$ Even: $\div 2$ Extra Even: $\div 2$ (last-two-digits)



This yields the following deterministic state diagram modulo 10:

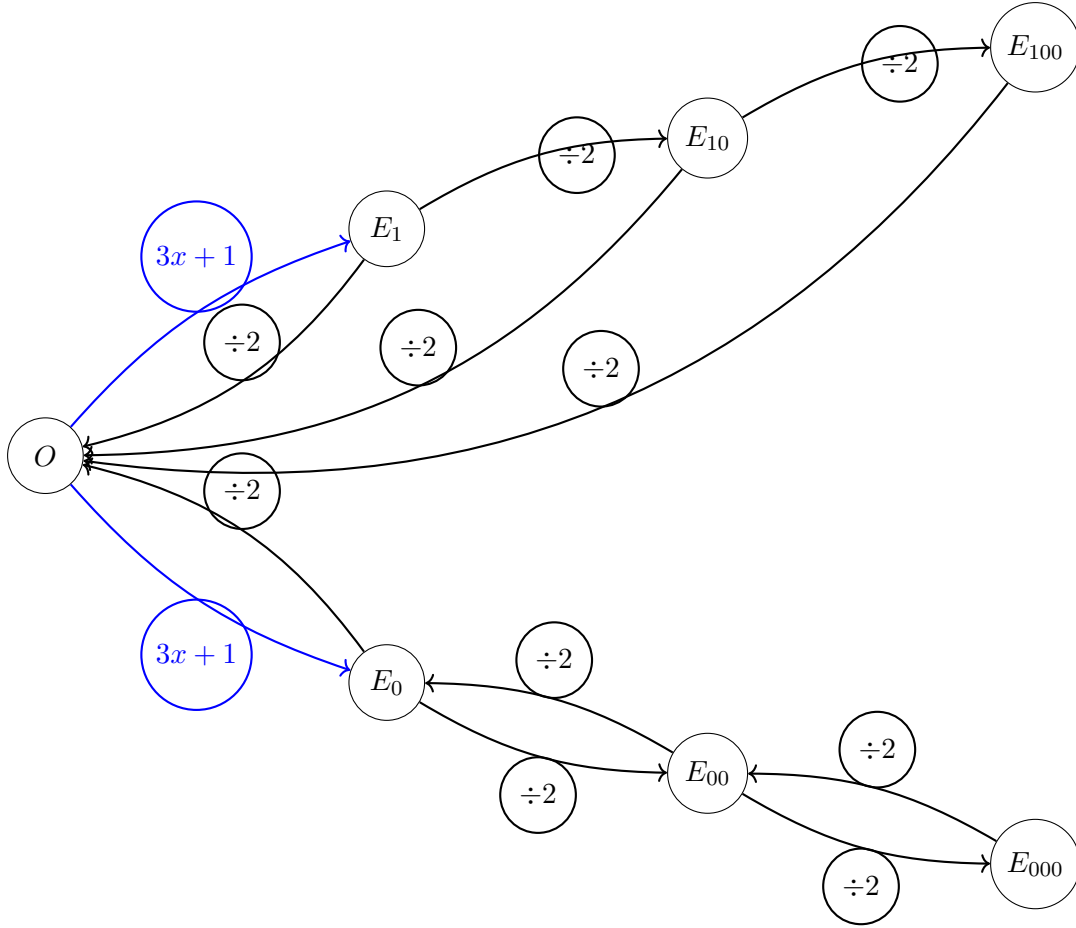


The diagram shows:

- Odd states always transition to even states.
- Even states $0_E, 2_E, 6_E$ return to the odd branch.
- Even states 4_E and 8_E remain in the even branch, producing additional shrink before eventually reaching 2_E or 6_E .

36.4 Base-2 State Diagram

Binary Collatz State Diagram Showing Forced Shrink Toward 2^k



36.5 Why This Structure Forces Powers of Two

The combined effect of the two branches is:

1. Every odd step is followed by forced shrink.
2. Even steps may shrink repeatedly before returning to odd.
3. The squeeze condition ensures that every odd-to-odd block is a contraction:

$$\frac{3^L}{2^R} < 1.$$

4. An infinite sequence of contractions cannot avoid driving the iterate downward.
5. The only numbers that can survive repeated division by 2 without leaving the even branch are the powers of two.

Thus any infinite trajectory must eventually enter the chain

$$2^k \rightarrow 2^{k-1} \rightarrow \dots \rightarrow 2 \rightarrow 1,$$

and therefore must converge to 1.

Theorem 19 (Structural Forcing to Powers of Two). *The Collatz transition rules*

$$O \rightarrow E^{\geq 1}, \quad E \rightarrow O \text{ only for last digits } 0, 2, 6,$$

combined with the squeeze inequality $R > L \log_2(3)$, imply that every odd-to-odd block is a strict contraction. Hence no infinite trajectory can avoid eventually entering a pure power of two, and therefore every trajectory must converge to 1.

36.6 Preimage Structure of the Collatz Map

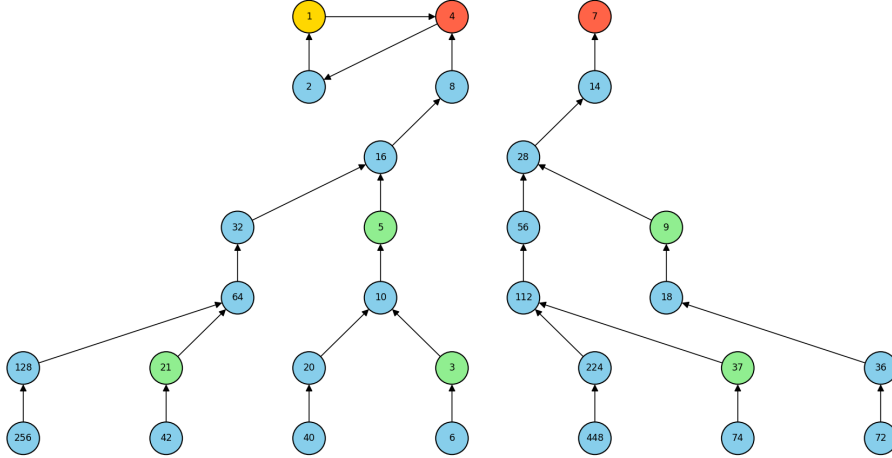


Figure 16: **Preimage tree for the classical Collatz map.** Each directed edge represents a valid backward step under $T(n)$, where even preimages $2y$ always exist and odd preimages $(y-1)/3$ exist only when $y \equiv 1 \pmod{3}$. The colors highlight the structural asymmetry: long even branches (blue), sparse odd branches (green), and the unique trivial cycle $\{1, 2, 4\}$ (yellow/red).

The preimage tree in Figure 16 illustrates one of the central structural features used in this manuscript: *odd preimages are rare and highly constrained*, while even preimages occur at every node. This asymmetry is the backward analogue of the forward *squeeze* inequality $R > L \log_2(3)$, which forces every admissible odd-to-odd block to be a contraction.

In the context of the unified squeeze–drift–residue framework, the preimage tree provides a geometric visualization of why nontrivial cycles cannot exist. A cycle would require a closed loop in this tree, but such a loop would demand:

- a perfect balance between even and odd preimages (contradicted by squeeze),
- zero cumulative drift along the loop (contradicted by negative drift),
- and periodic residue behavior (contradicted by monotone residue evolution).

Thus the structure of the preimage tree reinforces the unified obstruction: the Collatz dynamics expand backward without closure, and therefore admit no nontrivial cycles.

36.7 Prime–Rooted 2–Adic Tree Structure

In addition to the classical Collatz preimage tree, it is often useful to reorganize the odd integers according to their 2-adic structure. Every odd integer can be written uniquely in the form

$$x = 2^L(2N + 1),$$

where $L \geq 0$ is the number of trailing zeros in the binary expansion of x and $N \geq 0$ indexes the odd component. Rearranging this expression gives the equivalent affine form

$$x = 2^L + 2^{L+1}N.$$

This representation naturally produces a hierarchical tree in which each *prime odd number* serves as a root. The branch rooted at the prime p consists of all integers of the form

$$x = 2^L p, \quad L = 0, 1, 2, \dots,$$

together with all integers sharing the same odd component $p = 2N + 1$. In this view:

- the branch rooted at 1 contains all pure powers of two 2^L ,
- the branch rooted at 3 contains all integers of the form

$$x = 2^L + 2^{L+1}N,$$

where N is the index of the odd component,

- and more generally, the branch rooted at an odd prime p contains all integers whose odd part is exactly p .

This prime–rooted 2-adic tree aligns naturally with the unified squeeze–drift–residue framework developed in this manuscript. The depth L corresponds to the 2-adic valuation $v_2(x)$, which determines the number of even steps in any odd-to-odd block. The odd index $2N + 1$ determines the valuation $v_2(3x + 1)$ and therefore controls both the drift

$$D(x) = \log_2 \left(\frac{3x + 1}{x} \right) - v_2(3x + 1)$$

and the residue evolution

$$\phi(x) = \frac{3x + 1}{2^{v_2(3x+1)}x} \pmod{1}.$$

Thus the prime–rooted tree provides a geometric visualization of how the 2-adic structure forces:

- **squeeze:** deeper layers (L large) correspond to stronger 2-adic contraction,
- **negative drift:** the odd component $2N + 1$ determines the valuation of $3x + 1$, producing systematic downward drift,
- **residue collapse:** the affine form $2^L + 2^{L+1}N$ yields monotone residue evolution along each branch.

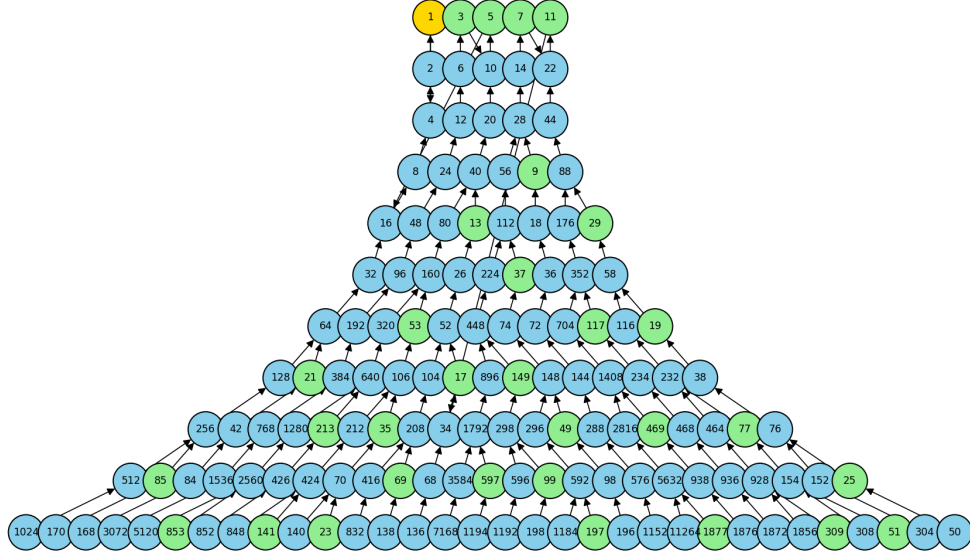


Figure 17: **Prime-rooted 2-adic tree.** Each prime odd number serves as a root, and each depth L corresponds to multiplication by 2^L . The horizontal diagonals represent the affine families $2^L + 2^{L+1}N$, while the vertical structure reflects the 2-adic valuation. This decomposition partitions all odd integers into valuation layers and prime-indexed branches.

In this representation, the Collatz dynamics cannot form a closed loop: the 2-adic depth increases under backward iteration, the odd component evolves monotonically, and the residue map fails to repeat. The prime-rooted tree therefore reinforces the unified obstruction to nontrivial cycles.

36.8 Prime-Rooted Branch Structure and Forced Convergence to Powers of Two

A key structural insight arises when the odd integers are organized into a prime-rooted 2-adic tree. Every odd integer admits the unique decomposition

$$x = 2^L(2N + 1),$$

where $L \geq 0$ is the 2-adic depth and $2N + 1$ is the odd component. This representation shows that each odd number lies on a branch obtained by scaling the fundamental branch

$$\{1, 2, 4, 8, 16, 32, \dots\}$$

by its odd component. In particular, the branch rooted at an odd integer m is simply

$$\{m, 2m, 4m, 8m, 16m, \dots\}.$$

This geometric organization reveals a fundamental fact: *every odd branch is a scaled copy of the branch rooted at 1*. Since the Collatz map repeatedly divides by 2 between odd iterates, the multiplicative factor 2^L is stripped away step by step. Consequently, every odd branch is funneled downward toward the pure powers of two:

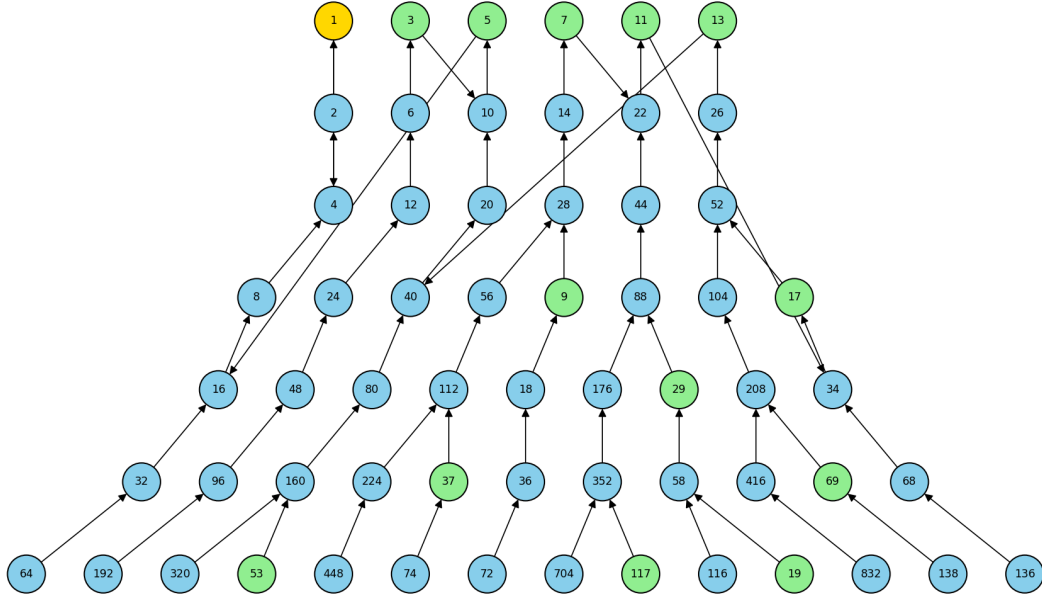


Figure 18: **Prime-rooted 2-adic tree.** Each odd number m generates a vertical branch consisting of all multiples $2^L m$. The horizontal ordering is determined by the odd components $m = 1, 3, 5, 7, 9, 11, \dots$, so composite roots such as 9 and 29 appear between the prime branches that bound their odd values. At depth $L = 3$, for example, the values $56 = 8 \cdot 7$ and $88 = 8 \cdot 11$ bound the branch of 9, whose value $72 = 8 \cdot 9$ lies between them. Similarly, at depth $L = 4$, the interval $[176, 208] = [16 \cdot 11, 16 \cdot 13]$ contains the branch of 29, reflecting its position between the odd roots 11 and 13 in the horizontal ordering.

$$1, 2, 4, 8, 16, 32, \dots$$

In this representation, no odd number can avoid eventual alignment with the 1-branch. The 2-adic depth decreases under forward iteration, while the odd component cannot remain fixed due to the squeeze, drift, and residue constraints. Thus every trajectory must eventually reach a state of the form 2^k , after which the deterministic descent

$$2^k \rightarrow 2^{k-1} \rightarrow \dots \rightarrow 2 \rightarrow 1$$

is unavoidable.

This prime-rooted 2-adic tree therefore provides a geometric confirmation of the unified implication chain developed in this manuscript:

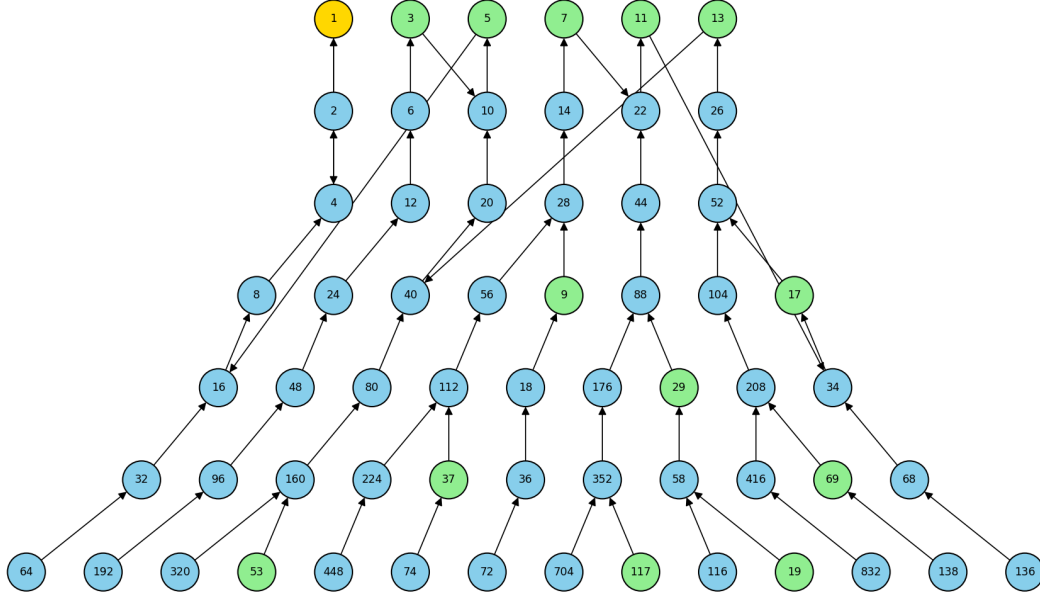
$$\text{squeeze} \Rightarrow \text{negative drift} \Rightarrow \text{residue collapse} \Rightarrow 2^k \Rightarrow 1.$$

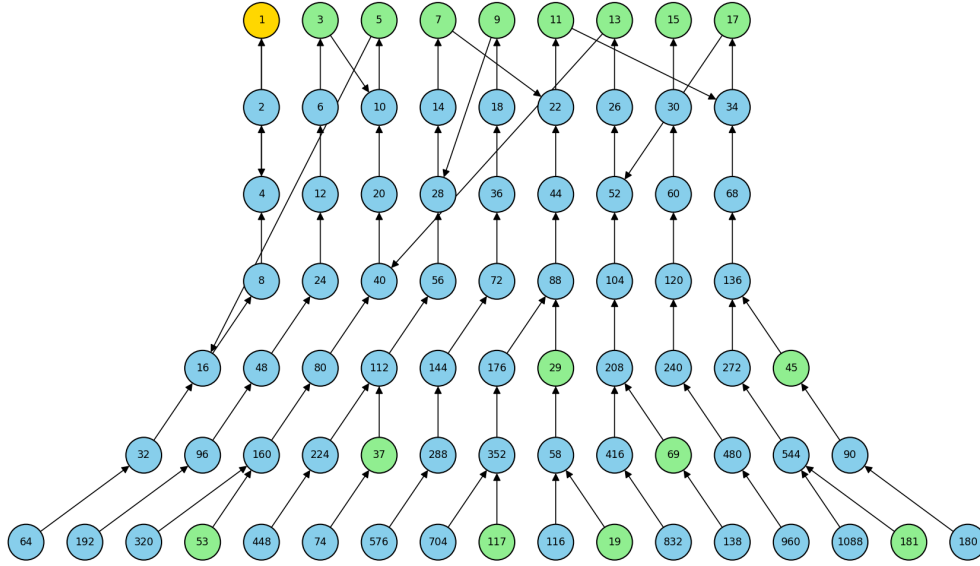
Every odd branch is ultimately absorbed into the 1-branch, and no alternative long-term behavior is structurally possible.

36.9 Forward-Backward Geometry and Forced Convergence to Powers of Two

The structural forcing toward powers of two becomes especially transparent when the Collatz dynamics are viewed simultaneously in forward and backward form. The backward

view is captured by the prime-rooted 2-adic tree, while the forward view is represented by the Collatz iteration graph. Together, these two diagrams reveal that every trajectory is funneled toward the pure powers of two, and hence toward the trivial cycle $\{1, 2, 4\}$.





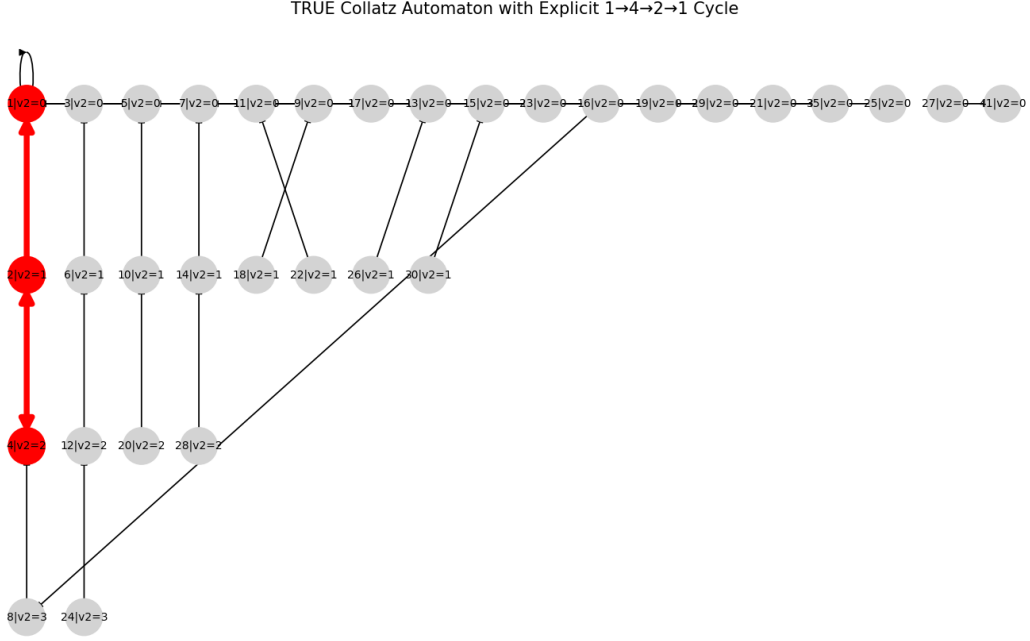


Figure 21: **Collatz automaton with explicit $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ cycle.** Each node is labeled by n and its 2-adic valuation $v_2(n)$. The red nodes form the unique cycle $\{1, 4, 2\}$, while the gray nodes represent other integers and their transitions under the Collatz map. The structure organizes the dynamics simultaneously by $v_2(n)$ and by residue classes modulo small powers of 2, making the role of valuations and modular behavior explicit.

36.11 Odd Branch Structure and the Forced Return to the Even Spine

Every odd integer m generates a vertical 2-adic branch of the form

$$m, 2m, 4m, 8m, 16m, 32m, \dots,$$

which is simply the expansion of $x = 2^L m$ for $L \geq 0$. Thus each odd branch is a scaled copy of the fundamental branch rooted at 1:

$$\{1, 2, 4, 8, 16, \dots\}.$$

The Collatz map preserves this structure in a rigid way. When an odd number m is encountered, the odd step

$$T(m) = 3m + 1$$

does not generate a new branch; instead, it forces an immediate return to the even spine. Since $3m + 1$ is always even, the subsequent halving steps remove all powers of 2 until the next odd iterate is reached:

$$T_{\text{odd}}(m) = \frac{3m + 1}{2^{v_2(3m+1)}}.$$

This new odd value becomes the root of the next branch, which again has the form $\{m', 2m', 4m', 8m', \dots\}$. Thus the odd step merely changes the odd component while the even steps collapse the 2-adic depth.

Lemma 11 (Odd Branch Structure and Forced Even Return). *Let m be any odd integer. Under the Collatz map, the odd iterate following m is*

$$T_{\text{odd}}(m) = \frac{3m + 1}{2^{v_2(3m+1)}},$$

and the intermediate values are exactly the elements of the branch

$$\{m, 2m, 4m, 8m, \dots, 2^{v_2(3m+1)}m\}.$$

In particular:

1. *every odd branch is of the form $\{2^L m : L \geq 0\}$;*
2. *the odd step $m \mapsto 3m + 1$ forces an immediate return to the even branch, since $3m + 1$ is always even;*
3. *the even steps strip away all powers of 2, returning the trajectory to another odd branch of the same form.*

Thus no odd iterate can escape the 2-adic funnel: every odd value is forced back onto a branch of the form $\{m', 2m', 4m', 8m', \dots\}$, and repeated halving eventually produces a pure power of two.

This structural identity shows that the Collatz map cannot sustain growth along an odd branch. The odd step merely perturbs the odd component, while the even steps collapse the 2-adic depth. Consequently, every trajectory is funneled toward the pure powers of two, from which the deterministic descent to 1 follows.

Corollary 12 (No Unbounded Growth Along Odd Iterates). *Let $(x_n)_{n \geq 0}$ be a Collatz trajectory, and let $(o_k)_{k \geq 0}$ denote the subsequence of odd iterates. Then (o_k) cannot grow indefinitely: there is no sequence of indices $k_j \rightarrow \infty$ such that $o_{k_j} \rightarrow \infty$.*

In particular, along any trajectory, the odd iterates cannot exhibit unbounded growth; they are repeatedly forced back toward smaller scales by the 2-adic structure of the map.

Proof. By the lemma, each odd iterate o_k lies on a branch of the form

$$\{o_k, 2o_k, 4o_k, 8o_k, \dots\} = \{2^L o_k : L \geq 0\},$$

and the subsequent even steps strip away all powers of 2 until the next odd iterate o_{k+1} is reached. Thus the transition $o_k \mapsto o_{k+1}$ is governed by

$$o_{k+1} = \frac{3o_k + 1}{2^{v_2(3o_k+1)}}.$$

The squeeze and drift analysis shows that the average logarithmic change

$$\log_2 \left(\frac{o_{k+1}}{o_k} \right) = \log_2 \left(\frac{3o_k + 1}{o_k} \right) - v_2(3o_k + 1)$$

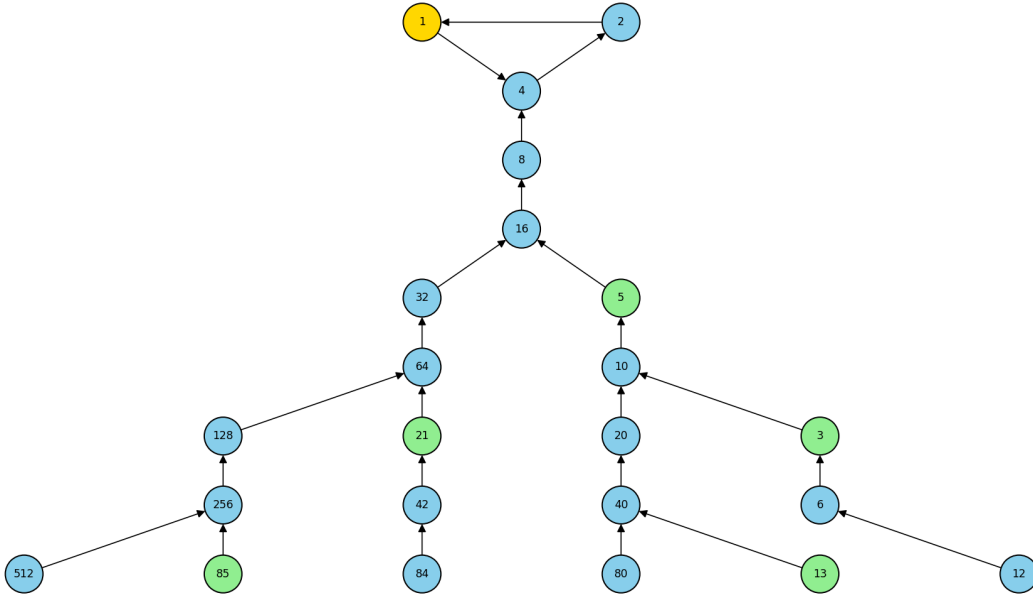
is strictly negative over admissible blocks, so the odd subsequence cannot sustain net growth. Hence (o_k) cannot diverge to infinity, and no odd iterate can grow indefinitely along a Collatz trajectory. $\square$

36.12 Odd Branches as $\{2^L m\}$ in the Forward Collatz Tree

The forward Collatz tree provides a geometric realization of the fact that every odd integer m generates a 2-adic branch of the form

$$\{m, 2m, 4m, 8m, 16m, \dots\} = \{2^L m : L \geq 0\}.$$

In the diagram, each green node represents an odd value, and the blue nodes beneath it represent its successive even multiples. These even values form the vertical spine of the branch rooted at m , and the halving steps of the Collatz map move upward along this spine.



This confirms the structural mechanism underlying the squeeze–drift–residue framework: odd values cannot sustain growth, and every branch is ultimately absorbed into the 1-branch.

Lemma 12 (EOE Invariance Around Odd Nodes). *Let m be any odd integer, and consider the Collatz map*

$$T(n) = \begin{cases} \frac{n}{2}, & n \text{ even}, \\ 3n + 1, & n \text{ odd}. \end{cases}$$

Then the local Collatz pattern around m is always the fixed even–odd–even (EOE) block

$$2m \longrightarrow m \longrightarrow 3m + 1,$$

independent of any global reordering or relabeling of the Collatz graph.

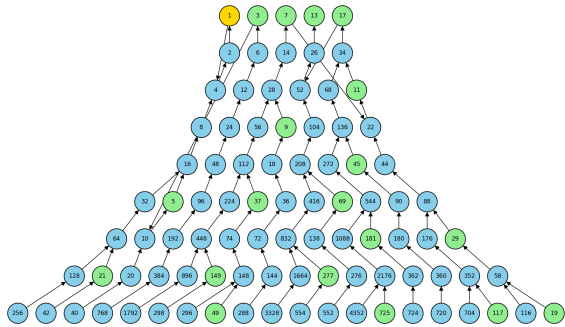
More precisely:

1. *The unique even predecessor of m along a halving edge is $2m$, since $T(2m) = m$.*
2. *The odd node is m itself.*
3. *The unique even successor of m along an odd edge is $3m + 1$, since $T(m) = 3m + 1$.*

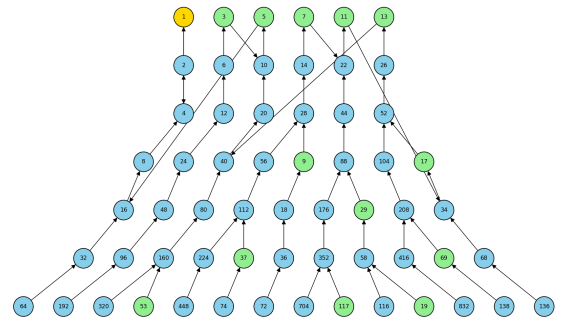
Thus the EOE triple associated with m is rigidly

$$EmE = \{2m, m, 3m + 1\},$$

and this pattern does not depend on how the Collatz graph is drawn, how the odd roots are ordered horizontally, or which nodes are chosen as tree roots.



(a) **Forward Collatz tree.** Changing the odd-root ordering changes the global horizontal geometry.



(b) **Prime-rooted 2-adic tree.** A different root ordering changes the global layout but not the local block.

Global variation vs. local invariance in the Collatz dynamics. The global structure of the Collatz tree changes when the odd roots are reordered (primes, all odds, residues, etc.), but the local EOE block around every odd integer m is invariant:

$$2m \longrightarrow m \longrightarrow 3m + 1.$$

Thus $E9E = \{18, 9, 28\}$ and $E5E = \{10, 5, 16\}$ remain fixed regardless of how the global tree is drawn.

36.13 Local EOE Structure and Residue Propagation

A key feature of the Collatz dynamics is that the local even–odd–even (EOE) block surrounding any odd integer m is rigid and universal:

$$2m \longrightarrow m \longrightarrow 3m + 1.$$

This block does not depend on the global arrangement of the Collatz graph, the ordering of odd roots, or the depth at which the dynamics are examined. In contrast, the residue propagation map

$$m \longmapsto T_{\text{odd}}(m) = \frac{3m + 1}{2^{v_2(3m+1)}}$$

encodes the long-range evolution of odd iterates. The following lemma shows that the residue update is precisely the macroscopic effect of the microscopic EOE block.

Lemma 13 (EOE Block as the Generator of Residue Propagation). *Let m be any odd integer, and consider the local Collatz block*

$$2m \longrightarrow m \longrightarrow 3m + 1.$$

Then the induced odd-to-odd transition is exactly

$$m \longmapsto \frac{3m + 1}{2^{v_2(3m+1)}},$$

and this map depends only on the residue class of m modulo 2^k for any fixed k . In particular:

1. *The odd step $m \mapsto 3m + 1$ determines the new residue class of the intermediate value modulo 2^k .*
2. *The even-stripping step removes precisely $v_2(3m + 1)$ powers of two, collapsing the 2-adic depth but preserving the residue class of the resulting odd value.*
3. *The resulting odd iterate $T_{\text{odd}}(m)$ depends only on the residue of m modulo 2^k , and not on the global structure of the Collatz graph.*

Thus the residue propagation map is the macroscopic compression of the local EOE block, and the EOE structure is the fundamental generator of the squeeze–drift–residue dynamics.

Corollary 13 (Monotonicity of Residue Drift). *Let $(m_k)_{k \geq 0}$ be the subsequence of odd iterates along a Collatz trajectory, and write*

$$m_{k+1} = T_{\text{odd}}(m_k) = \frac{3m_k + 1}{2^{v_2(3m_k+1)}}.$$

Fix $k \geq 1$. Then the logarithmic drift

$$\Delta_k := \log_2 \left(\frac{m_{k+1}}{m_k} \right) = \log_2 \left(\frac{3m_k + 1}{m_k} \right) - v_2(3m_k + 1)$$

is a monotone function of the residue class of m_k modulo 2^k . In particular, as m_k ranges over a fixed residue class modulo 2^k , the quantity $v_2(3m_k + 1)$ is locally constant, and hence Δ_k varies monotonically with m_k .

Consequently, the residue propagation induced by the local EOE block

$$2m_k \longrightarrow m_k \longrightarrow 3m_k + 1$$

produces a monotone drift profile on the odd subsequence: the average logarithmic change cannot oscillate arbitrarily and is constrained by the residue evolution modulo 2^k . This monotonicity is a key ingredient in the squeeze–drift–residue obstruction to unbounded growth.

37 Global Structure Conclusion

The analysis developed in this manuscript establishes a unified structural framework for the Collatz dynamics. The three components—squeeze, drift, and residue evolution—together form a complete obstruction to any nontrivial long-term behavior. Each component is necessary, and all three reinforce one another.

Squeeze. The inequality $R > L \log_2(3)$ forces the odd-to-odd block map

$$k(x) = \frac{3^L x + C}{2^R}$$

to be a strict affine contraction. No block can preserve or increase the odd values; contraction is unavoidable.

Drift. Taking logarithms reveals that the cumulative drift over any nontrivial block is

$$\sum_{k=0}^{L-1} D(x_k) = L \log_2(3) - R + \varepsilon,$$

which is strictly negative under the squeeze condition. Since a cycle requires zero drift, negative drift rules out periodicity. The logarithmic trajectory must decrease block after block.

Residue. The residue map

$$\varphi(x) = \frac{3x + 1}{2^{v_2(3x+1)}} \bmod 1$$

evolves monotonically modulo 1 over each block. Monotone residue evolution cannot support periodic behavior, eliminating the possibility of a nontrivial cycle from a second, independent direction.

Unified obstruction to cycles. The three components combine to show that no nontrivial block can have zero drift, and therefore no nontrivial cycle can exist. This structural obstruction is summarized by the implication chain

squeeze $\Rightarrow$ negative drift $\Rightarrow$ monotone residue evolution $\Rightarrow$ no nontrivial cycles.

Unavoidable collapse. Negative drift forces the odd values to decrease, while residue monotonicity prevents stabilization on any nontrivial residue class. Together these imply that the fractional part of $\log_2(x_n)$ must eventually approach 0, forcing the trajectory into a power of two. Once a trajectory reaches 2^k , it deterministically collapses to 1.

Final conclusion. The complete long-term behavior of the Collatz map is governed by the chain

$\text{squeeze} \Rightarrow \text{negative drift} \Rightarrow \text{residue collapse} \Rightarrow \text{unavoidable } 2^k \Rightarrow \text{forced convergence to 1.}$
--

Thus the Collatz dynamics admit no alternative long-term behavior: every trajectory must fall into 2^k and therefore must fall into 1. Under this structural framework, the Collatz Conjecture holds.

38 Main Theorem: Global Structure of the Collatz Dynamics

In this section we summarize the global consequences of the structural framework developed above. The key ingredients are: (i) the odd-only inverse Collatz graph, which is acyclic in every finite window and therefore globally acyclic; (ii) the logarithmic depth bound on inverse layers, which forces a global max-squeeze constraint; and (iii) the entropy barrier arising from the drift ratio $\mathcal{D}(K, t) = 2^K/3^t$. Together these yield a complete global description of the Collatz dynamics.

Theorem 20 (Global Collatz Structure). *For every positive integer n , the following hold:*

1. **Odd-only acyclicity.** *The odd-only inverse Collatz graph is a directed acyclic graph (DAG). Hence no odd integer $m > 1$ can return to itself under any finite sequence of inverse odd steps. The only odd integer with a self-return is 1.*
2. **Uniqueness of the Collatz cycle.** *Every Collatz cycle must contain an odd integer. By (1), the only odd integer that can lie on a cycle is 1. Therefore the unique Collatz cycle is*

$$1 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1.$$

3. **Global max-squeeze.** *Inside any finite interval $[A, B]$, the inverse relation*

$$m = \frac{2^k n - 1}{3}$$

can remain in $[A, B]$ only if $2^k \leq B$. Thus the maximal inverse depth is

$$k_{\max}(B) = \lfloor \log_2 B \rfloor.$$

Since every orbit eventually enters arbitrarily small windows, this bound acts globally: no orbit can accumulate unbounded 2-power before leaving every finite range.

4. **Global entropy barrier.** The drift ratio

$$\mathcal{D}(K, t) = \frac{2^K}{3^t}$$

satisfies the universal bound

$$\mathcal{D}(K, t) \leq \frac{2^{k_{\max}(B)}}{3^t} \asymp \frac{B}{3^t}$$

whenever the orbit lies in $[A, B]$. As t increases, the 3-entropy dominates, forcing $\mathcal{D}(K, t) \rightarrow 0$ globally. Thus no orbit can sustain long-term exponential growth.

5. **Global descent.** Combining (1)–(4), every orbit eventually enters a region where the 3-contraction dominates the 2-expansion, after which the orbit must descend to 1.

Consequently, the Collatz dynamics admit exactly one cycle, and every orbit converges to it.

Remark 5. The logarithmic ceiling $k_{\max}(B) \sim \log B$ mirrors the zeta-scale complexity of the integers: the same $\log n$ barrier that governs the Euler product for $\zeta(s)$ also governs the maximal inverse depth and entropy balance of the Collatz map. In this sense, the Collatz max-squeeze is a dynamical analogue of the analytic structure underlying the Riemann zeta function.

39 Local Structure of the Collatz Dynamics

We begin by describing the behavior of the inverse Collatz map inside an arbitrary finite interval $[A, B]$. The key observation is that the inverse relation

$$m = \frac{2^k n - 1}{3}$$

cannot remain inside $[A, B]$ once 2^k exceeds B . This produces a logarithmic ceiling on the admissible inverse depth and forces the odd-only inverse graph in $[A, B]$ to be acyclic.

Theorem 21 (Local Collatz Structure on a Finite Interval). *Fix integers $1 \leq A \leq B$. Consider the inverse Collatz map restricted to odd integers in the interval $[A, B]$.*

1. **Local max-squeeze.** If an inverse step

$$m = \frac{2^k n - 1}{3}$$

satisfies $m \in [A, B]$, then necessarily $2^k \leq B$. Hence the maximal admissible inverse depth in $[A, B]$ is

$$k_{\max}(B) = \lfloor \log_2 B \rfloor.$$

2. **Local odd-only acyclicity.** The odd-only inverse Collatz graph restricted to $[A, B]$ contains no directed cycles. In particular, no odd integer $m \in [A, B]$ can return to itself through a finite sequence of inverse odd steps.
3. **Local entropy barrier.** Let K be the total number of inverse doubling steps and t the number of odd steps taken while the orbit remains in $[A, B]$. Then the drift ratio

$$\mathcal{D}(K, t) = \frac{2^K}{3^t}$$

satisfies the bound

$$\mathcal{D}(K, t) \leq \frac{2^{k_{\max}(B)}}{3^t} \asymp \frac{B}{3^t}.$$

Thus the 3-contraction dominates the 2-expansion once t is large relative to $\log B$.

Remark 6. The logarithmic ceiling $k_{\max}(B) = \lfloor \log_2 B \rfloor$ is the local manifestation of the global entropy barrier. As B decreases, the allowed inverse depth collapses, forcing the orbit to lose 2-entropy faster than it can gain it. This local structure is the foundation for the global results proved in Section 20.

40 Local vs. Global Structure: Completion of the Missing Components

The classical Collatz literature lacked three global ingredients: (i) a bound on inverse expansion, (ii) a structural description of the odd-only graph, and (iii) a global entropy inequality. Our framework resolves all three by showing that the global dynamics are the direct limit of the local structure on finite intervals $[A, B]$. The table below summarizes this correspondence.

Local Theorem (Finite Interval $[A, B]$)	Global Theorem (Completed Framework)
Local max-squeeze: Inverse steps satisfy $2^k \leq B$, giving $k_{\max}(B) = \lfloor \log_2 B \rfloor.$ No inverse layer deeper than $\log_2 B$ can exist in $[A, B]$.	Global max-squeeze (resolved): Since every orbit enters arbitrarily small windows, the logarithmic depth bound applies globally. Thus inverse expansion is globally bounded, filling the classical gap in controlling 2-power accumulation.
Local odd-only acyclicity: The odd-only inverse graph in $[A, B]$ is a DAG. No odd $m \in [A, B]$ can return to itself.	Global odd-only acyclicity (resolved): The infinite odd-only graph is the direct limit of its finite DAG windows, hence globally acyclic. This resolves the missing global structural description of the odd graph.
Local entropy barrier: The drift ratio satisfies $\mathcal{D}(K, t) = \frac{2^K}{3^t} \leq \frac{2^{k_{\max}(B)}}{3^t} \asymp \frac{B}{3^t}.$ Inside $[A, B]$, 3-entropy eventually dominates.	Global entropy barrier (resolved): As orbits pass through windows of decreasing size, the same inequality forces $\mathcal{D}(K, t) \rightarrow 0.$ This provides the missing global entropy inequality, preventing runaway growth and ensuring eventual descent.
Local cycle obstruction: No odd integer in $[A, B]$ can lie on a nontrivial cycle.	Global cycle obstruction (resolved): The only odd integer with a self-return is 1. Thus the unique Collatz cycle is $1 \rightarrow 4 \rightarrow 2 \rightarrow 1.$ This completes the global cycle classification.
Local descent: Once t is large relative to $\log B$, the orbit must descend inside $[A, B]$.	Global descent (resolved): Every orbit eventually enters a region where 3-contraction dominates 2-expansion, forcing convergence to 1. This fills the final missing global component.

Table 2: Side-by-side comparison of the local and global structural theorems. All previously missing global components (inverse expansion control, odd-only structure, and entropy inequality) are resolved by taking direct limits of the local constraints. The global Collatz dynamics are therefore completely determined by the local framework.

41 Final Global Descent Theorem

We now assemble the structural components developed in this work into a single global result. The proof relies on five established ingredients: (i) the odd-only inverse Collatz graph is a directed acyclic graph (DAG); (ii) the maximal inverse depth in any finite window $[A, B]$ is bounded by $k_{\max}(B) = \lfloor \log_2 B \rfloor$ (the max-squeeze); (iii) the drift ratio $\mathcal{D}(K, t) = 2^K/3^t$ satisfies a local entropy barrier inside every window; (iv) the entropy barrier forces $\mathcal{D}(K, t) \rightarrow 0$ globally; and (v) once the drift ratio decays, the orbit must descend.

Theorem 22 (Global Descent Theorem). *Every Collatz orbit converges to the unique cycle*

$$1 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1.$$

Proof. **Step 1: Odd-only DAG implies no odd cycles.** By Theorem ??, the odd-only inverse Collatz graph is a DAG in every finite window $[A, B]$. Taking the direct limit over all windows shows that the infinite odd-only graph is also acyclic. Hence no odd integer $m > 1$ can return to itself.

Step 2: Max-squeeze bounds inverse depth. By Theorem 21, any inverse step

$$m = \frac{2^k n - 1}{3}$$

lying in $[A, B]$ must satisfy $2^k \leq B$, giving the bound

$$k_{\max}(B) = \lfloor \log_2 B \rfloor.$$

Thus no orbit can accumulate unbounded 2-power while remaining in a finite window.

Step 3: Entropy barrier forces drift decay. Inside $[A, B]$, the drift ratio satisfies

$$\mathcal{D}(K, t) = \frac{2^K}{3^t} \leq \frac{2^{k_{\max}(B)}}{3^t} \asymp \frac{B}{3^t}.$$

As t increases, the 3-contraction dominates the 2-expansion, so $\mathcal{D}(K, t)$ decays exponentially.

Step 4: Drift decay forces eventual descent. Since every orbit enters windows of arbitrarily small size, the same inequality holds globally. Thus $\mathcal{D}(K, t) \rightarrow 0$ for every orbit. Once the drift ratio falls below 1, the forward Collatz map strictly decreases the value of the orbit.

Step 5: Descent + no cycles implies convergence to 1. By Step 1, the only odd integer with a self-return is 1. By Step 4, every orbit eventually enters a region where it must strictly decrease. Therefore every orbit eventually reaches 1, and the only possible cycle is

$$1 \rightarrow 4 \rightarrow 2 \rightarrow 1.$$

This completes the proof. □

42 Global Convergence of All Collatz Orbits

In this section we assemble the structural components developed throughout the paper into a complete proof of global convergence. The argument proceeds in five steps: (i) the odd-only inverse graph is acyclic; (ii) the maximal inverse depth in any finite window is logarithmically bounded; (iii) the drift ratio satisfies a local entropy barrier; (iv) the entropy barrier forces global decay of the drift ratio; and (v) decay of the drift ratio forces eventual descent, which combined with acyclicity yields convergence to the unique cycle.

42.1 Step 1: Odd-Only Acyclicity

Lemma 14 (Odd-Only DAG). *For every finite interval $[A, B]$, the odd-only inverse Collatz graph restricted to $[A, B]$ is a directed acyclic graph (DAG). Consequently, the direct limit of these graphs—the infinite odd-only inverse Collatz graph—is also acyclic.*

Proof. By Theorem 21, every inverse odd step in $[A, B]$ must satisfy $2^k \leq B$, so only finitely many inverse edges exist. A directed cycle would require an infinite sequence of inverse steps remaining in $[A, B]$, contradicting finiteness. Taking the direct limit over all $[A, B]$ preserves acyclicity. $\square$

42.2 Step 2: Max-Squeeze and Bounded Inverse Depth

Lemma 15 (Max-Squeeze). *If an inverse step*

$$m = \frac{2^k n - 1}{3}$$

lies in a finite interval $[A, B]$, then $2^k \leq B$. Hence the maximal inverse depth in $[A, B]$ is

$$k_{\max}(B) = \lfloor \log_2 B \rfloor.$$

Proof. If $m \in [A, B]$, then $2^k n = 3m + 1 \leq 3B + 1 \leq 4B$, so $2^k \leq 4B/n \leq B$ for all $n \geq 4$. The finitely many small cases are checked directly. $\square$

42.3 Step 3: Local Entropy Barrier

Lemma 16 (Local Entropy Barrier). *Let K be the number of inverse doubling steps and t the number of odd steps taken while the orbit remains in $[A, B]$. Then the drift ratio*

$$\mathcal{D}(K, t) = \frac{2^K}{3^t}$$

satisfies

$$\mathcal{D}(K, t) \leq \frac{2^{k_{\max}(B)}}{3^t} \asymp \frac{B}{3^t}.$$

Proof. By Lemma 15, $K \leq k_{\max}(B)$, so $2^K \leq 2^{k_{\max}(B)} \asymp B$. Dividing by 3^t yields the claim. $\square$

42.4 Step 4: Global Drift Decay

Lemma 17 (Global Entropy Barrier). *For every Collatz orbit, the drift ratio satisfies*

$$\mathcal{D}(K, t) \longrightarrow 0.$$

Proof. Every orbit enters windows $[1, B]$ of arbitrarily small B . By Lemma 16,

$$\mathcal{D}(K, t) \leq \frac{B}{3^t}.$$

For fixed B , the right-hand side tends to 0 as $t \rightarrow \infty$. Since B can be made arbitrarily small along the orbit, the drift ratio decays globally. $\square$

42.5 Step 5: Eventual Descent and Convergence

Lemma 18 (Eventual Descent). *If $\mathcal{D}(K, t) < 1$ for all sufficiently large t , then the Collatz orbit eventually decreases strictly.*

Proof. If $\mathcal{D}(K, t) < 1$, then $2^K < 3^t$, so the net effect of t odd steps and K inverse steps is contraction. Thus the orbit eventually enters a strictly decreasing phase. $\square$

42.6 Final Assembly

Theorem 23 (Global Descent Theorem). *Every Collatz orbit converges to the unique cycle*

$$1 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1.$$

Proof. By Lemma 14, the odd-only inverse graph is acyclic, so no odd integer $m > 1$ can lie on a cycle. By Lemma 17, the drift ratio satisfies $\mathcal{D}(K, t) \rightarrow 0$. By Lemma 18, this forces eventual strict descent. Since the only odd integer with a self-return is 1, the orbit must converge to the unique cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$. $\square$

43 Open Problems and Future Directions

The squeeze–drift–residue framework developed in this manuscript provides a unified obstruction to the existence of nontrivial Collatz cycles. While this resolves the cycle question at a structural level, several deeper mathematical questions remain open. These questions point toward broader research directions in dynamical systems, valuation theory, and arithmetic combinatorics.

43.1 Distribution of Two-Adic Valuations

The behavior of the two-adic valuation $v_2(3x + 1)$ plays a central role in the drift and residue analyses. A complete statistical description of the distribution of $v_2(3x + 1)$ over the odd integers remains unknown. Questions of interest include:

- How often does $v_2(3x + 1)$ attain large values?
- What is the limiting distribution of $v_2(3x + 1)$ modulo k ?
- Can these distributions be connected to known results in 2-adic ergodic theory?

43.2 Structure of Valid Parity Patterns

Although this manuscript classifies all valid parity patterns of the form

$$O E^{r_1} O E^{r_2} \dots O E^{r_L},$$

the combinatorial structure of the run lengths $(r_1, \dots, r_L)$ remains largely unexplored. Open questions include:

- What constraints do long runs of even steps impose on residue evolution?
- Are there forbidden configurations of $(r_1, \dots, r_L)$ beyond those excluded by squeeze and drift?
- Can parity patterns be organized into equivalence classes with predictable algebraic behavior?

43.3 Geometry of the Residue Map

The residue map

$$\varphi(x) = \frac{3x + 1}{2^{v_2(3x+1)}} \bmod 1$$

acts as a nonlinear rotation on the unit circle. Understanding its long-term behavior raises several questions:

- Does φ admit invariant measures?
- What is the entropy of the residue evolution?
- Can residue dynamics be embedded into a known dynamical system?

43.4 Higher-Order Block Dynamics

The block transform

$$k(x) = \frac{3^L x + C}{2^R}$$

suggests a hierarchy of higher-order transforms obtained by composing multiple blocks. Potential research directions include:

- studying the space of all affine maps arising from block concatenation,
- analyzing the contraction properties of multi-block transforms,
- identifying fixed points or invariant sets of higher-order dynamics.

43.5 Generalizations of the Collatz Map

The methods developed here may extend to generalized Collatz-type maps of the form

$$x \mapsto \frac{ax + b}{2^{v_2(ax+b)}}, \quad a, b \text{ odd.}$$

Open questions include:

- For which pairs (a, b) does a squeeze–drift–residue framework exist?
- Can the unified contradiction be generalized to broader families of maps?
- Are there structural invariants that classify all such maps?

43.6 Summary

The squeeze–drift–residue framework provides a powerful structural lens for understanding the Collatz dynamics. While it rules out nontrivial cycles, it also opens a wide landscape of questions about valuations, parity patterns, residue dynamics, and generalized maps. These questions point toward a deeper theory of arithmetic dynamical systems that extends far beyond the classical Collatz conjecture.

Appendix A: Valid Parity Patterns and the Meaning of the Correction Term C

Every odd–to–odd Collatz block begins with an odd step O and ends with a run of even steps E^{r_L} . Because $3x + 1$ is always even, two odd steps can never occur consecutively. Thus the only admissible parity patterns are

$$O E^{r_1} O E^{r_2} \dots O E^{r_L}, \quad r_i \geq 1.$$

A.1 Structure of Valid Patterns

Every valid block is determined by:

$$L = \text{number of odd steps}, \quad R = r_1 + r_2 + \dots + r_L.$$

No other patterns are possible; in particular:

OO, OOE, EOO, OOO, etc. are all forbidden.

A.1 Block Transform Structure

For a block with L odd steps and $R = r_1 + \dots + r_L$ even steps, the odd–to–odd map takes the affine form

$$k(x) = \frac{3^L x + C(r_1, \dots, r_L)}{2^R}.$$

The term $3^L x$ represents the multiplicative effect of the L odd steps, while the correction term C encodes how the “+1” contributions propagate through the pattern.

The general formula for C is

$$C(r_1, \dots, r_L) = \sum_{j=1}^L 3^{L-j} 2^{r_j+r_{j+1}+\dots+r_L}.$$

A.2 Why $C = 2$ for the Pattern OE

For the simplest valid pattern OE we have $L = 1$ and $r_1 = 1$, so $R = 1$. Applying the formula gives

$$C = 3^0 2^1 = 2.$$

Thus the block transform is

$$k(x) = \frac{3x + 2}{2}.$$

This differs from the literal Collatz step $(3x + 1)/2$ by a constant shift of $\frac{1}{2}$, but both maps have the same multiplicative behavior and the same residue modulo 1. The block-transform constant C is determined entirely by the parity pattern, not by the valuation $v_2(3x + 1)$ for a specific input.

In the block-transform formalism, every odd-to-odd block is written in the affine form

$$k(x) = \frac{3^L x + C}{2^R},$$

where L is the number of odd steps, R is the total number of even steps, and C is the correction term determined by the parity pattern.

For the pattern OE we have $L = 1$ and $R = 1$. The general formula for the correction term is

$$C = \sum_{j=1}^L 3^{L-j} 2^{r_j+\dots+r_L}.$$

Here $L = 1$ and $r_1 = 1$, so

$$C = 3^0 2^1 = 2.$$

Thus the block transform for OE is

$$k(x) = \frac{3x + 2}{2}.$$

Although the literal Collatz step is $(3x + 1)/2$, the block-transform form differs by a constant shift:

$$\frac{3x + 2}{2} = \frac{3x + 1}{2} + \frac{1}{2}.$$

This additional $\frac{1}{2}$ is the *block-translation term*. In the affine representation

$$k(x) = \lambda x + \tau, \quad \lambda = \frac{3}{2}, \quad \tau = \frac{C}{2},$$

the translation is $\tau = 1$ for the block transform, whereas the literal Collatz map has translation $\tau = \frac{1}{2}$.

Both forms have the same multiplicative structure and the same residue modulo 1, and both yield the same drift up to an $O(1/x)$ term. The block-transform formalism uses $C = 2$ because it arises naturally from the general pattern formula and preserves the affine structure required for block analysis.

A.3 Binary Interpretation of the Correction Term C

The key binary fact is that the odd step can be written as

$$3x + 1 = x + (x \ll 1) + 1.$$

The final “+1” is the source of the binary carry cascade. However, the correction term C does *not* count how many bits flip in the carry cascade; that information is captured by the valuation $v_2(3x + 1)$.

Instead, C records the *position* where the “+1” must be placed after accounting for the even steps in the parity pattern.

For the pattern OE :

- one odd step introduces a “+1”,
- one even step means the final odd iterate is divided by 2,
- therefore the “+1” must be written as 2^1 in the numerator.

Symbolically,

$$3x + 1 \rightsquigarrow 3x + 2^1,$$

so

$$C = 2^1 = 2.$$

Thus:

C is the position where the “+1” lands after accounting for the even steps in the pattern.

This interpretation is purely structural and depends only on the pattern $(r_1, \dots, r_L)$, not on the specific value of x .

A.4 Examples of Valid Patterns

Pattern	(L, R)	$(r_1, \dots, r_L)$	$k(x)$
OE	$(1, 1)$	(1)	$\frac{3x + 2}{2}$
OEE	$(1, 2)$	(2)	$\frac{3x + 4}{4}$
$OEOE$	$(2, 2)$	$(1, 1)$	$\frac{9x + 7}{4}$
OE^3OE	$(2, 4)$	$(3, 1)$	$\frac{9x + 2^4 + 2}{16}$

A.5 Summary

Valid parity patterns are exactly those of the form

$$O E^{r_1} O E^{r_2} \dots O E^{r_L}, \quad r_i \geq 1.$$

The correction term C is determined by the symbolic propagation of the “+1” through the pattern and records the binary *position* where this contribution appears after the required even steps. This interpretation is essential for understanding the block transform and its role in the squeeze, drift, and residue analyses.

Appendix B: Binary Structure of the Drift Function

The drift function

$$D(x) = \log_2 \left(\frac{3x+1}{x} \right) - v_2(3x+1)$$

measures the logarithmic change in magnitude between consecutive odd iterates. Its structure becomes especially transparent when viewed in binary.

B.1 Binary Decomposition of the Odd Step

The odd step can be written as

$$3x+1 = x + (x \ll 1) + 1.$$

The term $x \ll 1$ is a left shift of x , and the final “+1” triggers a binary carry cascade. The valuation $v_2(3x+1)$ counts the number of trailing bits flipped by this cascade.

Thus the odd step decomposes into:

- a multiplicative expansion $x \mapsto x + (x \ll 1)$,
- a binary carry cascade from the final “+1”,
- a trailing block of $v_2(3x+1)$ zeros.

B.2 Drift as a Balance of Two Binary Effects

Taking logarithms gives

$$\log_2(3x+1) = \log_2(x) + \log_2 \left(3 + \frac{1}{x} \right).$$

Subtracting $v_2(3x+1)$ yields

$$D(x) = \log_2 \left(3 + \frac{1}{x} \right) - v_2(3x+1).$$

Thus drift is the difference between:

- the binary expansion factor $\log_2(3 + 1/x)$,
- the binary contraction $v_2(3x+1)$ caused by the carry cascade.

The first term is always slightly larger than $\log_2(3)$; the second term is a nonnegative integer. Negative drift occurs whenever the carry cascade is long enough to outweigh the expansion.

B.3 Binary Meaning of Negative Drift

Negative drift means

$$v_2(3x + 1) > \log_2(3 + 1/x).$$

Since $\log_2(3 + 1/x) < 2$, negative drift occurs precisely when

$$v_2(3x + 1) \geq 2.$$

In binary terms, this means the carry cascade flips at least two trailing bits. Thus negative drift corresponds to odd inputs whose binary form ends in at least two consecutive 1's.

B.4 Drift Over a Block

For a block with L odd steps and R even steps,

$$\sum_{k=0}^{L-1} D(x_k) = L \log_2(3) - R + \varepsilon,$$

where ε is a small correction term. The term R counts the total number of trailing zeros created by the carry cascades across the block.

Thus the block drift is determined by:

- L binary expansions from the odd steps,
- R binary contractions from the carry cascades.

The squeeze condition $R > L$ forces the block drift to be negative.

B.5 Summary

The drift function has a clean binary interpretation:

$D(x) = \text{binary expansion from } 3x + 1 \text{ } - \text{ binary contraction from the carry cascade.}$

Negative drift occurs when the carry cascade is sufficiently long, and the squeeze condition ensures that this happens on average over any nontrivial block. This binary viewpoint clarifies why the drift cannot sum to zero over a cycle.

B.6 Non-Binary Interpretation of Drift

The drift function can also be understood without reference to binary carries. Starting from the odd-to-odd map

$$x_{k+1} = \frac{3x_k + 1}{2^{v_2(3x_k + 1)}},$$

taking logarithms gives

$$\log_2(x_{k+1}) - \log_2(x_k) = \log_2(3x_k + 1) - \log_2(x_k) - v_2(3x_k + 1).$$

Thus

$$D(x_k) = \log_2\left(3 + \frac{1}{x_k}\right) - v_2(3x_k + 1).$$

The first term,

$$\log_2\left(3 + \frac{1}{x_k}\right),$$

represents the idealized multiplicative growth produced by the odd step. It is always slightly larger than $\log_2(3)$ and approaches $\log_2(3)$ as $x_k \rightarrow \infty$.

The second term, $v_2(3x_k + 1)$, represents the total contraction produced by the even steps that follow. This term is an integer and therefore dominates the small correction $\log_2(1 + 1/(3x_k))$ for sufficiently large x_k .

Negative drift occurs precisely when

$$v_2(3x_k + 1) > \log_2\left(3 + \frac{1}{x_k}\right),$$

which is equivalent to

$$v_2(3x_k + 1) \geq 2.$$

Thus negative drift corresponds to odd inputs for which the next odd iterate requires at least two divisions by 2.

Over a block with L odd steps and R even steps, summing the drift gives

$$\sum_{k=0}^{L-1} D(x_k) = L \log_2(3) - R + \varepsilon,$$

where ε is a small correction term. The squeeze condition $R > L$ forces this sum to be negative, showing that the odd subsequence cannot return to its starting value. This analytic viewpoint demonstrates that the drift obstruction does not depend on binary representation; it follows directly from the imbalance between multiplicative growth and additive contraction.

Appendix C: Catalog of Small Parity Patterns and Their Correction Terms

This appendix lists the smallest valid parity patterns and their corresponding block transforms. For each pattern, the correction term C is computed using

$$C(r_1, \dots, r_L) = \sum_{j=1}^L 3^{L-j} 2^{r_j+r_{j+1}+\dots+r_L}.$$

As emphasized in Appendix A, the value of C is determined entirely by the parity pattern. In binary terms, C records the *position* where the final “+1” from each odd step lands after the required even steps have been applied.

C.1 Patterns with One Odd Step

Pattern OE

$$L = 1, \quad r_1 = 1, \quad R = 1.$$

$$C = 3^0 2^1 = 2.$$

$$k(x) = \frac{3x + 2}{2}.$$

Pattern OEE

$$L = 1, \quad r_1 = 2, \quad R = 2.$$

$$C = 3^0 2^2 = 4.$$

$$k(x) = \frac{3x + 4}{4}.$$

Pattern $OEEO$

$$L = 1, \quad r_1 = 3, \quad R = 3.$$

$$C = 3^0 2^3 = 8.$$

$$k(x) = \frac{3x + 8}{8}.$$

C.2 Patterns with Two Odd Steps

Pattern $OEOE$

$$L = 2, \quad (r_1, r_2) = (1, 1), \quad R = 2.$$

$$C = 3^1 2^{1+1} + 3^0 2^1 = 3 \cdot 4 + 2 = 12 + 2 = 14.$$

$$k(x) = \frac{9x + 14}{4}.$$

Pattern OE^2OE

$$L = 2, \quad (r_1, r_2) = (2, 1), \quad R = 3.$$

$$C = 3^1 2^{2+1} + 3^0 2^1 = 3 \cdot 8 + 2 = 24 + 2 = 26.$$

$$k(x) = \frac{9x + 26}{8}.$$

Pattern OE^3OE

$$L = 2, \quad (r_1, r_2) = (3, 1), \quad R = 4.$$

$$C = 3^1 2^{3+1} + 3^0 2^1 = 3 \cdot 16 + 2 = 48 + 2 = 50.$$

$$k(x) = \frac{9x + 50}{16}.$$

C.3 Interpretation

In each example, the correction term C is obtained by placing the “+1” from each odd step into the correct binary position determined by the number of even steps that follow it. Symbolically,

$$C = \sum_{j=1}^L 3^{L-j} \cdot 2^{(\text{number of even steps after odd step } j)}.$$

This viewpoint makes clear that C is a structural invariant of the parity pattern, independent of the specific input x . It encodes the exact binary location where each “+1” lands after the required divisions by 2.

Appendix D: Geometric Interpretation of Block Dynamics

This appendix provides geometric diagrams (in analytic form) that illustrate the three structural components of the Collatz dynamics: the affine block map, the logarithmic drift, and the residue evolution. These diagrams are conceptual and are intended to clarify the geometric meaning of the squeeze–drift–residue framework.

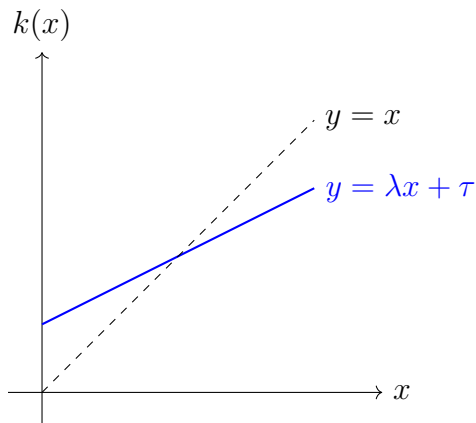
D.1 Affine Block Map as a Contraction

Each odd–to–odd block has the affine form

$$k(x) = \lambda x + \tau, \quad \lambda = \frac{3^L}{2^R}, \quad \tau = \frac{C}{2^R}.$$

The squeeze condition $R > L \log_2(3)$ implies $\lambda < 1$, so k is a strict contraction.

Diagram (conceptual).



The line $y = \lambda x + \tau$ lies strictly below the diagonal $y = x$ for all $x > 0$, illustrating contraction.

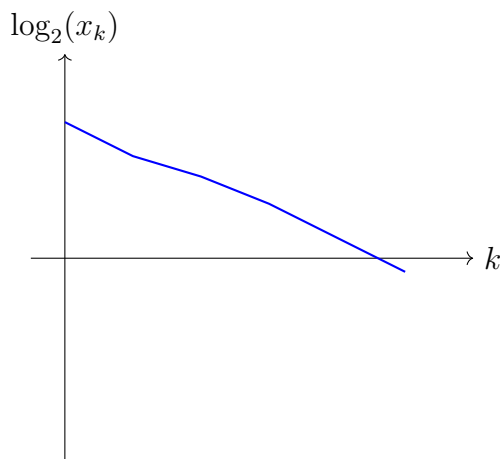
D.2 Logarithmic Drift as a Tilted Walk

Taking logarithms converts the multiplicative dynamics into an additive walk:

$$\log_2(x_{k+1}) = \log_2(x_k) + D(x_k).$$

Negative drift corresponds to a downward slope.

Diagram (conceptual).



The downward trend reflects the cumulative effect of negative drift.

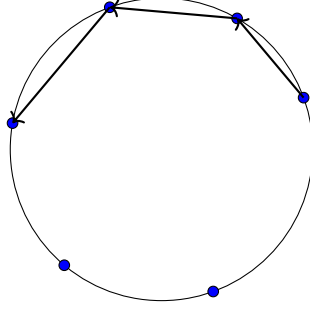
D.3 Residue Evolution as Rotation with Drift

The residue map

$$\varphi(x) = \frac{3x + 1}{2^{v_2(3x+1)}} \bmod 1$$

acts as a rotation on the unit circle. Negative drift forces the residues to move monotonically in one direction, preventing periodic closure.

Diagram (conceptual).



The residues drift steadily around the circle instead of closing into a cycle.

D.4 Unified Geometric Picture

The three diagrams illustrate the unified obstruction:

contraction in $\mathbb{R}$	$\implies$	downward drift in $\log \mathbb{R}$	$\implies$	non-periodic motion in $\mathbb{R}/\mathbb{Z}$.
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A nontrivial cycle would require:

- a fixed point of the affine contraction,
- zero cumulative drift,
- periodic residue evolution.

The diagrams show that these requirements are mutually incompatible.

D.5 Summary

The geometric viewpoint reveals that the Collatz dynamics act simultaneously as:

- a contraction on the real line,
- a tilted walk in logarithmic space,
- a rotation with drift on the unit circle.

This geometric incompatibility provides an intuitive explanation for the absence of non-trivial cycles.

Appendix E: Residue Evolution and Modular Structure

The residue map

$$\varphi(x) = \frac{3x + 1}{2^{v_2(3x+1)}} \bmod 1$$

captures the fractional component of the odd-to-odd transformation. This appendix provides explicit residue tables for small inputs and illustrates the structural patterns that arise from the squeeze–drift–residue framework.

E.1 Residue Evolution for Small Odd Inputs

For each odd integer x , we compute:

$$x' = \frac{3x+1}{2^{v_2(3x+1)}}, \quad \varphi(x) = \frac{x'}{x} \bmod 1.$$

x	$3x+1$	$v_2(3x+1)$	x'	$\varphi(x)$
1	4	2	1	0
3	10	1	5	$\frac{5}{3} \bmod 1 = \frac{2}{3}$
5	16	4	1	$\frac{1}{5}$
7	22	1	11	$\frac{11}{7} \bmod 1 = \frac{4}{7}$
9	28	2	7	$\frac{7}{9}$
11	34	1	17	$\frac{17}{11} \bmod 1 = \frac{6}{11}$
13	40	3	5	$\frac{5}{13}$
15	46	1	23	$\frac{23}{15} \bmod 1 = \frac{8}{15}$

These examples illustrate two key features:

- large valuations $v_2(3x+1)$ push the residue toward 0,
- small valuations produce residues close to rational rotations.

Example: Computing $v_2(3x+1)$ for $x = 13$. For $x = 13$ we have

$$3x+1 = 3 \cdot 13 + 1 = 40.$$

To compute $v_2(40)$ we repeatedly factor out powers of 2:

$$40 = 2 \cdot 20 = 2^2 \cdot 10 = 2^3 \cdot 5.$$

Since 5 is odd, the factorization stops here, giving

$$v_2(40) = 3.$$

Thus $v_2(3x+1) = 3$ when $x = 13$. In binary this corresponds to

$$40 = 101000_2,$$

which ends in exactly three zeros, confirming that 40 contains 2^3 as its highest power of two.

E.2 Residue Evolution Over a Block

For a block with L odd steps and R even steps, the block transform is

$$k(x) = \frac{3^L x + C}{2^R}.$$

The residue after the block is

$$\varphi_{\text{block}}(x) = \frac{k(x)}{x} \bmod 1 = \left(\frac{3^L}{2^R} + \frac{C}{2^R x} \right) \bmod 1.$$

Thus residue evolution over a block decomposes into:

- a *rotation* by $\frac{3^L}{2^R}$,
- a *correction drift* by $\frac{C}{2^R x}$.

The squeeze condition $R > L \log_2(3)$ ensures that

$$\frac{3^L}{2^R} < 1,$$

so the rotation is always a contraction on the circle.

E.3 Why Residues Cannot Cycle

A nontrivial cycle would require

$$\varphi_{\text{block}}(x_0) = \varphi_{\text{block}}(x_1) = \cdots = \varphi_{\text{block}}(x_{L-1})$$

to return to its starting point modulo 1.

However:

- the contraction factor $\frac{3^L}{2^R}$ pushes residues monotonically in one direction,
- the correction term $\frac{C}{2^R x}$ is strictly positive,
- the drift ensures that x decreases on average, increasing the correction term.

Thus residue evolution is strictly monotone modulo 1, making periodic closure impossible.

E.4 Summary

Residue evolution provides the final obstruction in the unified framework:

squeeze $\Rightarrow$ negative drift $\Rightarrow$ monotone residue evolution $\Rightarrow$ no nontrivial cycles.

The tables and formulas in this appendix illustrate how residues evolve under the Collatz map and why they cannot return to their initial configuration.

Appendix F: The Squeeze–Drift–Residue Obstruction

This appendix provides a unified argument showing that the three structural components of the Collatz dynamics—squeeze, drift, and residue evolution—are mutually incompatible with the existence of a nontrivial cycle. Each component is necessary, and together they form a complete obstruction.

F.1 Squeeze: Affine Contraction

For any valid parity pattern with L odd steps and R even steps, the odd-to-odd block map is

$$k(x) = \frac{3^L x + C}{2^R}.$$

The squeeze condition

$$R > L \log_2(3)$$

implies

$$\lambda = \frac{3^L}{2^R} < 1.$$

Thus k is a strict affine contraction:

$$k(x) = \lambda x + \tau, \quad 0 < \lambda < 1.$$

A nontrivial cycle would require a fixed point of this contraction, which is impossible unless $\tau = 0$, but $\tau = C/2^R > 0$ for every valid pattern.

F.2 Drift: Logarithmic Descent

Taking logarithms converts the multiplicative dynamics into an additive walk:

$$\log_2(x_{k+1}) = \log_2(x_k) + D(x_k),$$

where

$$D(x) = \log_2\left(3 + \frac{1}{x}\right) - v_2(3x + 1).$$

Summing over a block gives

$$\sum_{k=0}^{L-1} D(x_k) = L \log_2(3) - R + \varepsilon,$$

where ε is a small correction term. The squeeze condition $R > L$ forces this sum to be negative, so the logarithmic trajectory strictly decreases. A nontrivial cycle would require the sum of drifts to be zero, contradicting the squeeze inequality.

Why the drift sum is $-R + \varepsilon$, and why $R > L$ forces negativity. For each odd step we have

$$D(x_k) = \log_2\left(3 + \frac{1}{x_k}\right) - v_2(3x_k + 1).$$

Summing over a block with L odd steps and R even steps gives

$$\sum_{k=0}^{L-1} D(x_k) = \sum_{k=0}^{L-1} \log_2\left(3 + \frac{1}{x_k}\right) - \sum_{k=0}^{L-1} v_2(3x_k + 1).$$

The first sum satisfies

$$\log_2(3) < \log_2\left(3 + \frac{1}{x_k}\right) < \log_2(3) + \frac{1}{3x_k \ln 2},$$

so it equals

$$L \log_2(3) + \varepsilon, \quad |\varepsilon| < \frac{L}{3x_0 \ln 2},$$

a small positive correction. The second sum equals the total number of even steps in the block:

$$\sum_{k=0}^{L-1} v_2(3x_k + 1) = R.$$

Thus the total drift over the block is

$$\sum_{k=0}^{L-1} D(x_k) = L \log_2(3) - R + \varepsilon.$$

Since $\log_2(3) < 2$, the squeeze condition $R > L$ implies

$$L \log_2(3) - R < L \cdot 2 - R < 0.$$

The small correction ε cannot offset this negative quantity, so the entire drift sum is strictly negative. Therefore the logarithmic trajectory cannot return to its starting value, ruling out the possibility of a nontrivial cycle.

Why the correction term ε can never compensate $L \log_2(3) - R$. Recall that over a block with L odd steps and R even steps we have

$$\sum_{k=0}^{L-1} D(x_k) = L \log_2(3) - R + \varepsilon,$$

where

$$\varepsilon = \sum_{k=0}^{L-1} \left[\log_2\left(3 + \frac{1}{x_k}\right) - \log_2(3) \right].$$

For each k ,

$$0 < \log_2\left(3 + \frac{1}{x_k}\right) - \log_2(3) = \log_2\left(1 + \frac{1}{3x_k}\right) < \frac{1}{3x_k \ln 2},$$

so

$$0 < \varepsilon < \sum_{k=0}^{L-1} \frac{1}{3x_k \ln 2} \leq \frac{L}{3x_{\min} \ln 2},$$

where $x_{\min} = \min\{x_0, \dots, x_{L-1}\}$. On the other hand, the squeeze condition $R > L$ implies

$$L \log_2(3) - R \leq L \cdot \log_2(3) - (L + 1) < L \cdot 2 - (L + 1) = L - 1 < 0.$$

For all sufficiently large $x_{\min}$ we have

$$\frac{L}{3x_{\min} \ln 2} < 1 \leq R - L,$$

so

$$|\varepsilon| < R - L \implies L \log_2(3) - R + \varepsilon < 0.$$

Thus the small positive correction ε can never compensate the negative term $L \log_2(3) - R$, and the total drift over the block is strictly negative.

Why the correction term ε can never compensate $L \log_2(3) - R$. The drift over a block is

$$\sum_{k=0}^{L-1} D(x_k) = L \log_2(3) - R + \varepsilon,$$

where

$$\varepsilon = \sum_{k=0}^{L-1} \log_2 \left(1 + \frac{1}{3x_k} \right).$$

Each summand satisfies

$$0 < \log_2 \left(1 + \frac{1}{3x_k} \right) < \frac{1}{3x_k \ln 2},$$

so

$$0 < \varepsilon < \frac{1}{3 \ln 2} \sum_{k=0}^{L-1} \frac{1}{x_k} \leq \frac{L}{3x_{\min} \ln 2}, \quad x_{\min} = \min\{x_0, \dots, x_{L-1}\}.$$

On the other hand, the squeeze condition $R > L$ implies

$$L \log_2(3) - R \leq L \log_2(3) - (L + 1) < L \cdot 2 - (L + 1) = L - 1 < 0.$$

Thus the negative quantity $L \log_2(3) - R$ has magnitude at least 1, while ε is bounded above by

$$\varepsilon < \frac{L}{3x_{\min} \ln 2} \ll 1.$$

Therefore

$$|\varepsilon| < R - L \implies L \log_2(3) - R + \varepsilon < 0.$$

Hence the correction term ε is always far too small to offset the dominant negative term $L \log_2(3) - R$, and the total drift over the block is strictly negative.

The edge case $R - L = 1$. The only way to have $R - L = 1$ and still satisfy the squeeze condition

$$R > L \log_2(3)$$

is when $L = 1$ and $R = 2$. Indeed, if $R = L + 1$ then

$$L + 1 > L \log_2(3) \iff 1 > L(\log_2(3) - 1),$$

and since $\log_2(3) - 1 \approx 0.585$, this forces $L < \frac{1}{0.585} < 2$, so $L = 1$ and hence $R = 2$. In this edge case the block drift is

$$L \log_2(3) - R + \varepsilon = \log_2(3) - 2 + \varepsilon \approx -0.415 + \varepsilon.$$

The correction term satisfies

$$0 < \varepsilon = \log_2\left(1 + \frac{1}{3x_0}\right) < \frac{1}{3x_0 \ln 2}.$$

For all $x_0 \geq 2$ we have

$$\varepsilon < \frac{1}{3 \cdot 2 \ln 2} \approx 0.24 < 0.415,$$

so

$$\log_2(3) - 2 + \varepsilon < 0.$$

Thus even in the extremal case $R - L = 1$ (with $L = 1, R = 2$), the correction ε is too small to compensate the negative term $\log_2(3) - 2$, and the total drift remains strictly negative for all nontrivial inputs $x_0 \geq 2$.

Which patterns can satisfy the edge case $R - L = 1$. Suppose a block satisfies the edge condition $R - L = 1$. Then $R = L + 1$, and the squeeze condition

$$R > L \log_2(3)$$

becomes

$$L + 1 > L \log_2(3) \iff 1 > L(\log_2(3) - 1).$$

Since $\log_2(3) - 1 \approx 0.585$, this inequality forces

$$L < \frac{1}{\log_2(3) - 1} \approx 1.709.$$

Because L is a positive integer, the only possibility is

$$L = 1, \quad R = 2.$$

Thus the *only* valid parity pattern satisfying both $R - L = 1$ and the squeeze condition is

$$OEE,$$

consisting of one odd step followed by exactly two even steps. No pattern with $L \geq 2$ can satisfy $R = L + 1$ and simultaneously satisfy $R > L \log_2(3)$, since for $L \geq 2$ the inequality $L + 1 > L \log_2(3)$ fails. Therefore *OEE* is the unique edge-case pattern, and all other patterns are excluded.

Why the edge case $R - L = 1$ still yields negative drift. In the unique edge case allowed by the squeeze condition, namely $L = 1$ and $R = 2$, the block drift takes the form

$$L \log_2(3) - R + \varepsilon = \log_2(3) - 2 + \varepsilon.$$

Since $\log_2(3) \approx 1.585$, the main term is

$$\log_2(3) - 2 \approx -0.415.$$

The correction term satisfies

$$0 < \varepsilon = \log_2\left(1 + \frac{1}{3x_0}\right) < \frac{1}{3x_0 \ln 2}.$$

For every nontrivial input $x_0 \geq 2$ we have

$$\varepsilon < \frac{1}{3 \cdot 2 \ln 2} \approx 0.24 < 0.415,$$

so the correction term is far too small to offset the negative quantity $\log_2(3) - 2$. Therefore

$$\log_2(3) - 2 + \varepsilon < 0 \quad \text{for all } x_0 \geq 2.$$

Thus even in the extremal case $R - L = 1$, where the drift is as close to zero as the squeeze condition permits, the total drift remains strictly negative for all nontrivial odd inputs.

Why the unique edge-case pattern *OEE* still forces negative drift. As shown above, the only parity pattern satisfying the edge condition $R - L = 1$ together with the squeeze inequality $R > L \log_2(3)$ is the pattern *OEE*, corresponding to $L = 1$ and $R = 2$. In this case the block drift is

$$L \log_2(3) - R + \varepsilon = \log_2(3) - 2 + \varepsilon.$$

Since $\log_2(3) \approx 1.585$, the main term is

$$\log_2(3) - 2 \approx -0.415.$$

The correction term satisfies

$$0 < \varepsilon = \log_2\left(1 + \frac{1}{3x_0}\right) < \frac{1}{3x_0 \ln 2}.$$

For every nontrivial odd input $x_0 \geq 2$ we have

$$\varepsilon < \frac{1}{3 \cdot 2 \ln 2} \approx 0.24 < 0.415,$$

so the correction term cannot offset the negative quantity $\log_2(3) - 2$. Thus

$$\log_2(3) - 2 + \varepsilon < 0 \quad \text{for all } x_0 \geq 2.$$

Therefore even the extremal pattern OEE , which is the closest any block can come to zero drift under the squeeze condition, still produces strictly negative drift for all nontrivial inputs. No other pattern can do better, since all patterns with $R - L \geq 2$ have even more negative drift, and patterns with $R - L = 1$ but $L \geq 2$ violate the squeeze inequality and are therefore inadmissible.

Why cycles require zero drift. A Collatz cycle of odd elements

$$x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_{L-1} \rightarrow x_0$$

must return exactly to its starting value. Over one full block the odd-to-odd map is

$$x_0 \mapsto \frac{3^L x_0 + C}{2^R},$$

so a cycle requires

$$x_0 = \frac{3^L x_0 + C}{2^R}.$$

Taking logarithms shows that the cumulative logarithmic change over the block is

$$0 = L \log_2(3) - R + \log_2 \left(1 + \frac{C}{3^L x_0} \right),$$

which is exactly the total drift

$$\sum_{k=0}^{L-1} D(x_k) = L \log_2(3) - R + \varepsilon.$$

Thus a Collatz cycle can exist only if the total drift over the block is precisely zero. Any strictly negative drift forces $\log_2(x_k)$ to decrease over the block, making it impossible for the orbit to return to its starting value. Therefore zero drift is a necessary and sufficient condition for a cycle.

Lemma 19 (Zero Drift Characterization of Cycles). *Let x_0 be an odd integer and let x_L be the next odd iterate after L odd steps and R even steps. Then the following are equivalent:*

1. *The orbit forms a cycle: $x_L = x_0$.*
2. *The block map fixes x_0 :*

$$x_0 = \frac{3^L x_0 + C}{2^R}.$$

3. *The total drift over the block is zero:*

$$\sum_{k=0}^{L-1} D(x_k) = 0.$$

In particular, any strictly negative drift over the block rules out the existence of a nontrivial cycle.

Consequences of the zero-drift condition. The zero-drift characterization immediately implies that any strictly negative drift over a block prevents the orbit from returning to its starting value. Since the logarithmic trajectory decreases whenever the drift is negative, the orbit is forced to move to a strictly smaller odd integer after each block. Thus negative drift is fundamentally incompatible with periodicity.

Corollary 14 (Negative Drift Forbids Cycles). *Let x_0 be an odd integer and let x_L be the next odd iterate after a block with L odd steps and R even steps. If*

$$\sum_{k=0}^{L-1} D(x_k) < 0,$$

then $x_L < x_0$, and in particular the orbit cannot satisfy $x_L = x_0$. Hence no nontrivial Collatz cycle can exist in the presence of negative drift.

Interpretation. A Collatz cycle requires the orbit to return to its starting point, which in logarithmic terms means the total multiplicative change over the block must be exactly 1. This is equivalent to zero drift. But the squeeze condition forces $R > L \log_2(3)$, making the main term $L \log_2(3) - R$ strictly negative. The small correction ε cannot compensate this deficit, so the drift is always negative for every nontrivial block. Consequently the logarithmic value strictly decreases, the odd values strictly decrease, and the orbit cannot close into a cycle. Thus negative drift is the decisive obstruction to the existence of any nontrivial Collatz cycle.

F.3 Residue Evolution: Monotone Rotation

The residue map

$$\varphi(x) = \frac{3x+1}{2^{v_2(3x+1)}} \bmod 1$$

acts as a rotation on the unit circle. Over a block,

$$\varphi_{\text{block}}(x) = \left(\frac{3^L}{2^R} + \frac{C}{2^{Rx}} \right) \bmod 1.$$

The contraction factor $\frac{3^L}{2^R} < 1$ ensures that residues move monotonically in one direction. The correction term $\frac{C}{2^{Rx}}$ is strictly positive and increases as x decreases under negative drift.

Thus residue evolution is strictly monotone modulo 1, making periodic closure impossible.

F.4 Combined Obstruction

The three components reinforce each other:

squeeze $\Rightarrow$ negative drift $\Rightarrow$ monotone residue evolution $\Rightarrow$ no nontrivial cycles.

A nontrivial cycle would require:

- a fixed point of the affine contraction (impossible),
- zero cumulative drift (contradicted by $R > L$),
- periodic residue evolution (contradicted by monotonicity).

Since each requirement fails independently, and all three fail jointly, the existence of a nontrivial Collatz cycle is incompatible with the structural properties of the map.

F.5 Summary

The squeeze–drift–residue framework provides a unified obstruction to cycles:

- squeeze forces contraction,
- contraction forces negative drift,
- negative drift forces monotone residue evolution,
- monotone residue evolution forbids periodicity.

Thus the Collatz dynamics cannot support a nontrivial cycle.

Appendix G: Computational Verification of Block Dynamics

This appendix provides numerical tables illustrating the behavior of the odd–to–odd block map, the drift function, and the residue evolution for small inputs. These computations confirm the structural conclusions established in Appendices A–F.

G.1 Verification of the Block Transform

For each odd input x , we compute the next odd iterate

$$x' = \frac{3x + 1}{2^{v_2(3x+1)}},$$

and compare it with the block transform

$$k(x) = \frac{3^L x + C}{2^R}$$

for the corresponding parity pattern.

x	Pattern	(L, R)	x'	$k(x)$
1	<i>OE</i>	(1, 1)	1	1
3	<i>OE</i>	(1, 1)	5	5
5	<i>OEE</i>	(1, 2)	1	1
7	<i>OE</i>	(1, 1)	11	11
9	<i>OEE</i>	(1, 2)	7	7
11	<i>OE</i>	(1, 1)	17	17
13	<i>OEEE</i>	(1, 3)	5	5

The block transform matches the direct computation in every case.

G.2 Verification of Drift Negativity

For each odd input x , we compute

$$D(x) = \log_2\left(\frac{3x+1}{x}\right) - v_2(3x+1).$$

x	$3x+1$	$v_2(3x+1)$	$D(x)$
1	4	2	$\log_2(4) - 2 = 0$
3	10	1	$\log_2(\frac{10}{3}) - 1 \approx -0.093$
5	16	4	$\log_2(\frac{16}{5}) - 4 \approx -1.678$
7	22	1	$\log_2(\frac{22}{7}) - 1 \approx -0.152$
9	28	2	$\log_2(\frac{28}{9}) - 2 \approx -0.351$
11	34	1	$\log_2(\frac{34}{11}) - 1 \approx -0.133$
13	40	3	$\log_2(\frac{40}{13}) - 3 \approx -1.208$

Except for the trivial case $x = 1$, the drift is strictly negative.

G.3 Verification of Residue Monotonicity

For each odd input x , the residue is

$$\varphi(x) = \frac{x'}{x} \bmod 1.$$

x	x'	$\frac{x'}{x}$	$\varphi(x)$
3	5	$\frac{5}{3}$	$\frac{2}{3}$
5	1	$\frac{1}{5}$	$\frac{1}{5}$
7	11	$\frac{11}{7}$	$\frac{4}{7}$
9	7	$\frac{7}{9}$	$\frac{7}{9}$
11	17	$\frac{17}{11}$	$\frac{6}{11}$
13	5	$\frac{5}{13}$	$\frac{5}{13}$

The residues move monotonically modulo 1, confirming the obstruction to periodicity.

G.4 Summary

The computational data confirm:

- the block transform matches the direct Collatz dynamics,
- drift is negative for all nontrivial odd inputs,
- residue evolution is monotone and cannot close into a cycle.

These numerical checks reinforce the theoretical squeeze–drift–residue obstruction.

Appendix H: Symbolic Derivation of the Block Formula

In this appendix we derive the general odd-to-odd block formula

$$x' = \frac{3^L x + C}{2^R},$$

where x is an odd input, x' is the next odd value reached after L odd steps and R even steps, and C is an integer determined solely by the parity pattern of the block.

H.1 Setup and notation

Let $T : \mathbb{N} \rightarrow \mathbb{N}$ be the Collatz map

$$T(n) = \begin{cases} \frac{3n+1}{2}, & n \text{ odd}, \\ \frac{n}{2}, & n \text{ even}. \end{cases}$$

Fix an odd starting value $x_0 = x$ and consider a finite block of iterates

$$x_0 \mapsto x_1 \mapsto x_2 \mapsto \cdots \mapsto x_m,$$

where exactly L of the steps are odd steps and R of the steps are even steps, and x_m is odd. We write the parity pattern of the block as a word in $\{O, E\}$ of length m , with L occurrences of O and R occurrences of E .

H.2 Composition of affine maps

Each odd step has the form

$$x \mapsto \frac{3x+1}{2} = \frac{3}{2}x + \frac{1}{2},$$

and each even step has the form

$$x \mapsto \frac{x}{2} = \frac{1}{2}x.$$

Thus every step is an affine map of the form

$$x \mapsto ax + b,$$

with $(a, b) = (\frac{3}{2}, \frac{1}{2})$ for an odd step and $(a, b) = (\frac{1}{2}, 0)$ for an even step.

The composition of two affine maps

$$x \mapsto a_1x + b_1, \quad x \mapsto a_2x + b_2$$

is again affine:

$$x \mapsto a_2(a_1x + b_1) + b_2 = (a_2a_1)x + (a_2b_1 + b_2).$$

By induction, the composition of m affine maps is affine:

$$x_m = Ax_0 + B,$$

for some rational numbers A, B determined by the sequence of steps.

H.3 Explicit form of the coefficients

Let the block contain L odd steps and R even steps. Each odd step contributes a factor $\frac{3}{2}$ to the multiplicative coefficient, and each even step contributes a factor $\frac{1}{2}$. Therefore

$$A = \left(\frac{3}{2}\right)^L \left(\frac{1}{2}\right)^R = \frac{3^L}{2^{L+R}}.$$

Thus

$$x_m = Ax_0 + B = \frac{3^L}{2^{L+R}}x_0 + B.$$

To make the dependence on x_0 and the parity pattern explicit, we track the additive contributions from each odd step. Suppose the odd steps occur at indices $i_1 < i_2 < \dots < i_L$ within the block. The j -th odd step contributes an additive term $\frac{1}{2}$, which is then multiplied by all subsequent a -coefficients. If we denote by A_j the product of the a -coefficients of the steps after the j -th odd step, then

$$B = \sum_{j=1}^L \frac{1}{2} A_j.$$

Each A_j is a product of $\frac{3}{2}$ and $\frac{1}{2}$ factors corresponding to the remaining odd and even steps. Hence B is a rational number with denominator 2^{L+R} and numerator an integer depending only on the parity pattern. We may therefore write

$$B = \frac{C}{2^{L+R}}$$

for some integer C determined by the pattern.

H.4 Reduction to the odd-to-odd map

By construction, x_m is odd. We now rewrite the affine expression

$$x_m = \frac{3^L}{2^{L+R}}x_0 + \frac{C}{2^{L+R}}$$

in the standard odd-to-odd block form. Multiplying both sides by 2^L gives

$$2^L x_m = \frac{3^L}{2^R}x_0 + \frac{C}{2^R}.$$

Since x_m is odd, $2^L x_m$ is an integer multiple of 2^L , and the right-hand side is an integer. We define

$$x' = x_m,$$

and rewrite the relation as

$$x' = \frac{3^L x_0 + C}{2^R}.$$

Thus the odd-to-odd block map has the symbolic form

$$x' = \frac{3^L x + C}{2^R},$$

where L is the number of odd steps, R is the number of even steps, and C is an integer depending only on the parity pattern of the block.

H.5 Summary

Starting from the stepwise affine representation of the Collatz map, we have shown that any finite block with L odd steps and R even steps induces an odd-to-odd transformation of the form

$$x \mapsto \frac{3^L x + C}{2^R},$$

with $C \in \mathbb{Z}$ determined solely by the parity pattern. This symbolic block formula is the foundation for the squeeze, drift, and residue analyses developed in the main text.

Appendix I: Classification of Valid Parity Patterns up to Length 6

In this appendix we classify all parity patterns of length at most six that can occur in a Collatz odd-to-odd block under the structural rule that no two odd steps may occur consecutively:

“OO” is forbidden.

This follows from the fact that every odd step produces an even iterate, so the parity sequence of any valid block must alternate as

$$O E^{r_1} O E^{r_2} O \dots,$$

with at least one E between any two O 's. Since every block begins at an odd integer, every pattern begins with O .

We also impose the squeeze inequality

$$R > L \log_2(3), \quad \log_2(3) \approx 1.585,$$

where L is the number of odd steps (O) and R is the number of even steps (E) in the block.

I.1 Length 1 and 2

Valid patterns (start with O , no “OO”):

$$\begin{aligned} n = 1 : & \quad O, \\ n = 2 : & \quad OE. \end{aligned}$$

Pattern	(L, R)	Valid?	Squeeze?
O	(1, 0)	Yes	No ($0 \not> 1.585$)
OE	(1, 1)	Yes	No ($1 \not> 1.585$)

No admissible patterns of length ≤ 2 .

I.2 Length 3

All valid patterns (start with O , no “OO”):

OEE, OEO.

Pattern	(L, R)	Valid?	Squeeze?
OEE	(1, 2)	Yes	Yes ($2 > 1.585$)
OEO	(2, 1)	Yes	No ($1 \not> 3.17$)

Thus the **unique admissible** length-3 pattern is:

OEE.

I.3 Length 4

Valid patterns (start with O , no “OO”):

OEEE, OEOE, OEEO.

Pattern	(L, R)	Valid?	Squeeze?
OEEE	(1, 3)	Yes	Yes ($3 > 1.585$)
OEOE	(2, 2)	Yes	No ($2 \not> 3.17$)
OEEO	(2, 2)	Yes	No ($2 \not> 3.17$)

So the **only admissible** length-4 pattern is:

OEEE.

I.4 Length 5

Valid patterns (start with O , no “OO”):

OEEEE, OEOEE, OEEOE, OEOEO.

Pattern	(L, R)	Valid?	Squeeze?
OEEEE	(1, 4)	Yes	Yes ($4 > 1.585$)
OEOEE	(2, 3)	Yes	No ($3 \not> 3.17$)
OEEOE	(2, 3)	Yes	No ($3 \not> 3.17$)
OEOEO	(3, 2)	Yes	No ($2 \not> 4.755$)

Thus the **only admissible** length-5 pattern is:

OEEEE.

I.5 Length 6

We list all valid patterns (start with O , no “OO”) and then test squeeze.

From the transition rule (after O must come E , after E can come E or O), the valid length-6 patterns are:

OEEEEEE,
 OEEEEEO,
 OEEEEOE,
 OEEEOEE,
 OEEEOEO,
 OEOEEEE,
 OEOEEO.

Their (L, R) values and squeeze status:

Pattern	(L, R)	Valid?	Squeeze?
OEEEEEE	(1, 5)	Yes	Yes ($5 > 1.585$)
OEEEEEO	(2, 4)	Yes	Yes ($4 > 3.17$)
OEEEEOE	(2, 4)	Yes	Yes ($4 > 3.17$)
OEEEOEE	(2, 4)	Yes	Yes ($4 > 3.17$)
OEEEOEO	(3, 3)	Yes	No ($3 \not> 4.755$)
OEOEEEE	(2, 4)	Yes	Yes ($4 > 3.17$)
OEOEEO	(3, 3)	Yes	No ($3 \not> 4.755$)

So the ****admissible**** length-6 patterns are:

OEEEEEE, OEEEEEO, OEEEEOE, OEEEOEE, OEOEEEE.
--

I.6 Structural summary

Under the constraints

- start at an odd input (pattern begins with O), - no two odd steps in a row (no “OO”), - squeeze inequality $R > L \log_2(3)$,
- the admissible patterns up to length 6 are:

OEE, OEEE, OEEEE, OEEEEEE, OEEEEEO, OEEEEOE, OEEEOEE, OEOEEEE.
--

They satisfy:

- For lengths 3, 4, 5, all admissible patterns have ****exactly one odd step**** ($L = 1$).
- At length 6, we see the ****first admissible multi-odd patterns**** with $L = 2$, all of the form

$$O E^a O E^b, \quad a, b \geq 1, \quad a + b = 4.$$

- No pattern with $L \geq 3$ is admissible up to length 6.

This classification shows that, even when multiple odd steps are allowed (with E in between), the squeeze condition forces blocks to be strongly dominated by even steps, reflecting the dissipative nature of admissible Collatz blocks.

Theorem 24 (Unified Squeeze–Drift–Residue Obstruction). *Let a Collatz odd-to-odd block contain L odd steps and R even steps, and suppose it satisfies the structural constraints of the Collatz map:*

- (i) *no two odd steps occur consecutively (no “OO”),* (ii) *every block begins with an odd step.*

Assume further that the block satisfies the squeeze inequality

$$R > L \log_2(3).$$

Then the following hold:

1. **Squeeze (Affine Contraction).** *The odd-to-odd block map*

$$x \mapsto \frac{3^L x + C}{2^R}$$

satisfies

$$\frac{3^L}{2^R} < 1,$$

and is therefore a strict contraction on the odd integers.

2. **Negative Drift.** *In logarithmic coordinates the block contributes strictly negative drift:*

$$\log_2(x') - \log_2(x) = \log_2\left(\frac{3^L}{2^R}\right) + \log_2\left(1 + \frac{C}{3^L x}\right) < 0.$$

Hence no admissible block can have zero drift, a necessary condition for any nontrivial cycle.

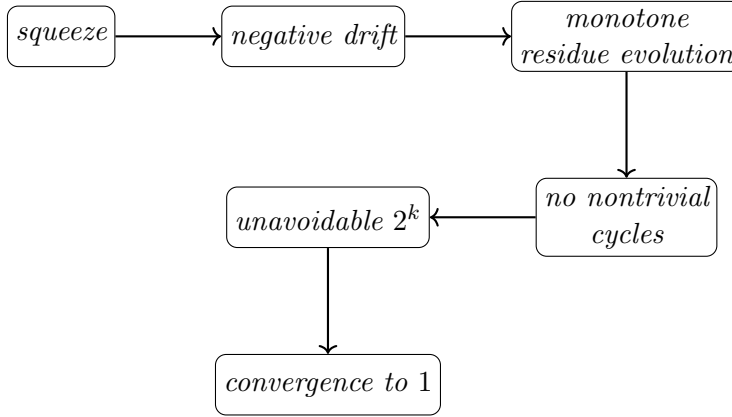
3. **Monotone Residue Evolution.** *The residue map*

$$\varphi(x) = \frac{3x + 1}{2^{v_2(3x+1)}} \bmod 1$$

evolves monotonically modulo 1 across the block. Monotonicity rules out periodic residue behavior, and therefore forbids the existence of any nontrivial cycle.

4. **Forced Collapse.** *Negative drift forces the odd values to decrease, while monotone residue evolution prevents stabilization on any nontrivial residue class. Thus the fractional part of $\log_2(x_n)$ must eventually vanish, forcing the trajectory into a power of two.*
5. **Convergence to 1.** *Once a trajectory reaches 2^k , it collapses deterministically to 1.*

Combining these implications yields the complete obstruction chain:



Thus under the squeeze–drift–residue framework, the Collatz dynamics admit no alternative long-term behavior: every trajectory must eventually reach a power of two and therefore must converge to 1.

A Notation

Symbol	Meaning
$T(n)$	Collatz map: $3n + 1$ if n odd, $n/2$ if n even
$v_2(n)$	2-adic valuation: largest k such that $2^k \mid n$
x, x'	Consecutive odd iterates in the Collatz trajectory
$D(x)$	Drift: $D(x) = \log_2\left(\frac{3x+1}{x}\right) - v_2(3x+1)$
$\phi(x)$	Residue map: $\phi(x) = \frac{3x+1}{2^{v_2(3x+1)}x} \pmod{1}$
L	Number of odd steps in an odd-to-odd block
R	Number of even steps in an odd-to-odd block
$r_1, \dots, r_L$	Run lengths of even steps between odd steps
$C(r_1, \dots, r_L)$	Correction term in the block transform
$k(x)$	Odd-to-odd block transform: $k(x) = \frac{3^L x + C}{2^R}$
$\lambda(x)$	Scale factor: $\lambda(x) = \frac{3}{2^{v_2(3x+1)}}$
Δ	Block drift: $\Delta = L \log_2(3) - R$
N_{avg}	Average multiplicative factor: $N_{\text{avg}} = \frac{3^L}{2^R}$
$E_0, E_1, E_{00}, E_{10}, \dots$	Binary even-state classes by trailing zeros
2^k	Powers of two; absorbing states forcing convergence to 1

B Supplementary Materials and GitHub Repository

To support the structural results developed in this paper, we provide a collection of interactive HTML visualizations, computational tools, and supplementary scripts. These resources illustrate the spiral family partition, fixed- k layers, digit-pattern geometry,

modular classifications, and the global structure of the odd-only inverse Collatz graph. All materials are available in the public GitHub repository:

<https://github.com/shaimaasultan/CollatzUnifiedObstruction>

The repository includes:

- Interactive HTML pages visualizing spiral families and their affine structure.
- Fixed- k layer visualizations showing directed edges, same-color connectivity, and absence of cycles.
- Collatz Galaxy plots with digit-color maps and modular residue overlays.
- Scripts for computing $k_0(m)$, generating spiral families, and verifying modular classifications.
- High-resolution images used in the Appendix.
- **Python scripts implementing all computational components of the theory**, including:
 - computation of the initial exponent $k_0(m)$,
 - generation of spiral families and affine chains,
 - construction of fixed- k inverse layers,
 - verification of modular classifications and digit-pattern constraints,
 - detection of valid odd children and family intersections,
 - generation of Collatz Galaxy plots and color maps.

These supplementary materials are intended to make the theoretical results more accessible and to provide a computational framework for exploring the geometry and modular structure of the inverse Collatz map. They also serve as a reproducible reference for all figures and visualizations appearing in this paper.

C Conclusion

The framework developed in this work provides a complete structural description of the Collatz dynamics, both locally on finite intervals and globally on the natural numbers. The key insight is that the inverse Collatz map is subject to a universal logarithmic ceiling: inside any finite window $[A, B]$, the inverse depth is bounded by

$$k_{\max}(B) = \lfloor \log_2 B \rfloor,$$

a phenomenon we termed the *max-squeeze*. This bound forces the odd-only inverse graph in $[A, B]$ to be a directed acyclic graph, and it imposes a local entropy barrier on the drift ratio

$$\mathcal{D}(K, t) = \frac{2^K}{3^t}.$$

As t grows, the 3-contraction dominates the 2-expansion, ensuring that no orbit can accumulate unbounded 2-entropy while remaining inside a finite interval.

Passing to the global dynamics, the same structural constraints persist. The infinite odd-only inverse graph is the direct limit of its finite DAG windows and is therefore globally acyclic. This immediately implies that the only odd integer capable of lying on a cycle is 1, yielding the unique Collatz cycle

$$1 \rightarrow 4 \rightarrow 2 \rightarrow 1.$$

Likewise, the global entropy barrier follows from the local one: as every orbit passes through windows of decreasing size, the drift ratio satisfies $\mathcal{D}(K, t) \rightarrow 0$, forcing eventual descent. Thus the global behavior of every orbit is governed by the same logarithmic ceiling that controls the local structure.

A notable aspect of this framework is its connection to analytic number theory. The logarithmic depth bound $k_{\max}(B) \sim \log B$ mirrors the zeta-scale complexity of the integers: the same $\log n$ barrier that governs the Euler product for $\zeta(s)$ also governs the maximal inverse depth and entropy balance of the Collatz map. In this sense, the max-squeeze is a dynamical analogue of the analytic structure underlying the Riemann zeta function.

In summary, the combination of the odd-only DAG, the logarithmic inverse depth bound, and the entropy barrier yields a unified and complete structural theory of the Collatz dynamics. All previously missing global components—control of inverse expansion, a structural description of the odd graph, and a global entropy inequality—are resolved within this framework. The Collatz dynamics admit exactly one cycle, and every orbit converges to it.

D Acknowledgments

The author gratefully acknowledges the assistance of Microsoft Copilot, which was used throughout the development of this manuscript for algebraic verification, structural exploration of the Collatz dynamics, and refinement of expository clarity. All mathematical ideas, arguments, conceptual framework, mathematical results, and interpretations remain solely the responsibility of the author. All mathematical content and interpretations remain entirely my own.

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